Exponential Quantum Advantage in Processing Massive Classical Data

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with Alexander Zlokapa, Hartmut Neven, Ryan Babbush,
John Preskill, Jarrod McClean, Hsin-Yuan (Robert) Huang

arXiv:2604.07639

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Will QC impact our daily life? How?

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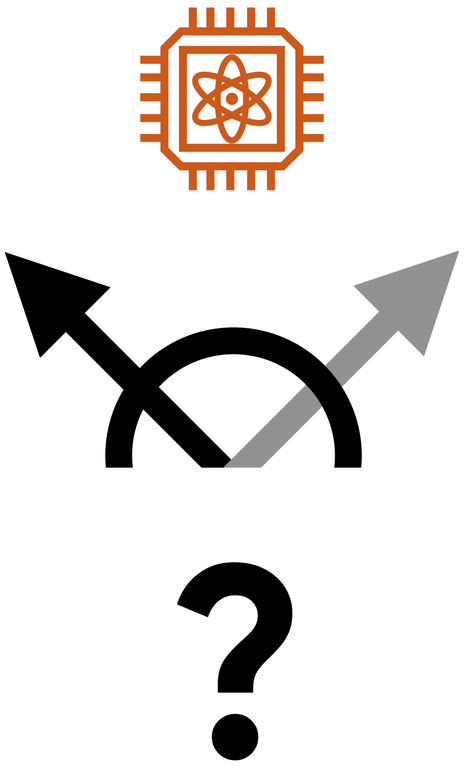
Utility of QC

Specialized devices
for niche tasks

General utility
in everyday life



Utility of QC

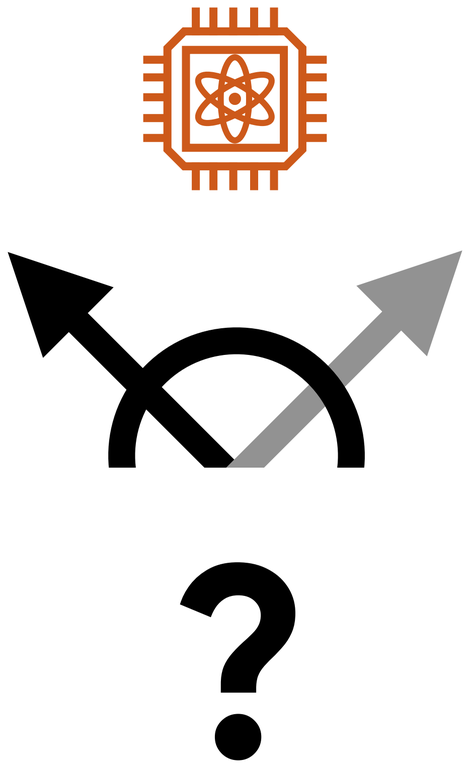
Specialized devices for niche tasks

Cryptanalysis

Shor'94

Quantum Simulation

Feynman'82, Lloyd'96



General utility in everyday life

?

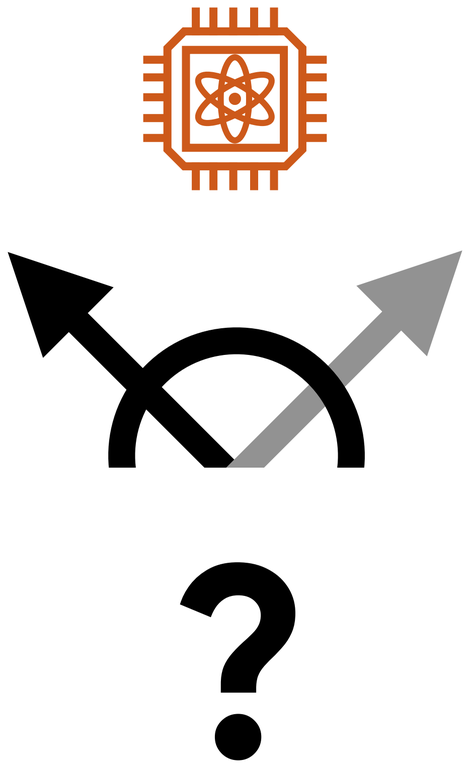
Utility of QC

Specialized devices for niche tasks

Special structures exploitable by QC but classically hard

e.g., inherently quantum

- Quantum Chemistry
- Condensed Matter Physics



General utility in everyday life

?

Utility of QC

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Special structures exploitable by QC but classically hard

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- Quantum Chemistry
- Condensed Matter Physics

General utility in everyday life

Unstructured massive classical data

AI

Healthcare

Finance

Media

Science

?

Engineering

Life in a classical world

Life in a classical world

Reviews

“One of the best movies.”

“Fantastic production.”

“Story makes no sense.”

Life in a classical world

Reviews

Classical Data

“One of the best movies.”

$\vec{x}_1$

“Fantastic production.”

$\vec{x}_2$

“Story makes no sense.”

$\vec{x}_3$

Life in a classical world

Reviews

Classical Data

“One of the best movies.”

$\vec{x}_1$

“Fantastic production.”

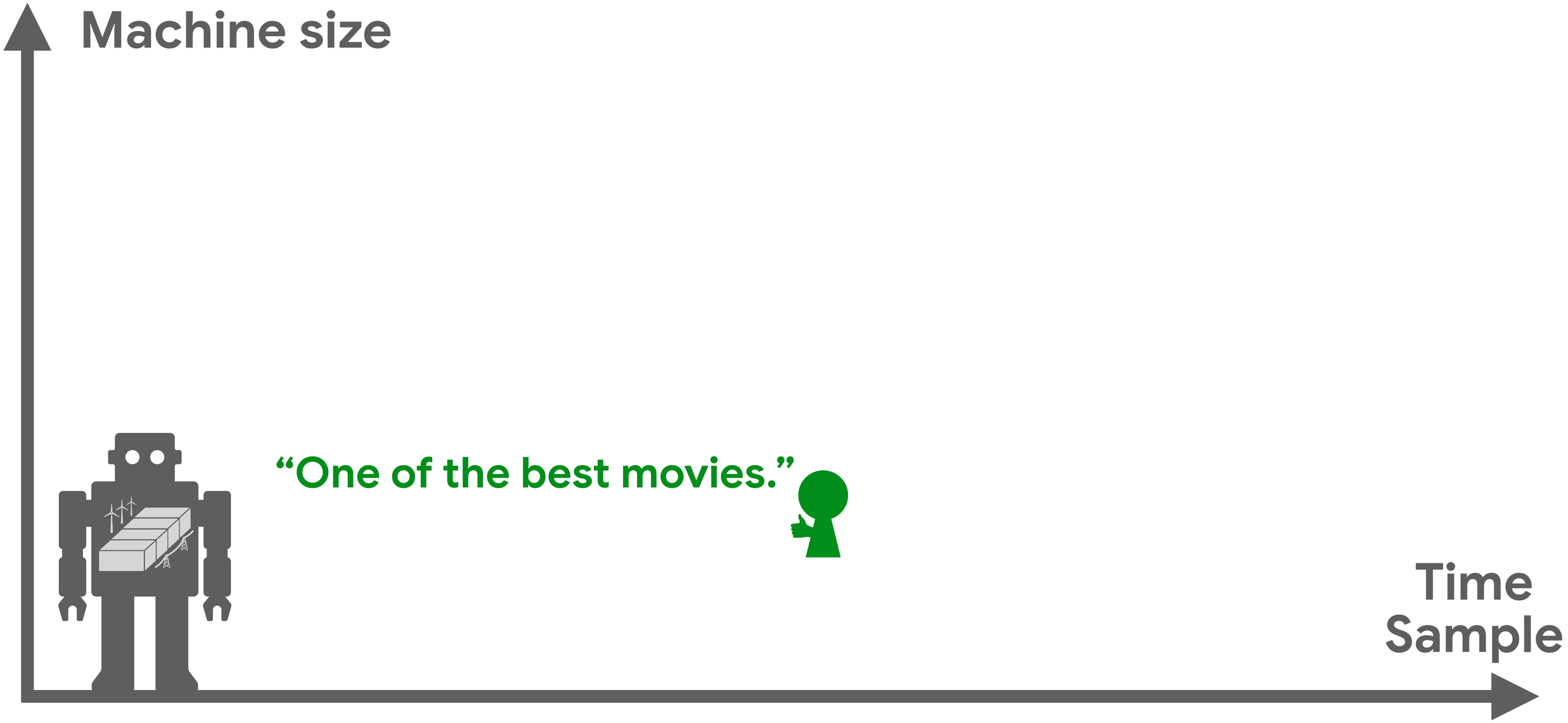
$\vec{x}_2$

“Story makes no sense.”

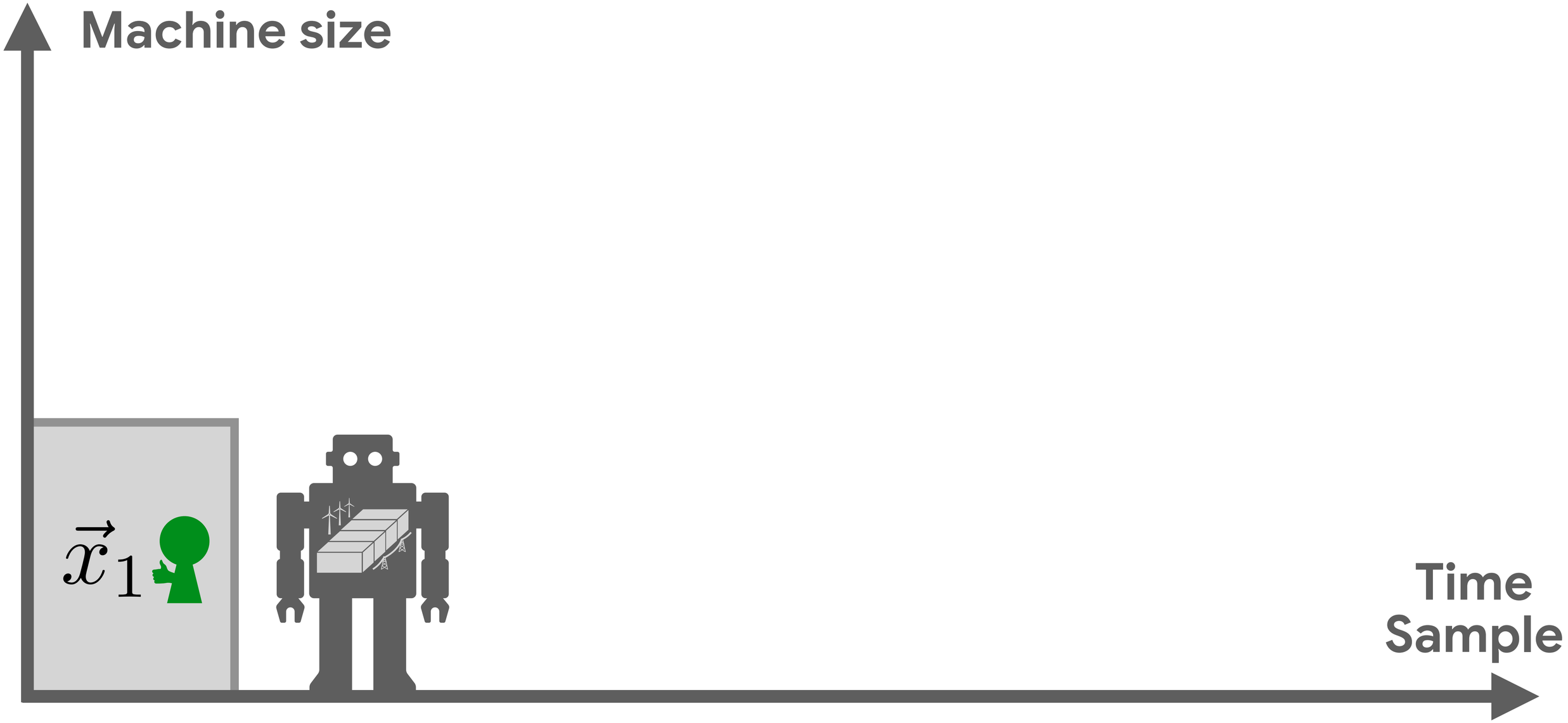
$\vec{x}_3$

Task: sentiment analysis $\vec{x}$

Classical ML



Classical ML



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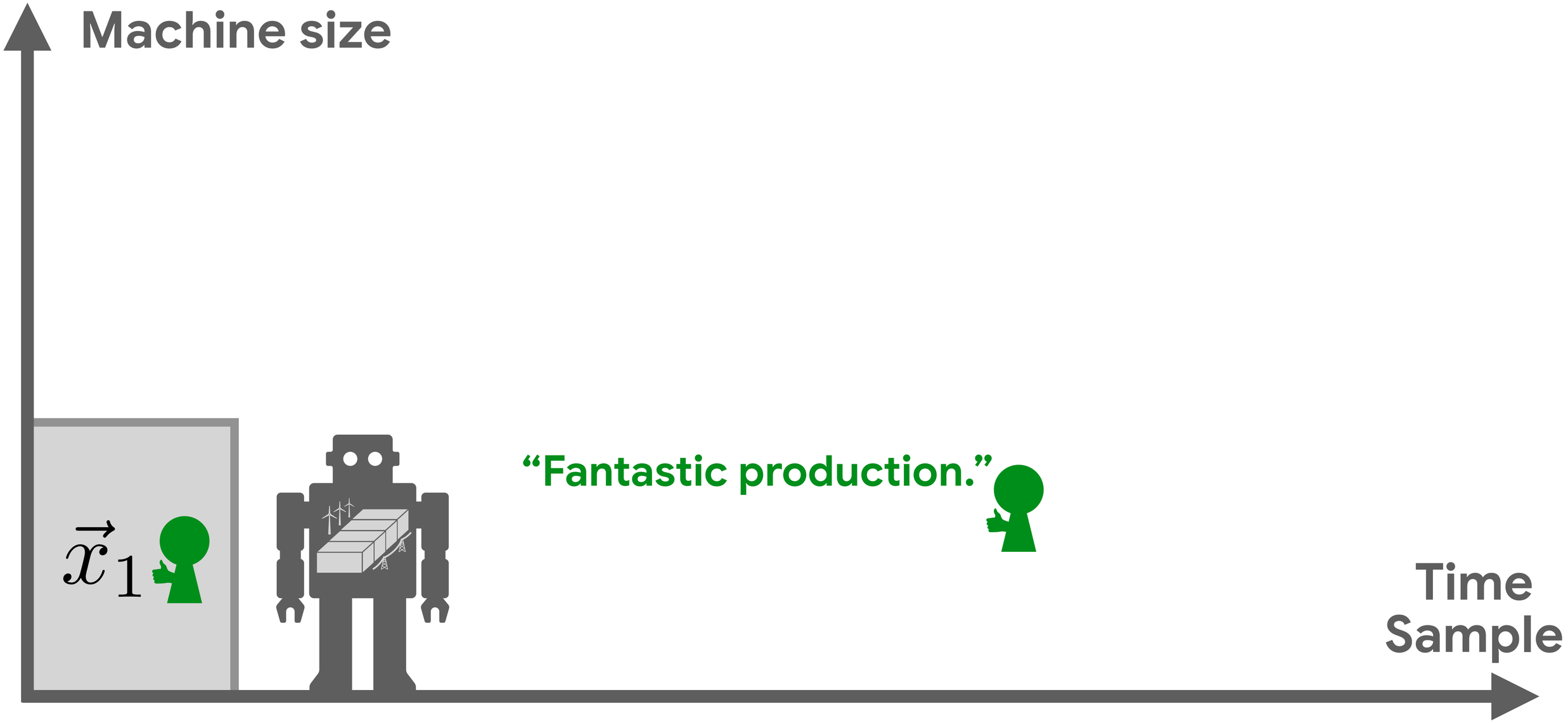

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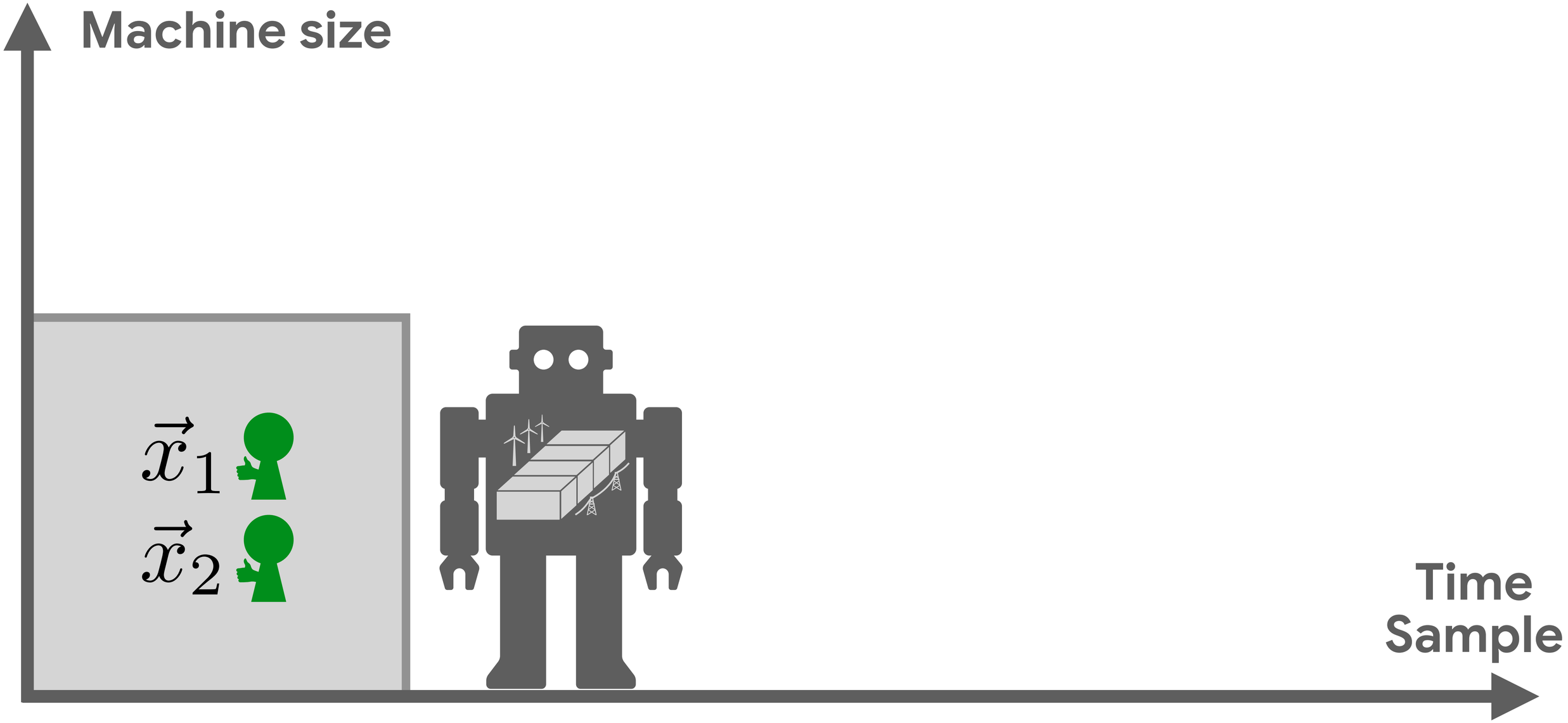

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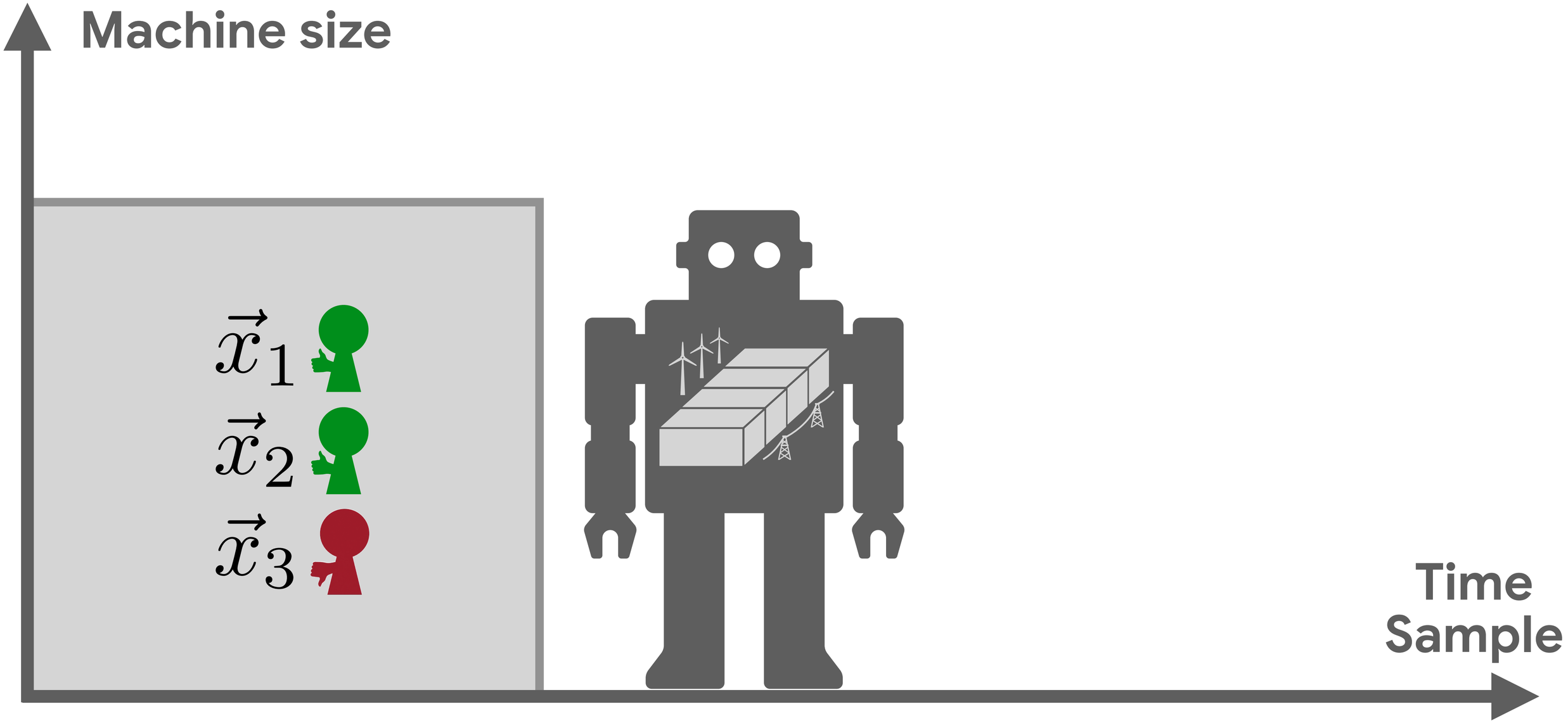
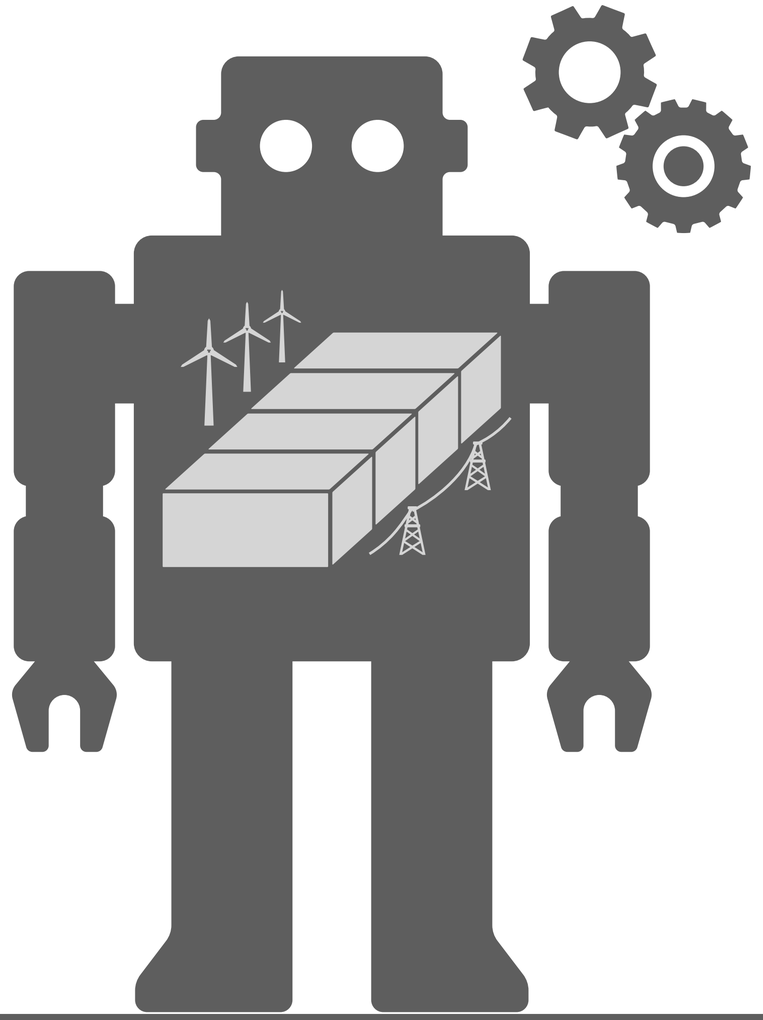


Classical ML

Machine size

Data Loading

Store the entire world.



Time
Sample

Classical ML

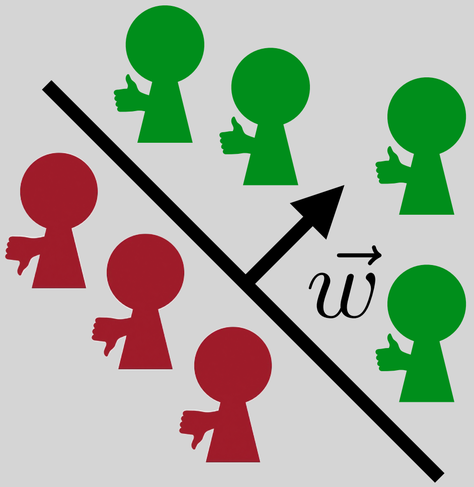
Machine size

Data Loading

Store the entire world.

Data Processing

Analyze the stored world.



Time
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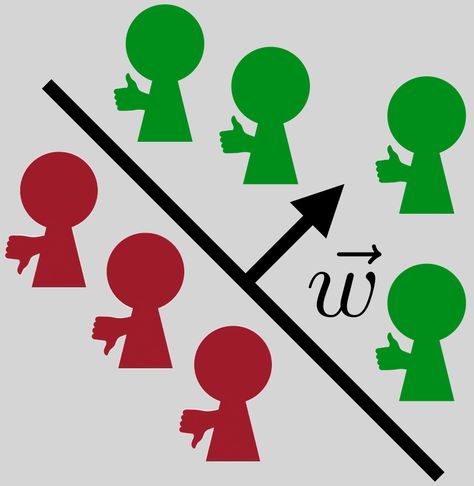


Testing

$\vec{x}$

predict

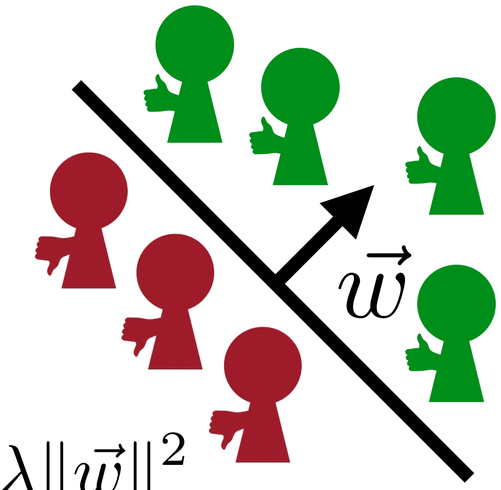
$\hat{y} = \text{sgn}(\vec{x} \cdot \vec{w})$

Time
Sample

Main Results

Classification: given data $(i, \vec{x}_i, y_i)$ randomly sampled from N examples, each with a D-dim sparse feature vector and a label, train a regularized D-dim linear classifier $\vec{w}$ such that we can predict the label of any classifiable sparse test vectors.

margin not too small



predict

$$\hat{y} = \text{sgn}(\vec{x} \cdot \vec{w})$$

$$\min_{\vec{w}} \|X\vec{w} - \vec{y}\|^2 + \lambda\|\vec{w}\|^2$$

Main Results

Classification: given data $(i, \vec{x}_i, y_i)$ randomly sampled from N examples, each with a D-dim sparse feature vector and a label, train a regularized D-dim linear classifier $\vec{w}$ such that we can predict the label of any classifiable sparse test vectors.

margin not too small

Theorem 1 (Classification)

With $\tilde{O}(N)$ samples, a quantum machine of $\mathrm{poly}(\log D)$ size can solve classification, whereas any classical machine of $O(D^{0.99})$ size cannot.

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margin not too small

Theorem 1 (Classification)

With $\tilde{O}(N)$ samples, a quantum machine of $\mathrm{poly}(\log D)$ size can solve classification, whereas any classical machine of $O(D^{0.99})$ size cannot.

Standard classical algorithm: $O(D)$ size and $O(N)$ samples

Main Results

Classification: given data $(i, \vec{x}_i, y_i)$ randomly sampled from N examples, each with a D-dim sparse feature vector and a label, train a regularized D-dim linear classifier $\vec{w}$ such that we can predict the label of any classifiable sparse test vectors.

margin not too small

Theorem 1 (Classification)

With $\tilde{O}(N)$ samples, a quantum machine of $\mathrm{poly}(\log D)$ size can solve classification, whereas any classical machine of $O(D^{0.99})$ size cannot.

Best known classical algorithm: $O(D^{0.99})$ size and $\mathrm{superpoly}(N)$ samples

[Numerical linear algebra in linear space SODA'26]

Main Results

Classification: often the training examples change over time, but the underlying classification rule remains the same.

Example: large particle colliders

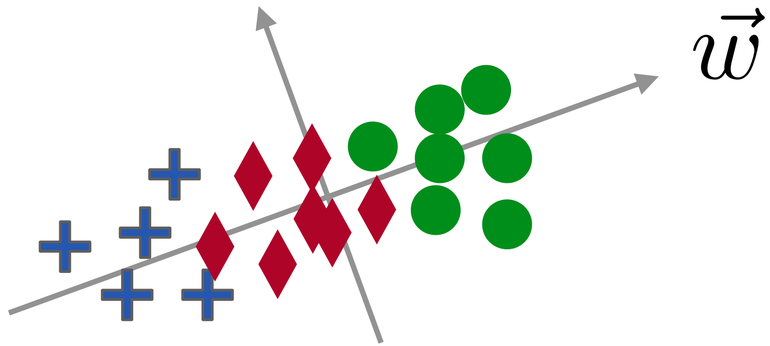
Theorem 2 (Dynamic Classification)

A quantum machine of $\mathrm{poly}(\log D)$ size can solve classification using only $\tilde{O}(N)$ samples, whereas any classical machine of $O(D^{0.99})$ size requires at least $\mathrm{superpoly}(N)$ samples.

Main Results

Dimension reduction: given data $(i, \vec{x}_i)$ sampled from N examples, each with a D-dim sparse feature vector, learn an identifiable D-dim principal component $\vec{w}$ such that we can predict the 1D representation of any new sparse test vectors.

eigenvalue gapped, non-negligible overlap with a prominent feature



unsupervised

Main Results

Dimension reduction: given data $(i, \vec{x}_i)$ sampled from N examples, each with a D-dim sparse feature vector, learn an identifiable D-dim principal component $\vec{w}$ such that we can predict the 1D representation of any new sparse test vectors.

eigenvalue gapped, non-negligible overlap with a prominent feature

Theorem 3 (Dim Reduction)

With $\tilde{O}(N)$ samples, a quantum machine of $\mathrm{poly}(\log D)$ size can solve dim. reduct., whereas any classical machine of $O(D^{0.99})$ size cannot.

Main Results

Dimension reduction: given data $(i, \vec{x}_i)$ sampled from N examples, each with a D-dim sparse feature vector, learn an identifiable D-dim principal component $\vec{w}$ such that we can predict the 1D representation of any new sparse test vectors.

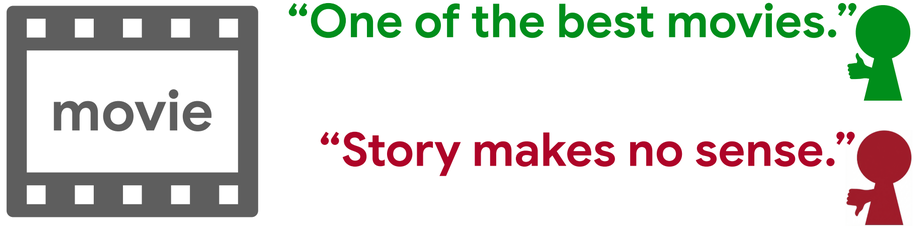
eigenvalue gapped, non-negligible overlap with a prominent feature

Theorem 4 (Dynamic Dim Reduction)

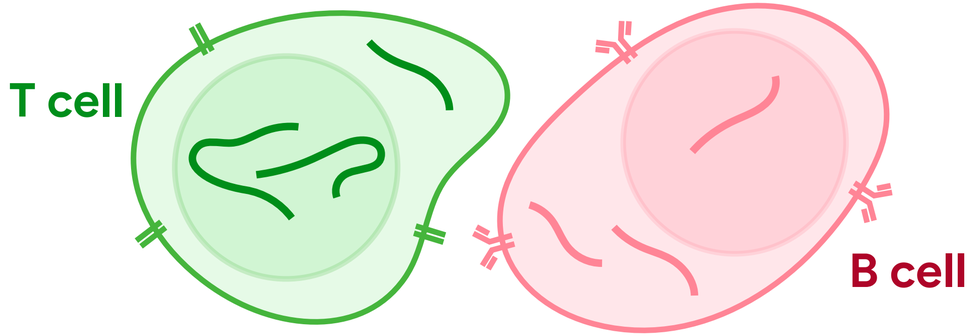
A quantum machine of $\mathrm{poly}(\log D)$ size can solve dim. reduct. using only $\tilde{O}(N)$ samples, whereas any classical machine of $O(D^{0.99})$ size requires at least $\mathrm{superpoly}(N)$ samples.

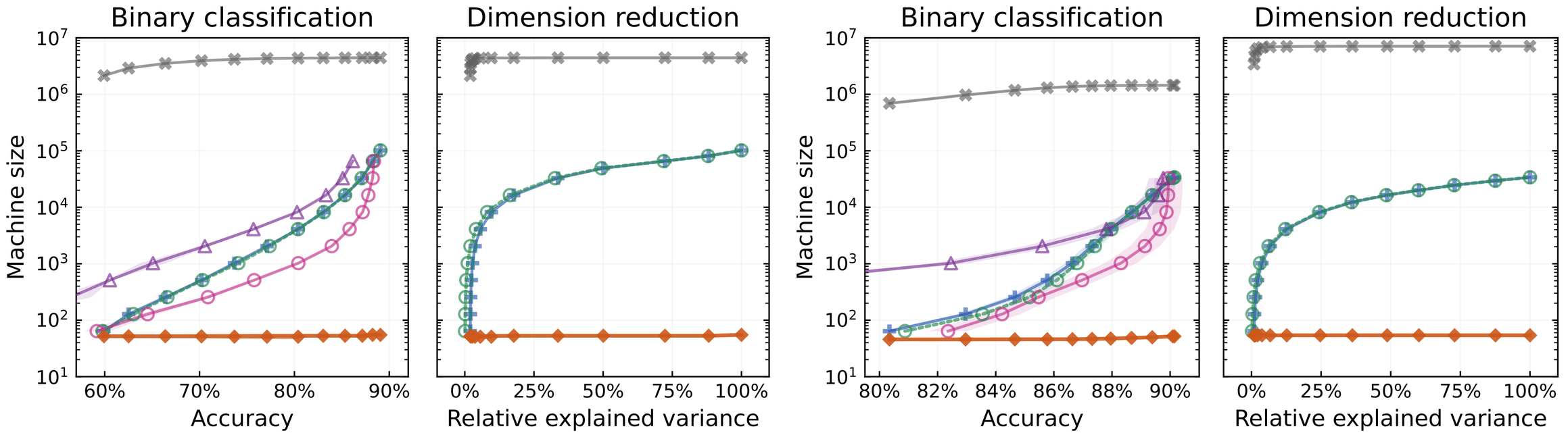
Real-world datasets

(a) Movie review sentiment analysis



(b) Single cell RNA sequencing





Sure, but...

Q1. Is memory a real bottleneck in reality? Compared to time?

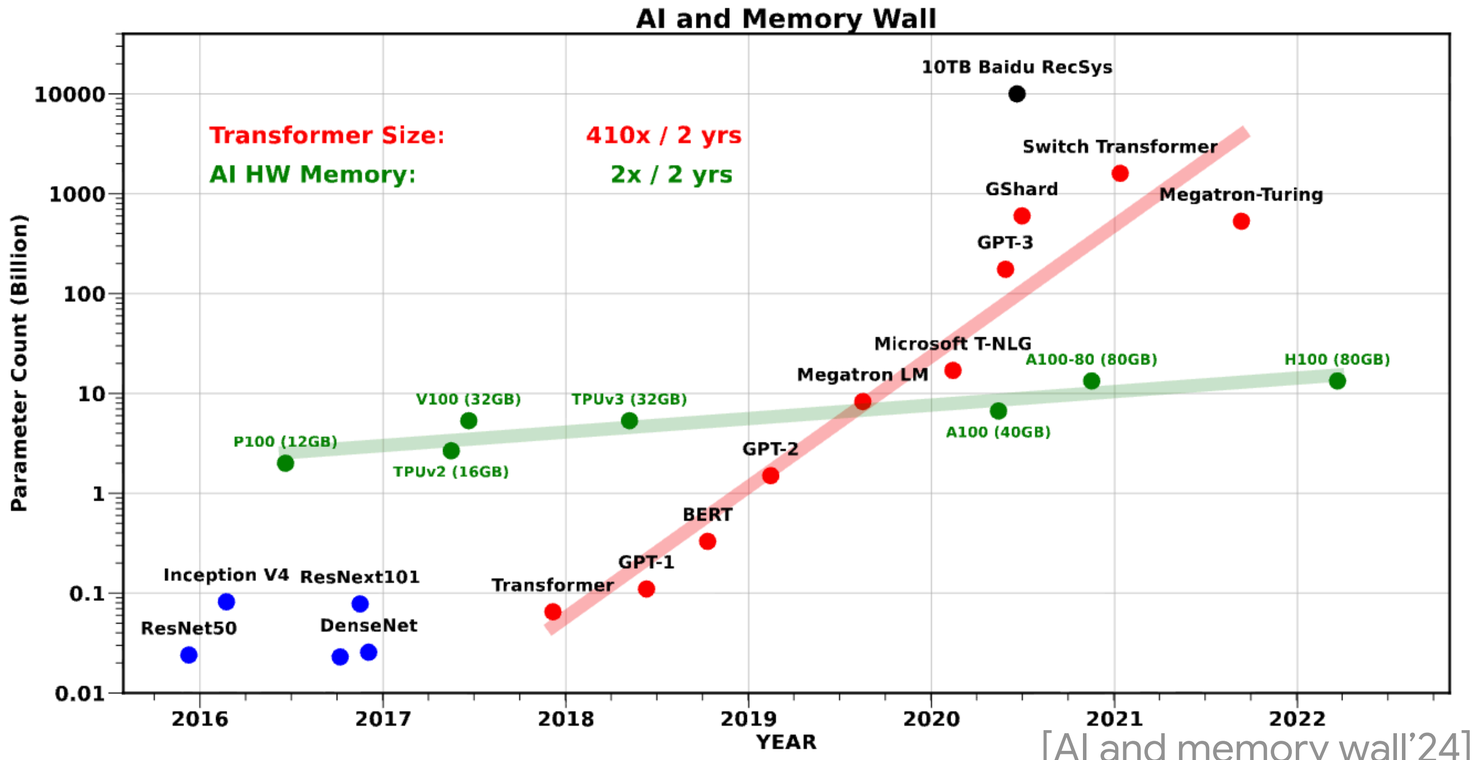
A1. [Hitting the memory wall'94], [AI and memory wall'24],
60%-90% of time is memory access / comm., not computing.

Hitting the Memory Wall: Implications of the Obvious

Wm. A. Wulf
Sally A. McKee

Department of Computer Science
University of Virginia
{wulf | mckee}@virginia.edu

December 1994



[AI and memory wall'24]

Sure, but...

Q1. Is memory a real bottleneck in reality? Compared to time?

A1. [Hitting the memory wall'94], [AI and memory wall'24].

Q2. Your classical lower bound uses constructed dataset; isn't this the same as embedding factoring/RCS into a contrived problem?

A2. Our q alg works for any datasets, not just constructed ones.

=> It does not rely on exploiting contrived structures in the dataset.

Sure, but...

Q1. Is memory a real bottleneck in reality? Compared to time?

A1. [Hitting the memory wall'94], [AI and memory wall'24].

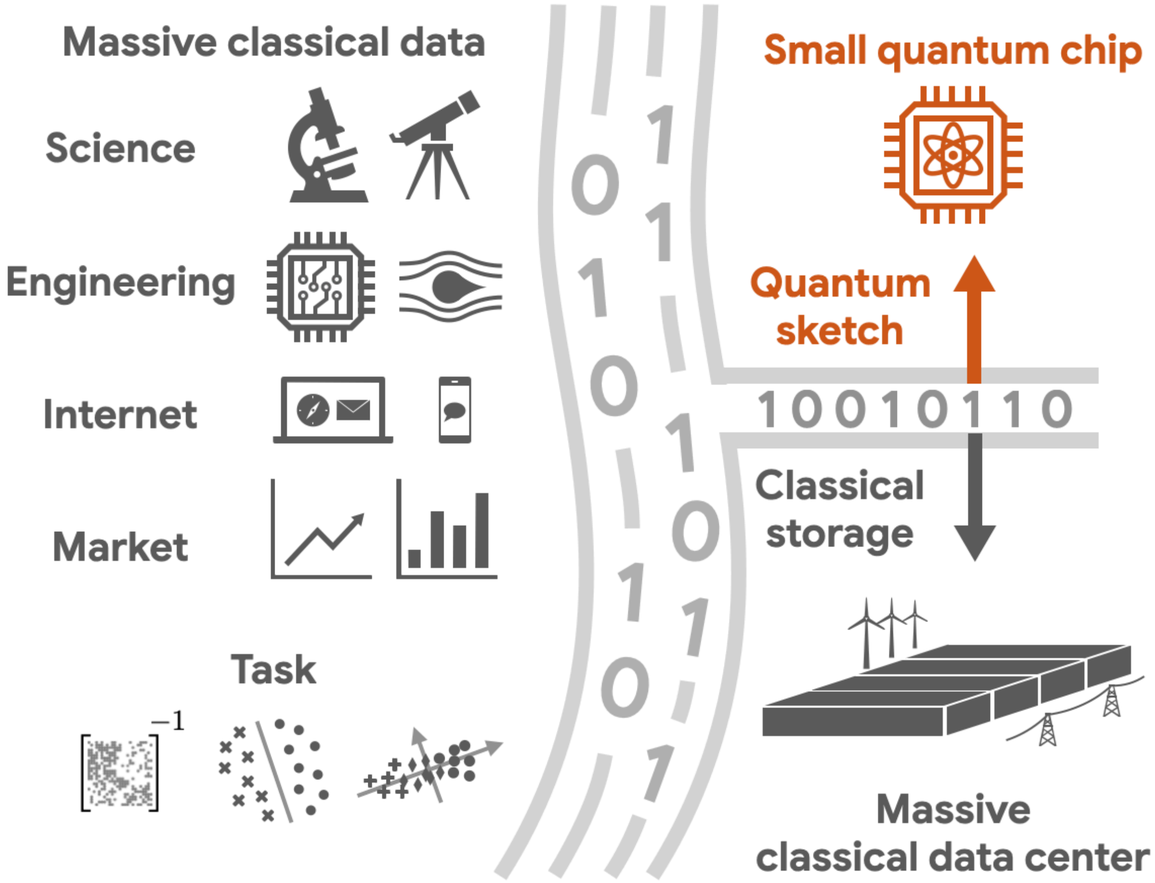
Q2. Your classical lower bound uses constructed dataset; isn't this the same as embedding factoring/RCS into a contrived problem?

A2. Our q alg works for any datasets, not just constructed ones.

Q3. To process ~PB data (10^3 TB, one Google data center), how long would it take?

A3. Back-of-envelope estimate: with reasonable parallelization (~50 x 10^3 logical qubits), 10^15 x 1us / 10^3 = 10^6 s = 10 days.

Main Results



Machine learning

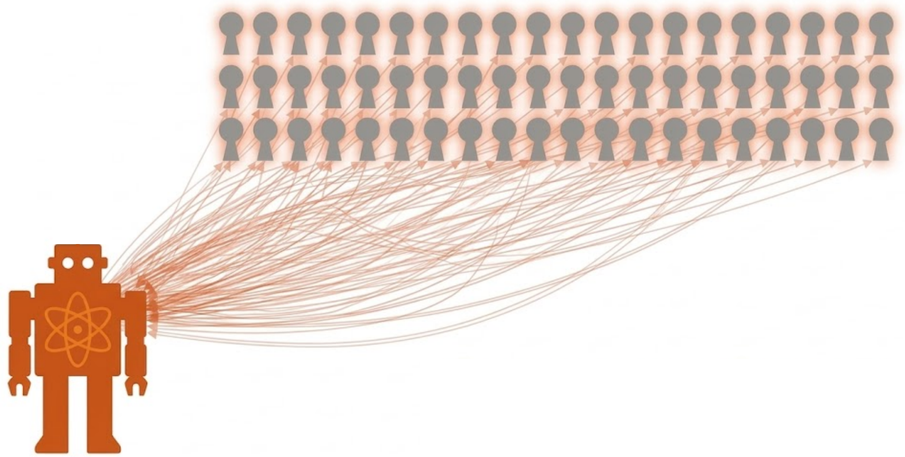
is a broad domain of

exponential quantum advantage.

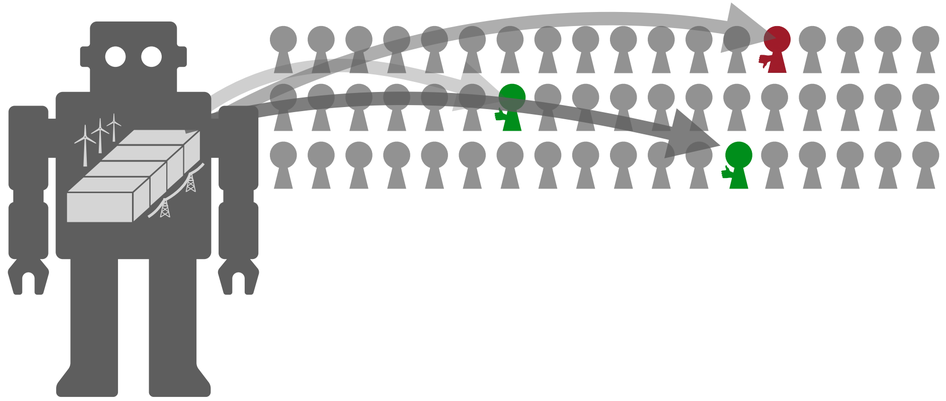
despite challenges in data loading, readout, dequantization, etc.

Origin of Advantage

Quantum Algorithm

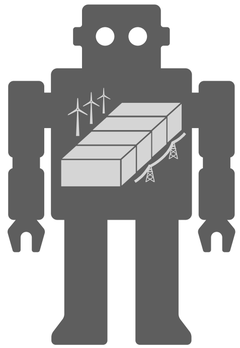


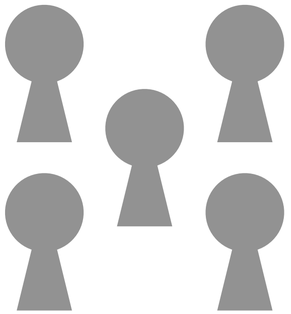
Classical Hardness



Quantum advantage?

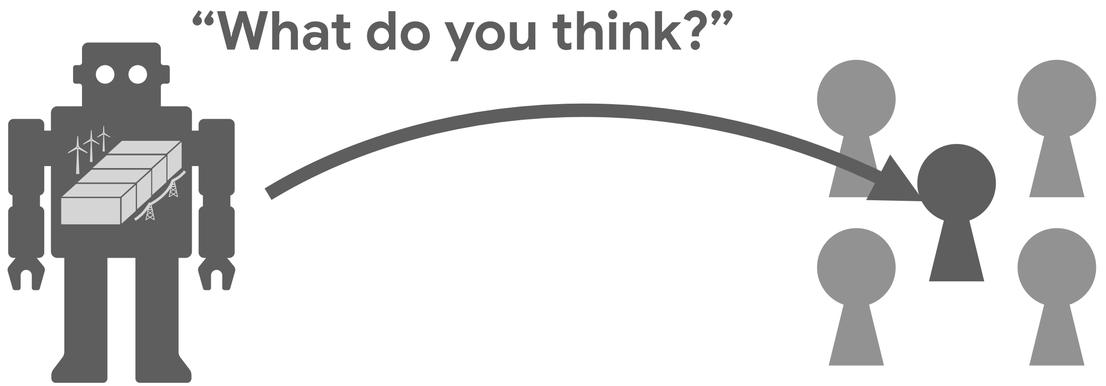
Classical ML





Quantum advantage?

Classical ML

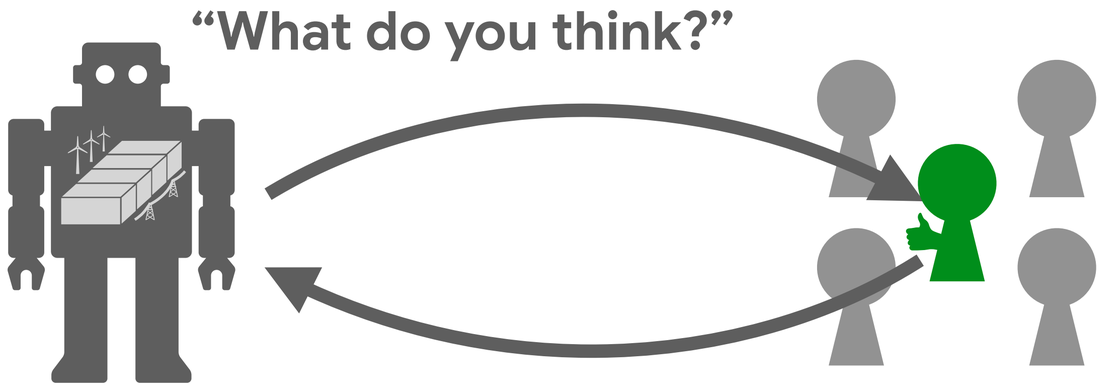


Quantum advantage?

Classical ML

Quantum advantage?

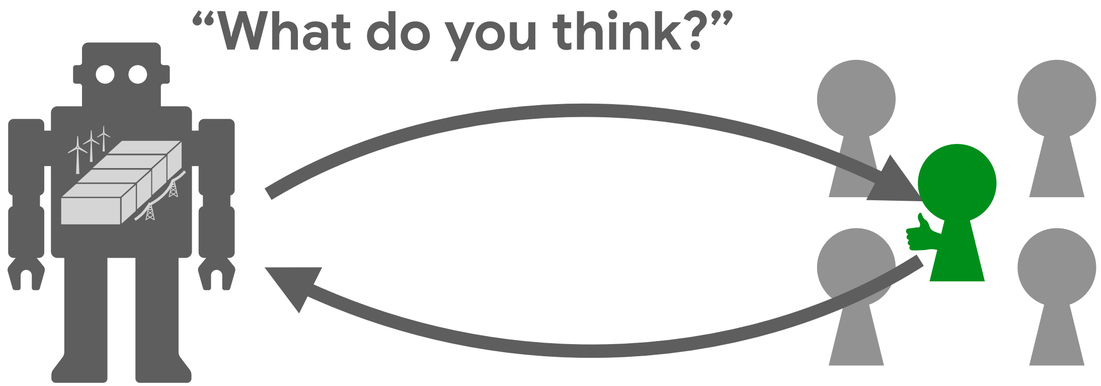
Classical ML



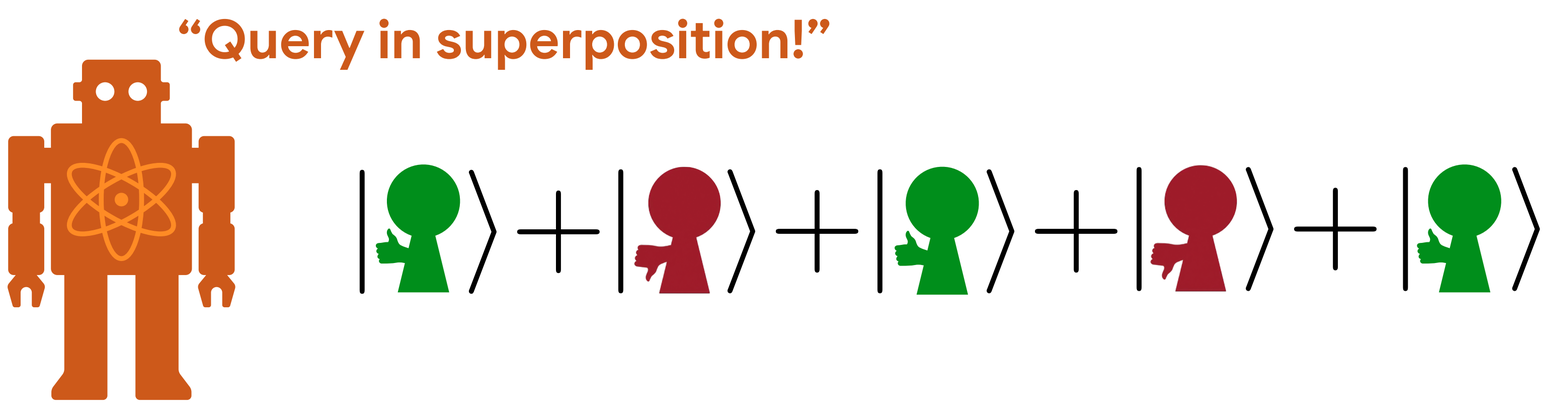
Quantum ML

Quantum advantage?

Classical ML

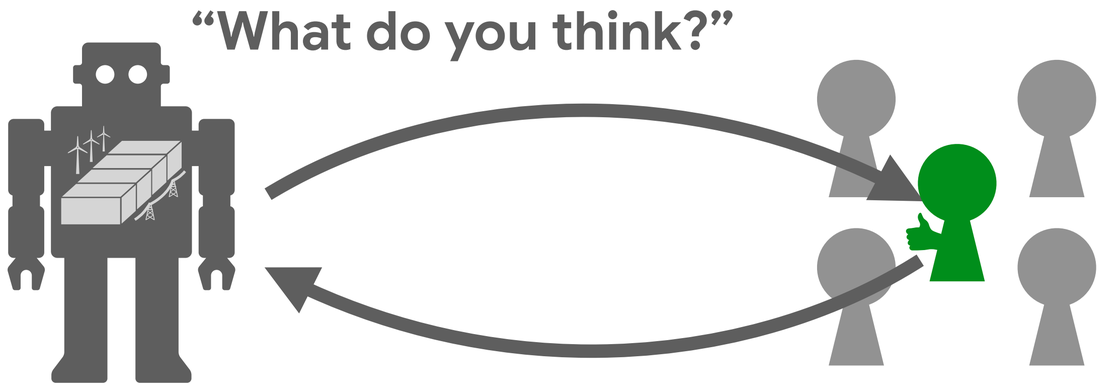


Quantum ML

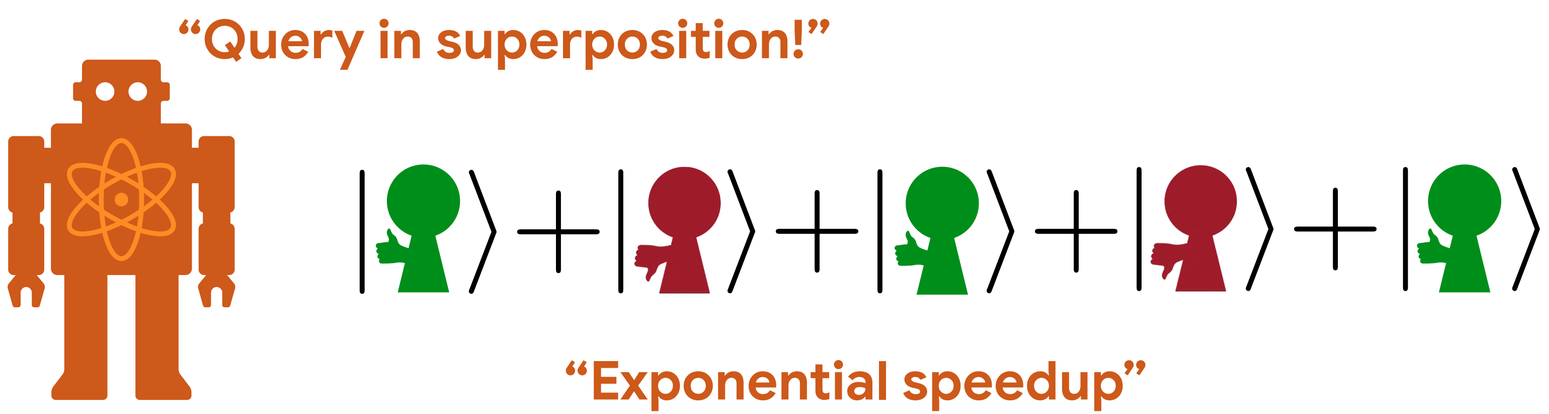


Quantum advantage?

Classical ML



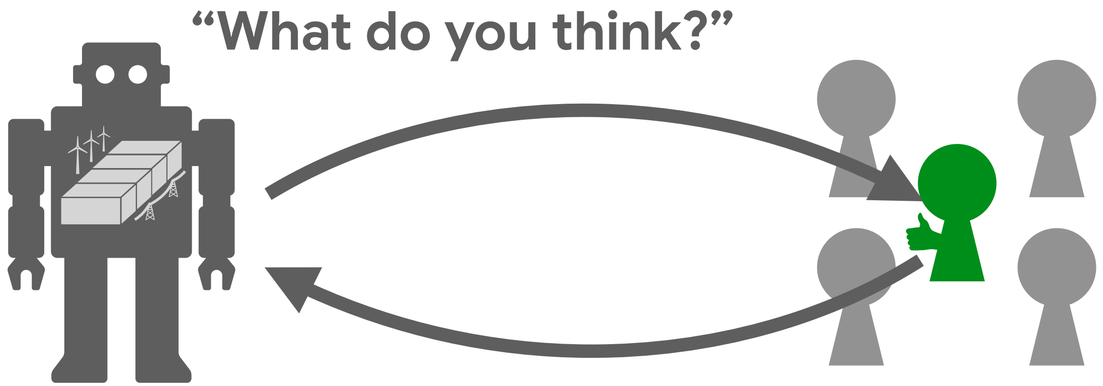
Quantum ML



[quantum oracle query algorithm]

Quantum advantage?

Classical ML

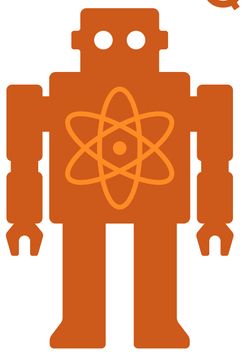


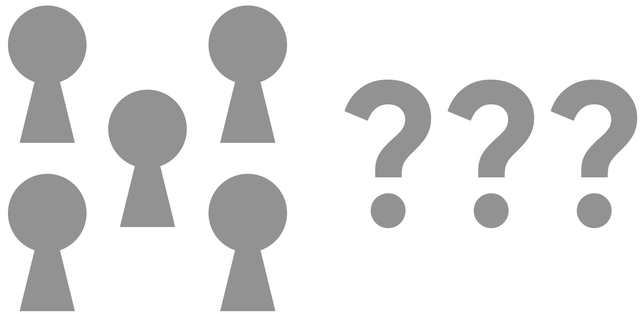
Quantum ML

Challenge

Can we access a classical world in quantum superposition?

“Query in superposition!”





We’re classical…

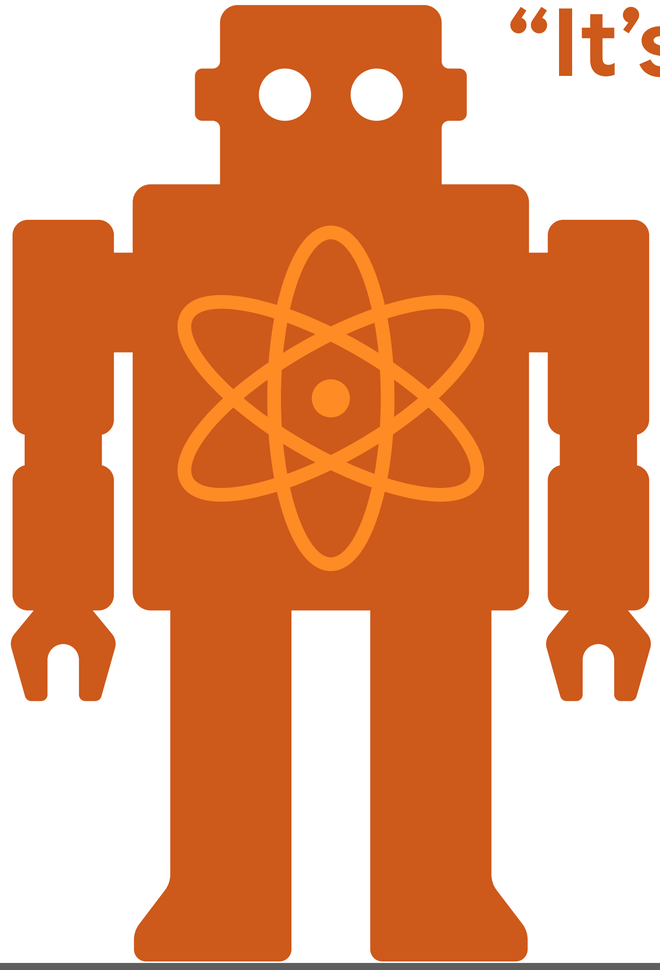
[Read the fine print! Aaronson’15]

Attempt: QRAM

Machine size

Data Loading

Store the entire world.



"It's easy!"

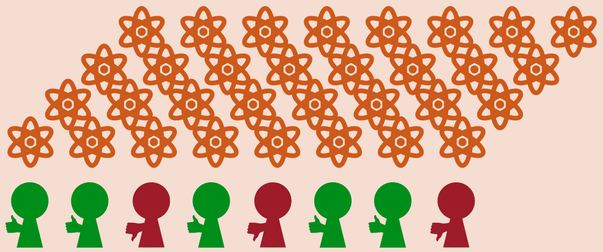
Time
Sample

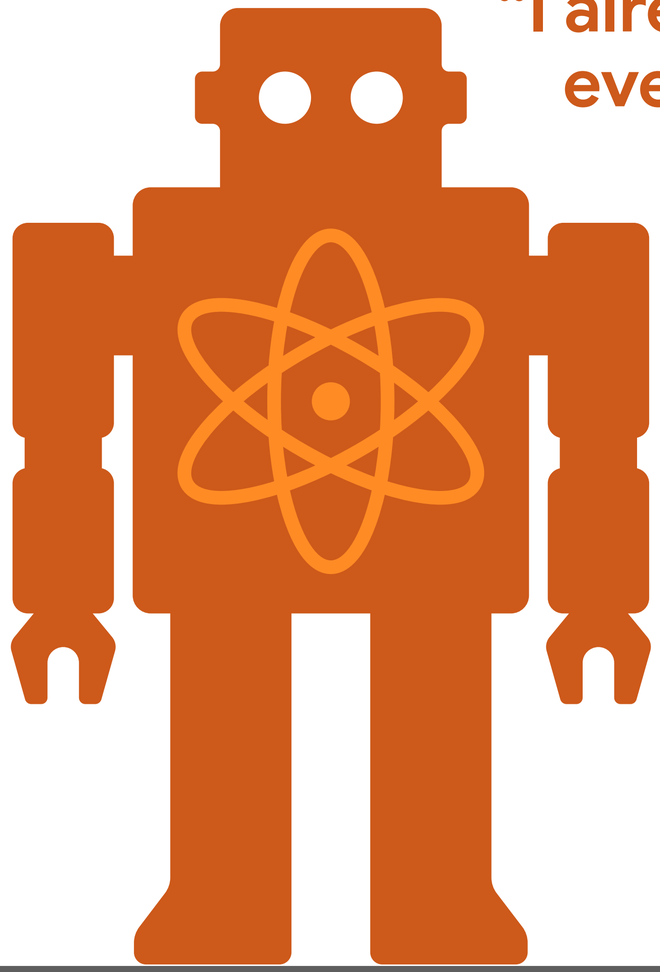
Attempt: QRAM

Machine size

Data Loading

Store the entire world
in a quantum RAM.





"I already stored
everything."

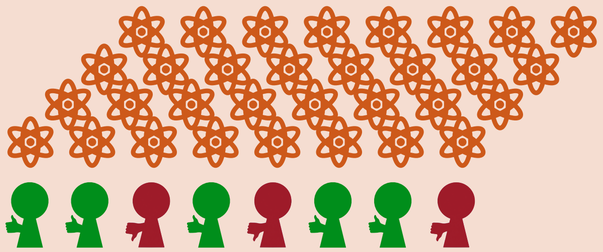
Time
Sample

Attempt: QRAM

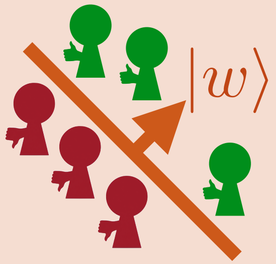
Machine size

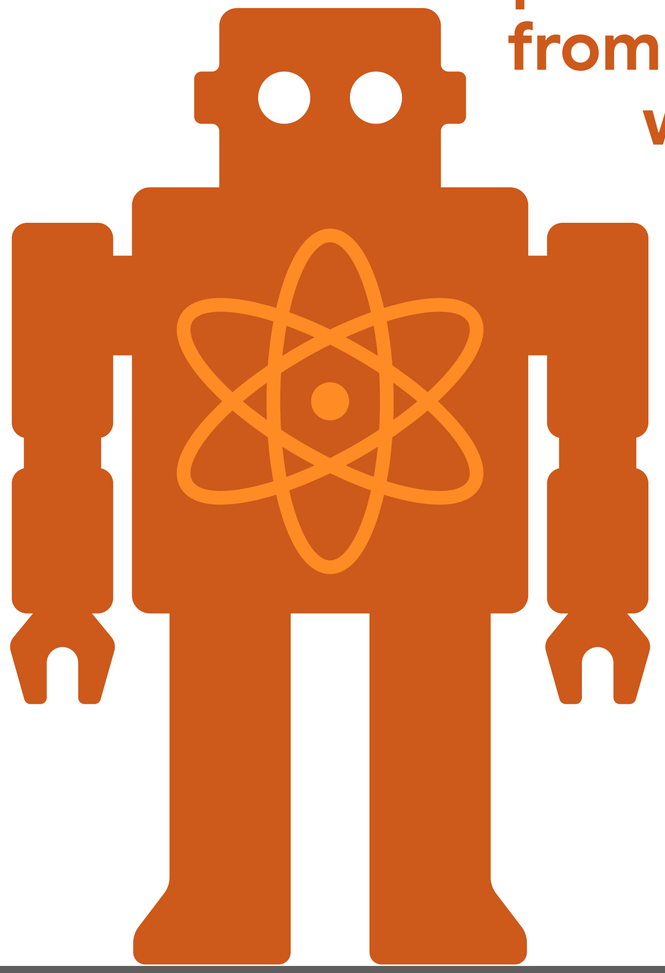
Data Loading

Store the entire world in a quantum RAM.



Data Processing with quantum query





"I can create quantum queries from the stored world."

[QRAM, PRL'08]

Time

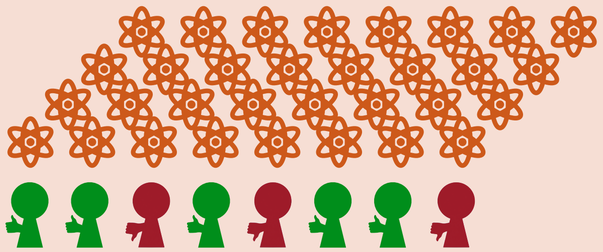
Sample

Attempt: QRAM

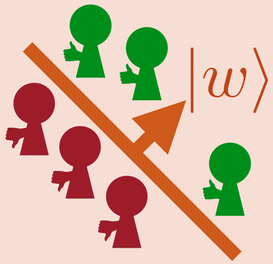
Machine size

Data Loading

Store the entire world in a quantum RAM.



Data Processing with quantum query



Classical ML

No end-to-end quantum advantage!

Time Sample

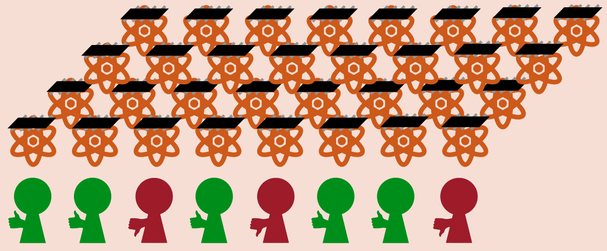
Attempt: QRAM

Machine size

Classical ML

Maintaining a coherent copy of the entire world.

Data loading + QEC.



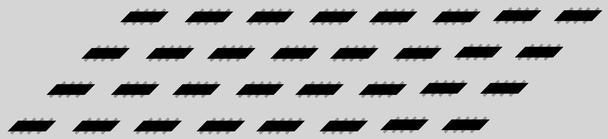
Time
Sample

Attempt: QRAM

Machine size

Classical ML

Classical co-processors meant for QEC
can solve the problem themselves.



Faster...

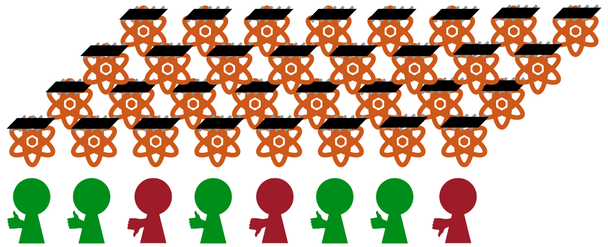
Time
Sample

Or, passive QRAM [2610.03700] + distillation-teleportation [2505.20265]?

Challenge

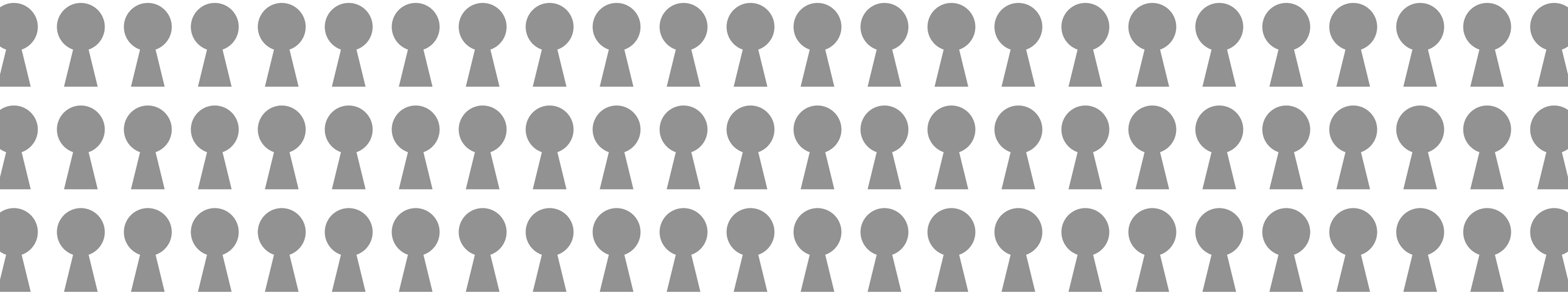
Can we access a classical world in quantum superposition?

(without storing the entire world)



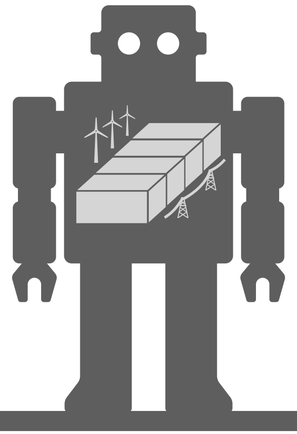
Challenge

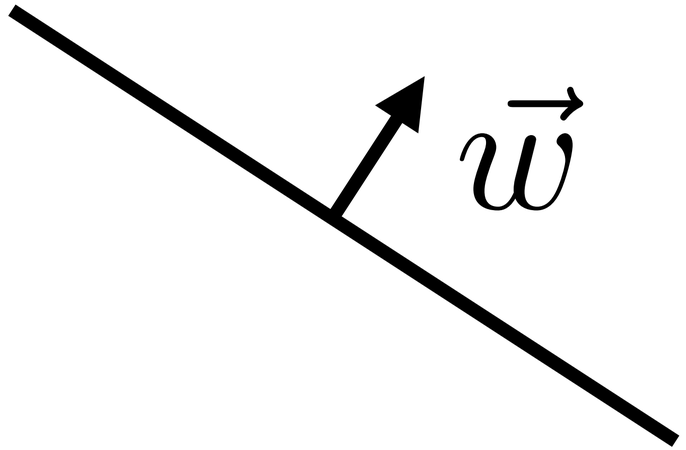
How to do ML without storing the entire dataset?



Train a model on the fly

Machine size

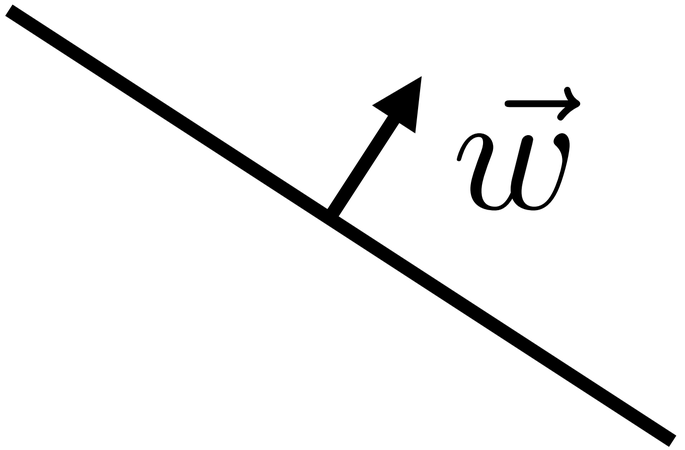


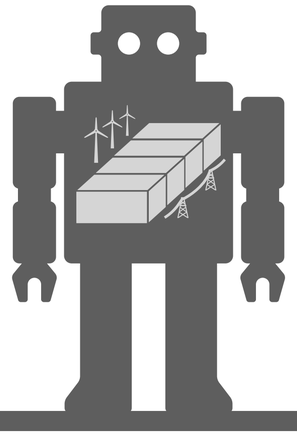


Time
Sample

Train a model on the fly

Machine size

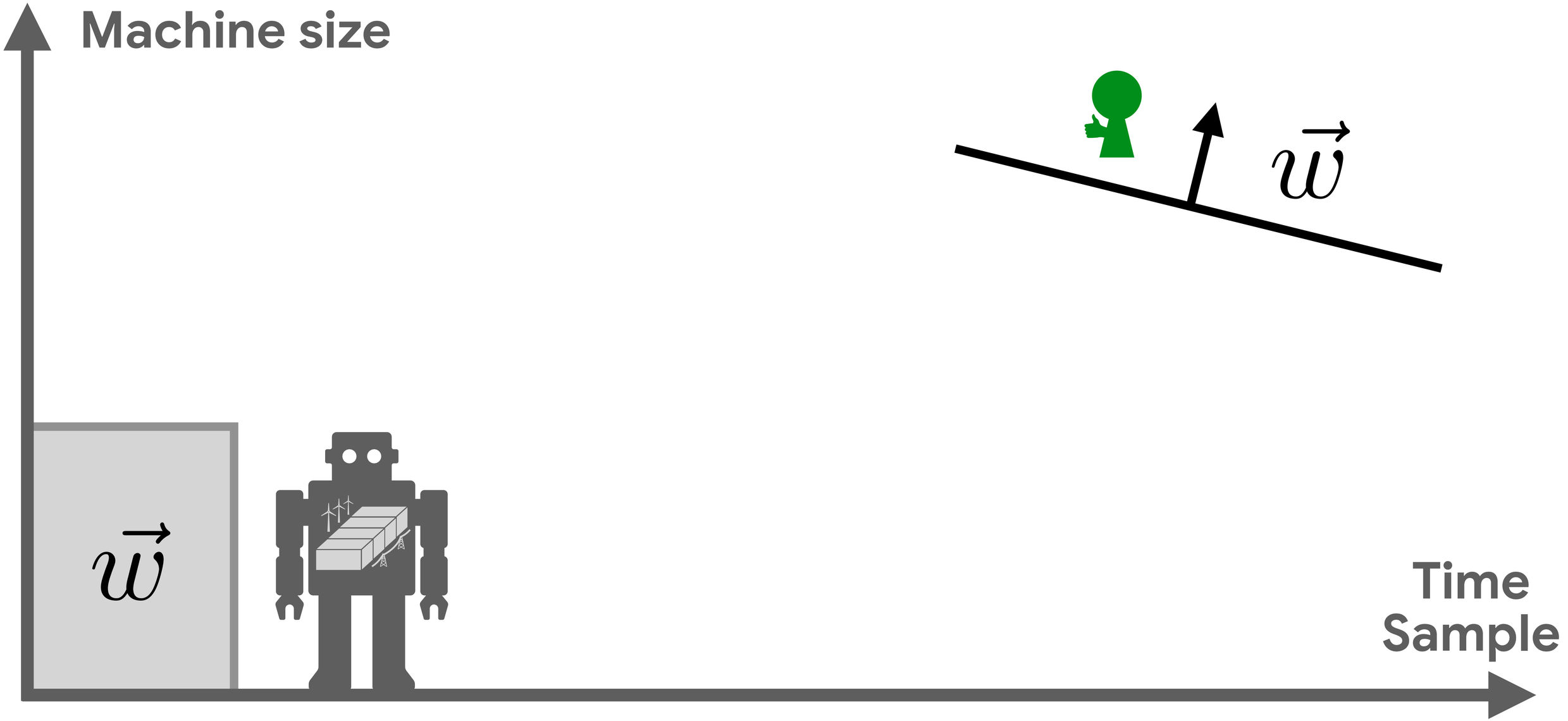




"One of the best movies."

Time
Sample

Train a model on the fly



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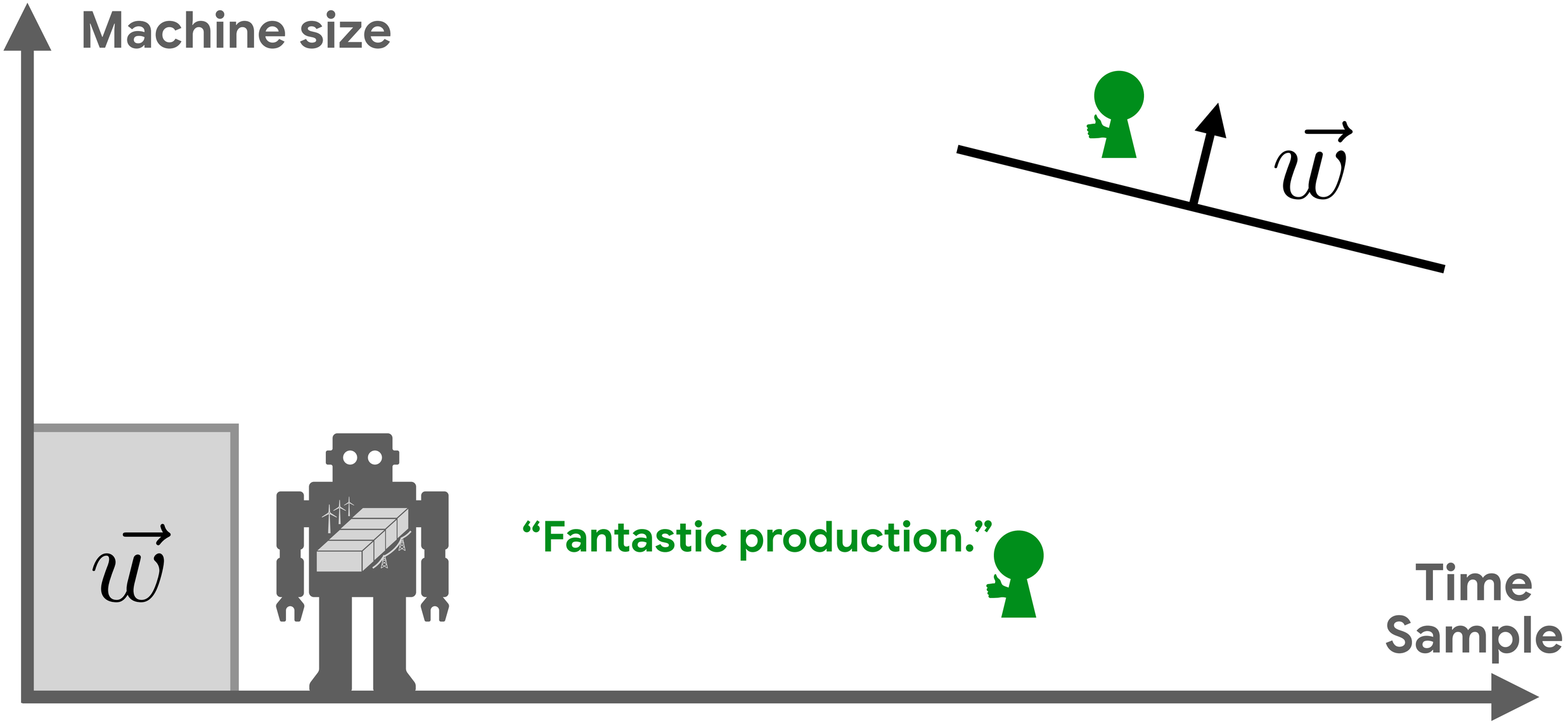
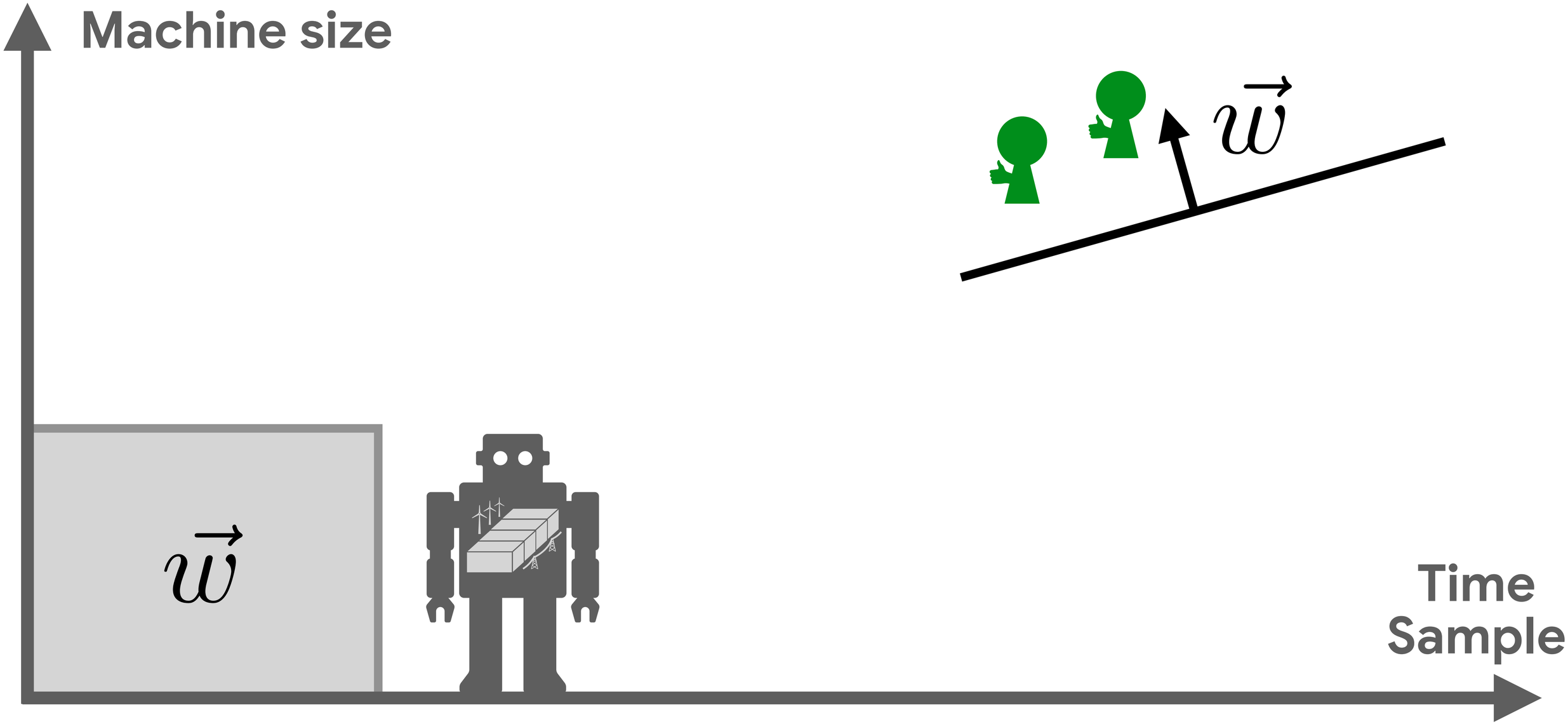
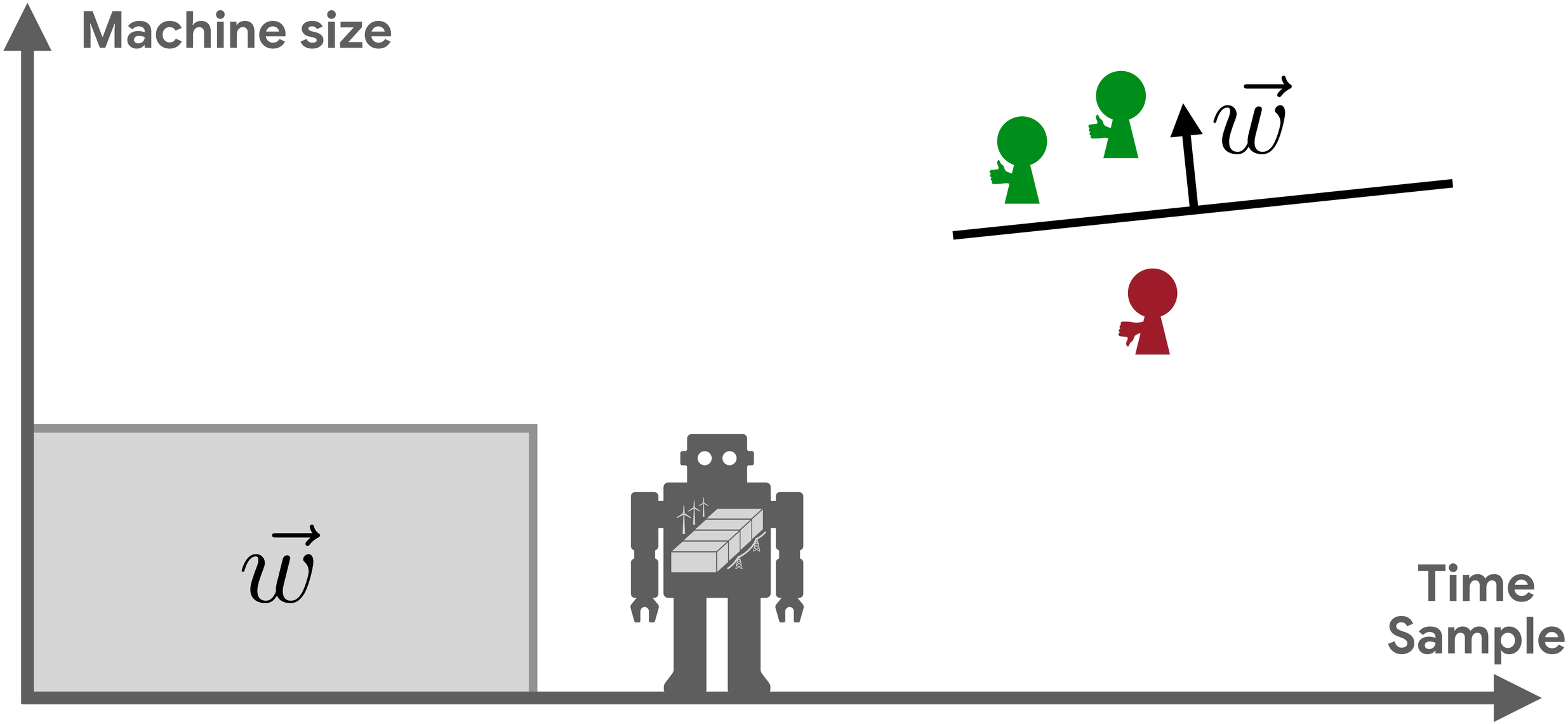


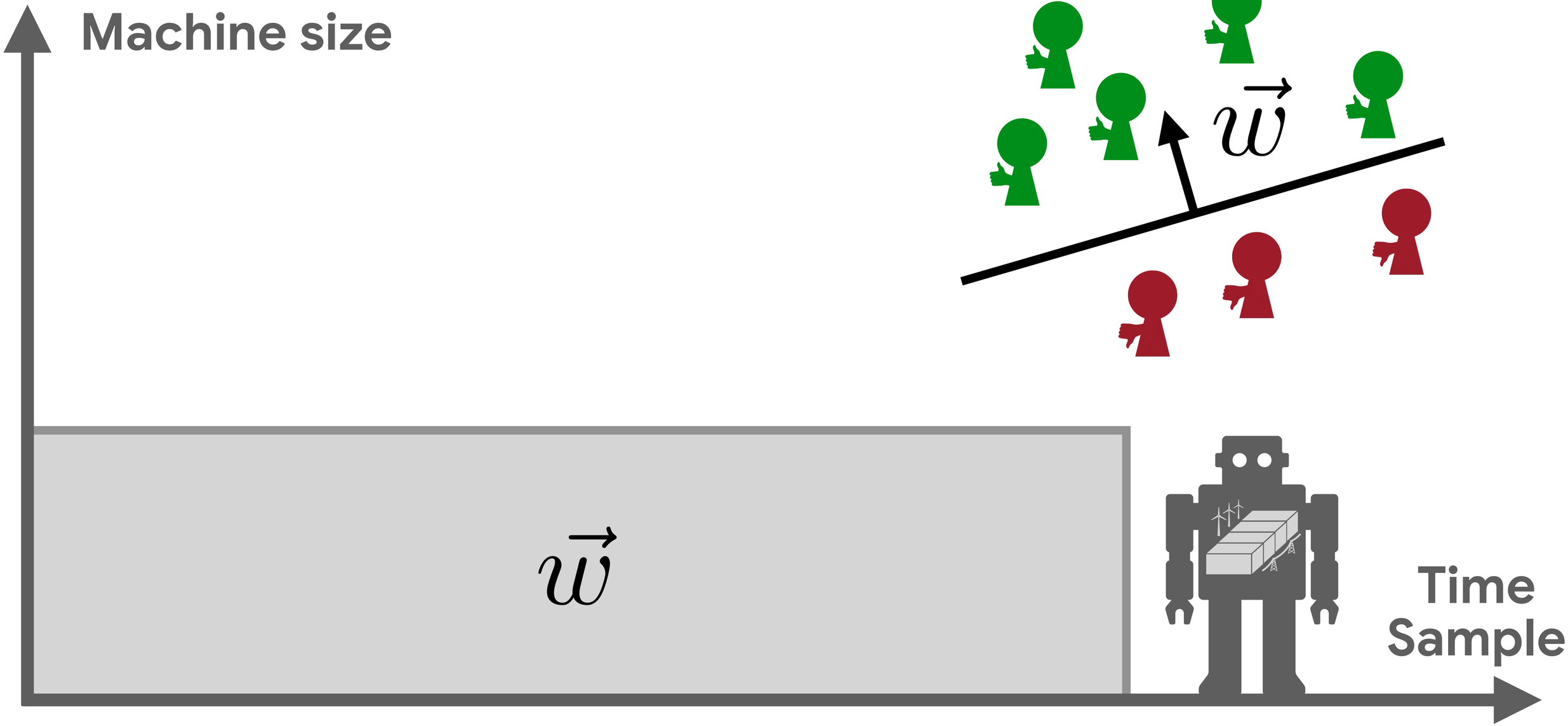
Train a model on the fly



Train a model on the fly

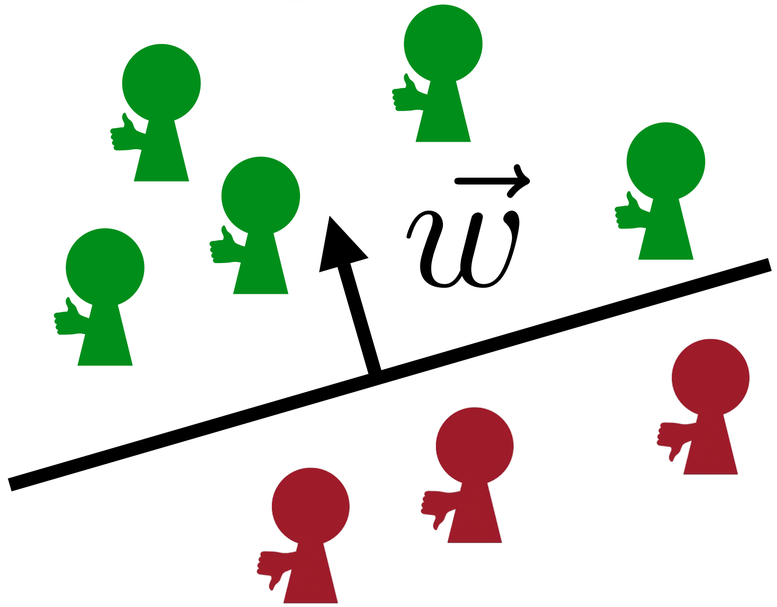


Train a model on the fly



Train a model on the fly

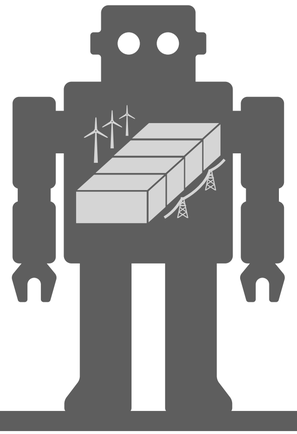
Machine size



Process data on the fly

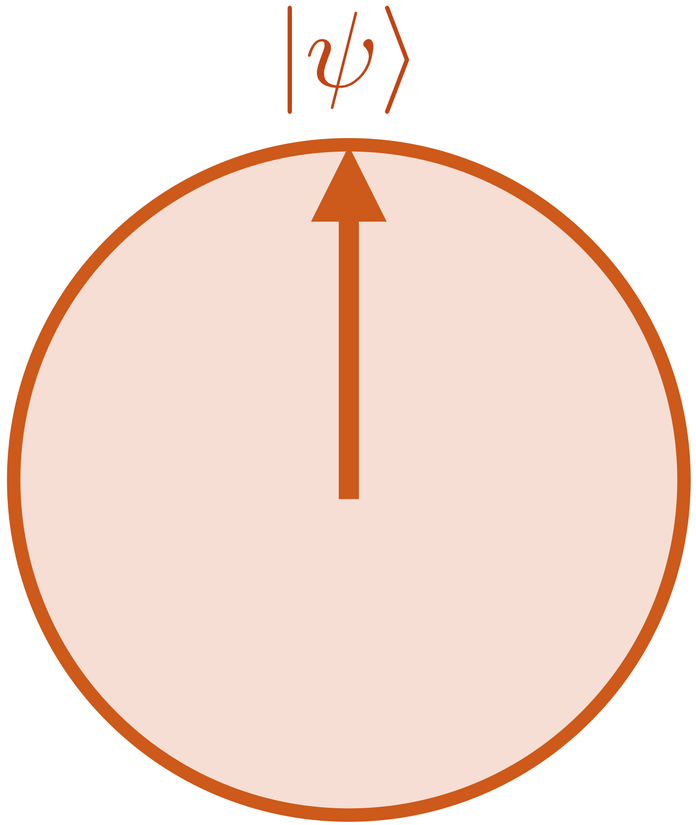
Incrementally update a world model.

[streaming, sketching, online learning]



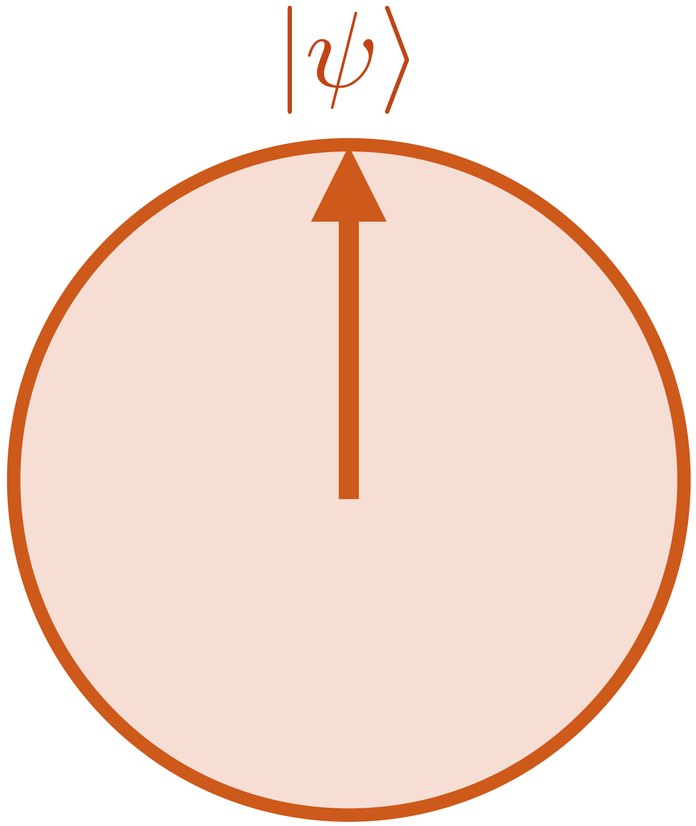
Time
Sample

Quantum oracle sketching



Maintain a quantum model

Quantum oracle sketching

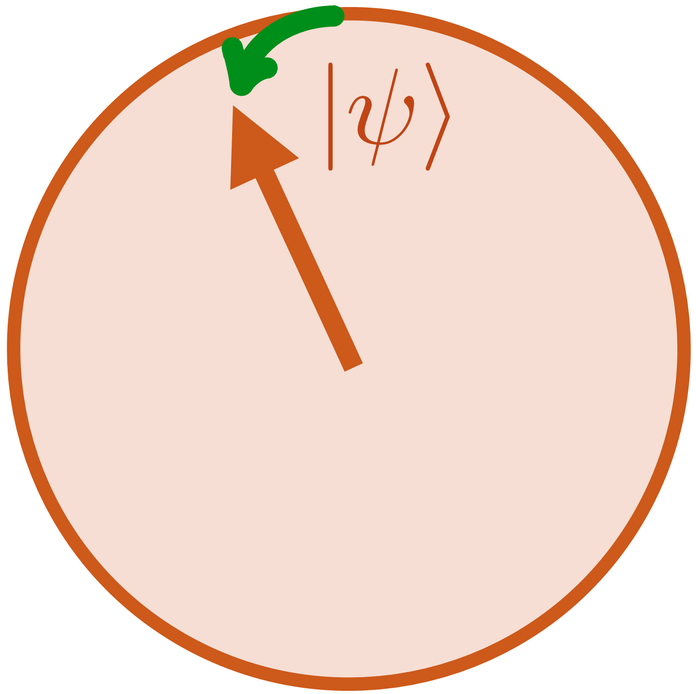


Maintain a quantum model

For each data sample z

“One of the best movies.”

Quantum oracle sketching



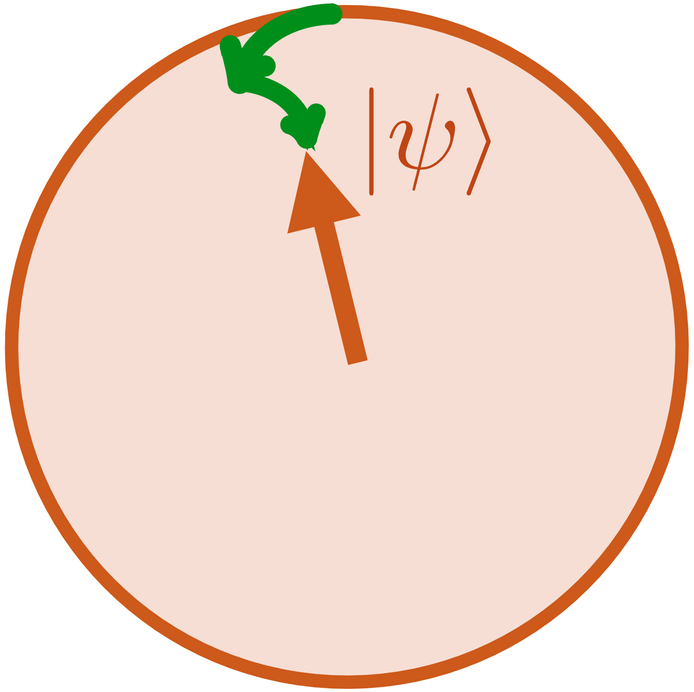
"One of the best movies."

Maintain a quantum model

For each data sample z

apply a small quantum rotation V_z

Quantum oracle sketching



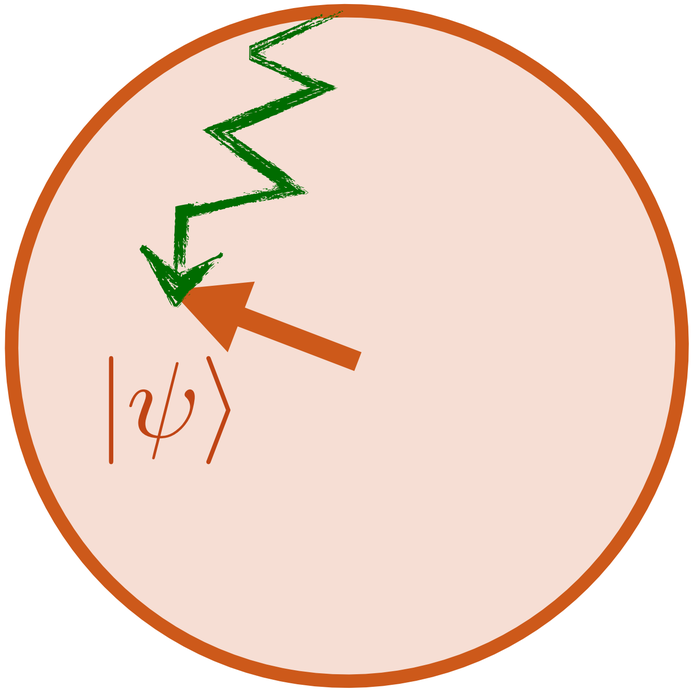
Maintain a quantum model

For each data sample z

apply a small quantum rotation V_z

Repeat for M samples

Quantum oracle sketching



a random gate sequence

$$V_{z_M} \cdots V_{z_1}$$

Maintain a quantum model

For each data sample z

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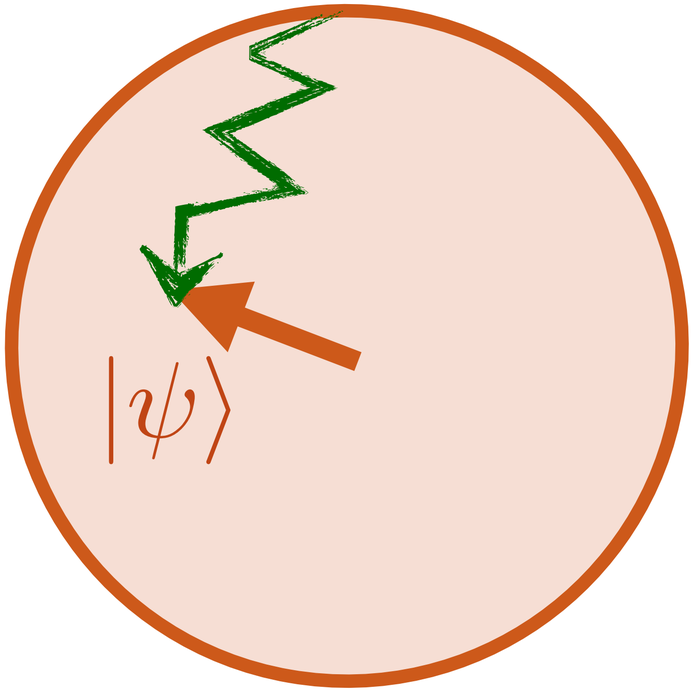
Repeat for M samples

Quantum oracle sketching



a random gate sequence

$$V_{z_M} \cdots V_{z_1}$$

Maintain a quantum model

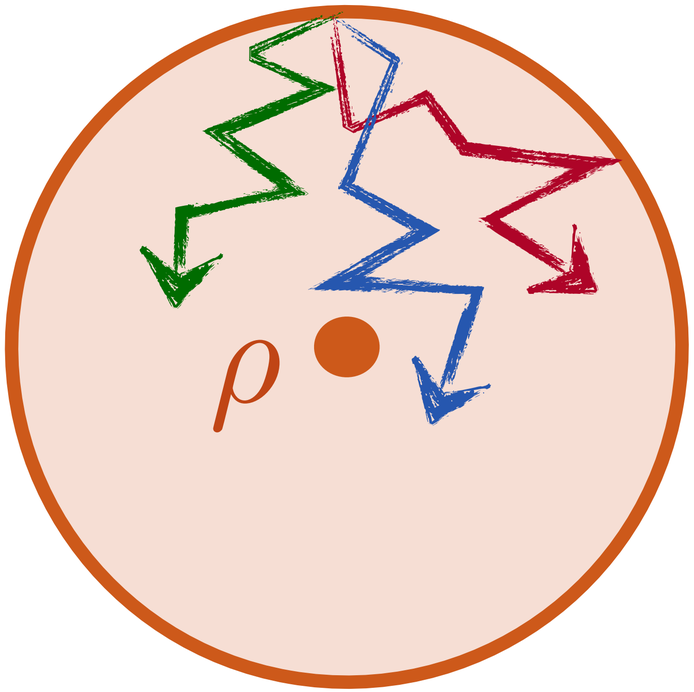
For each data sample z

apply a small quantum rotation V_z

Repeat for M samples

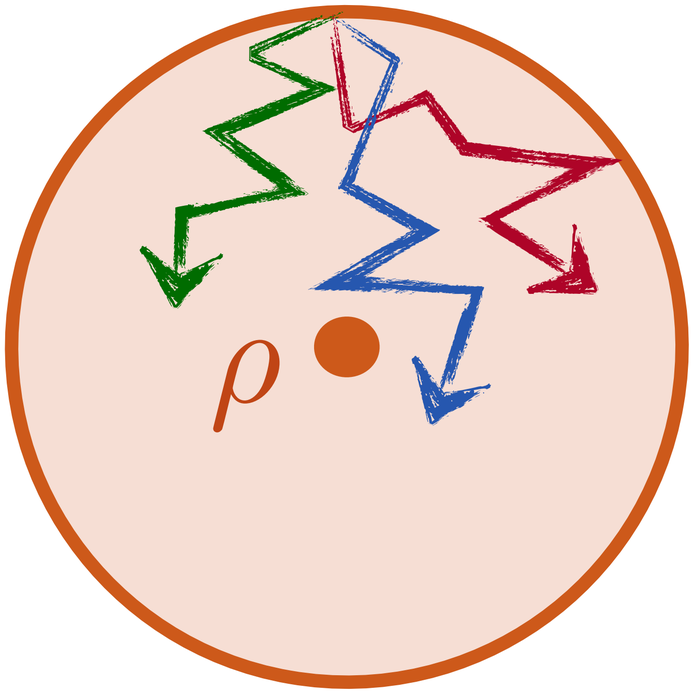
≈ a quantum oracle query

Quantum oracle sketching



Data randomness leads to decoherence for generic rotations $V_{z_M} \cdots V_{z_1}$

Quantum oracle sketching

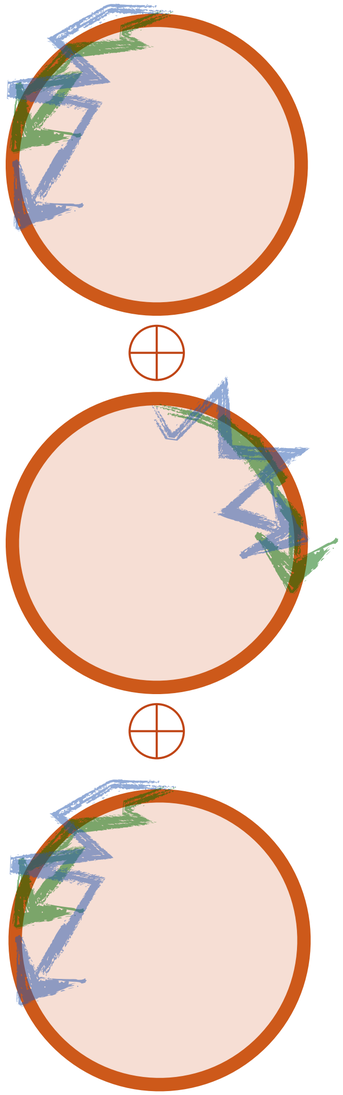


Data randomness leads to decoherence for generic rotations $V_{z_M} \cdots V_{z_1}$

✗ no quantum advantage

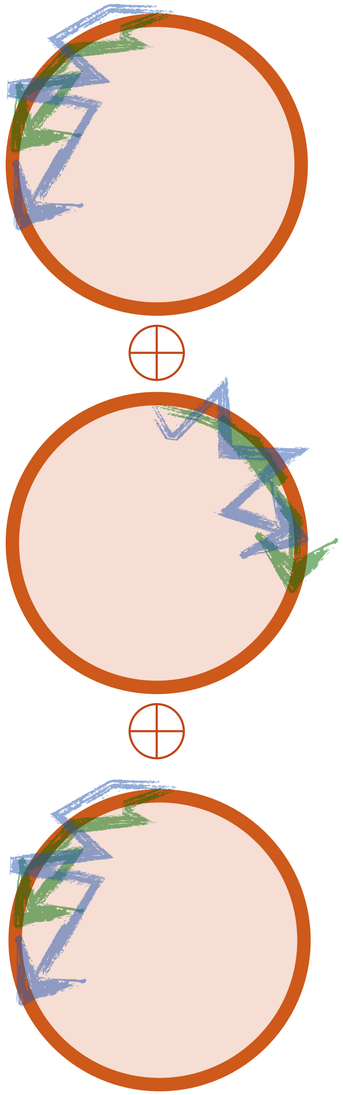
[randomized Hamiltonian simulation; qDrift]

Quantum oracle sketching



Carefully design the rotations to be nearly orthogonal to avoid decoherence.

Quantum oracle sketching



Carefully design the rotations to be nearly orthogonal to avoid decoherence.

Theorem (Quantum Oracle Sketch)

With $\Theta(NQ^2)$ samples, we can emulate Q quantum queries to an oracle of N items with no memory overhead.

Toy example

QRAM on the fly

Example

Goal: estimate property of $f : [N] \to \{0, 1\},$ given $\{x_i, f(x_i)\}_{i=1}^{M}, p(x) = 1/N$

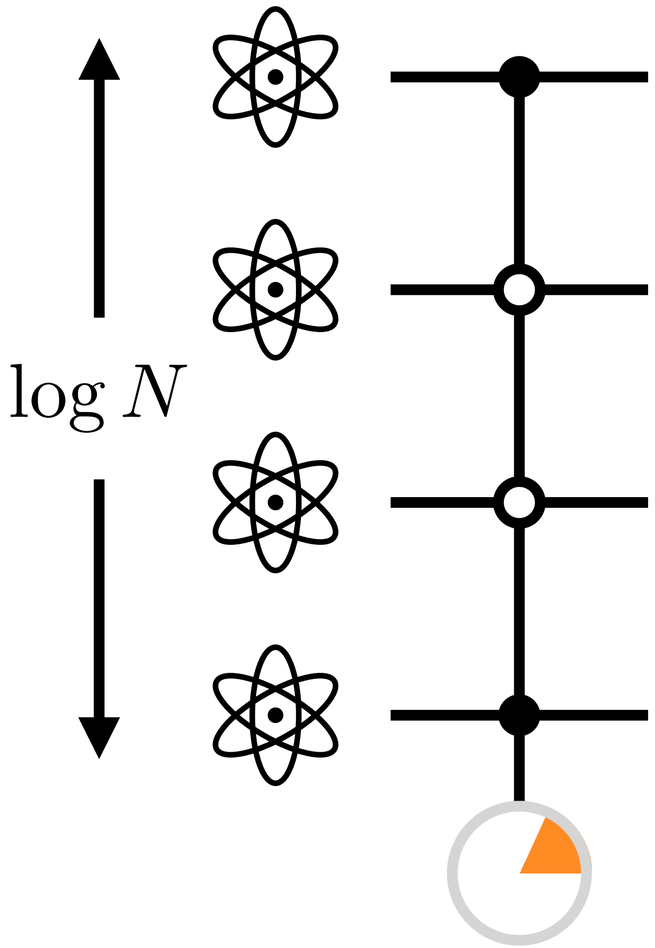
(desired property specified by an oracle query algorithm)

Toy example

QRAM on the fly

Example

Goal: estimate property of $f : [N] \to \{0, 1\}$, given $\{x_i, f(x_i)\}_{i=1}^{M}, p(x) = 1/N$



$$x_i = 1001$$

$$f(x_i) = 1$$

multi-controlled phase gate

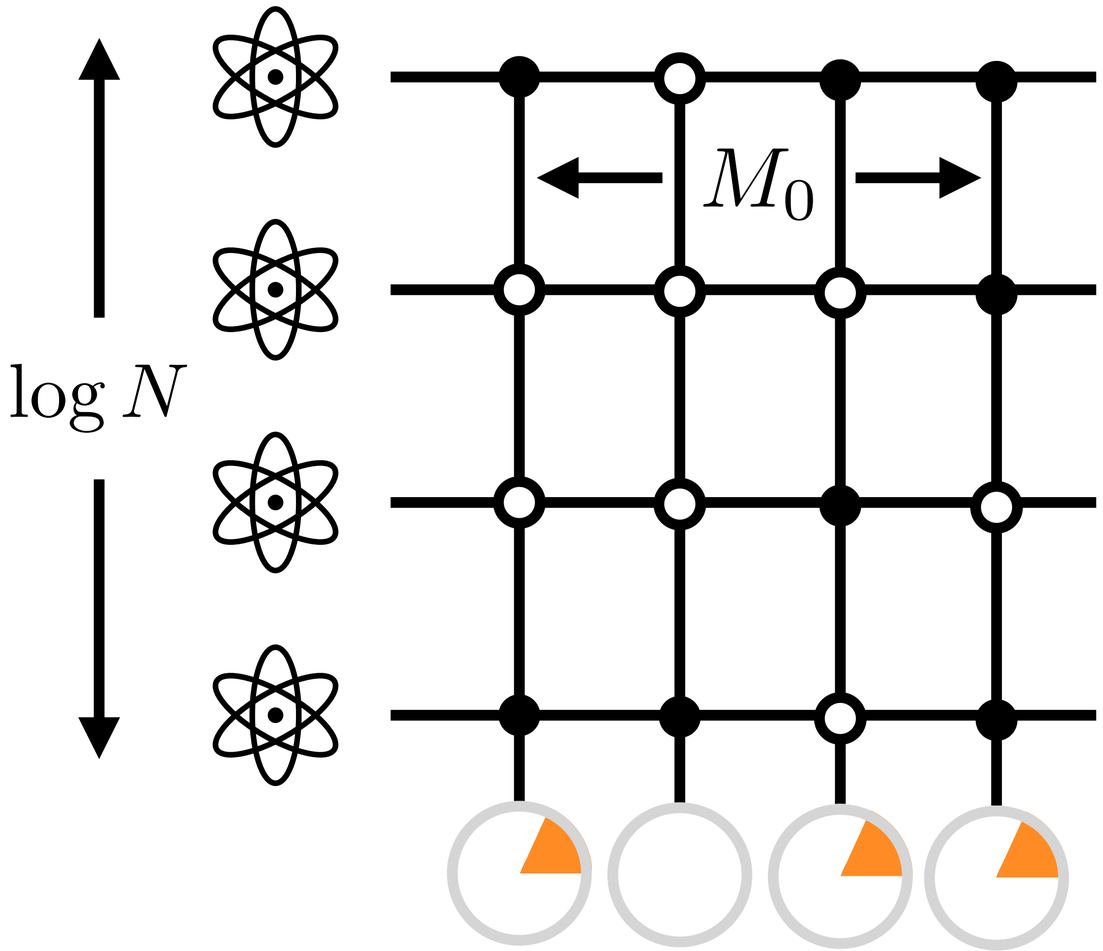
$$V_i = \exp\left(i\frac{t}{M_0} f(x_i)|x_i\rangle\langle x_i|\right)$$

Toy example

QRAM on the fly

Example

Goal: estimate property of $f : [N] \to \{0, 1\}$, given $\{x_i, f(x_i)\}_{i=1}^{M}, p(x) = 1/N$



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$$V_{M_0} \cdots V_1$$

$$= \exp\left(it \sum_x m_x f(x)|x\rangle\langle x|\right)$$

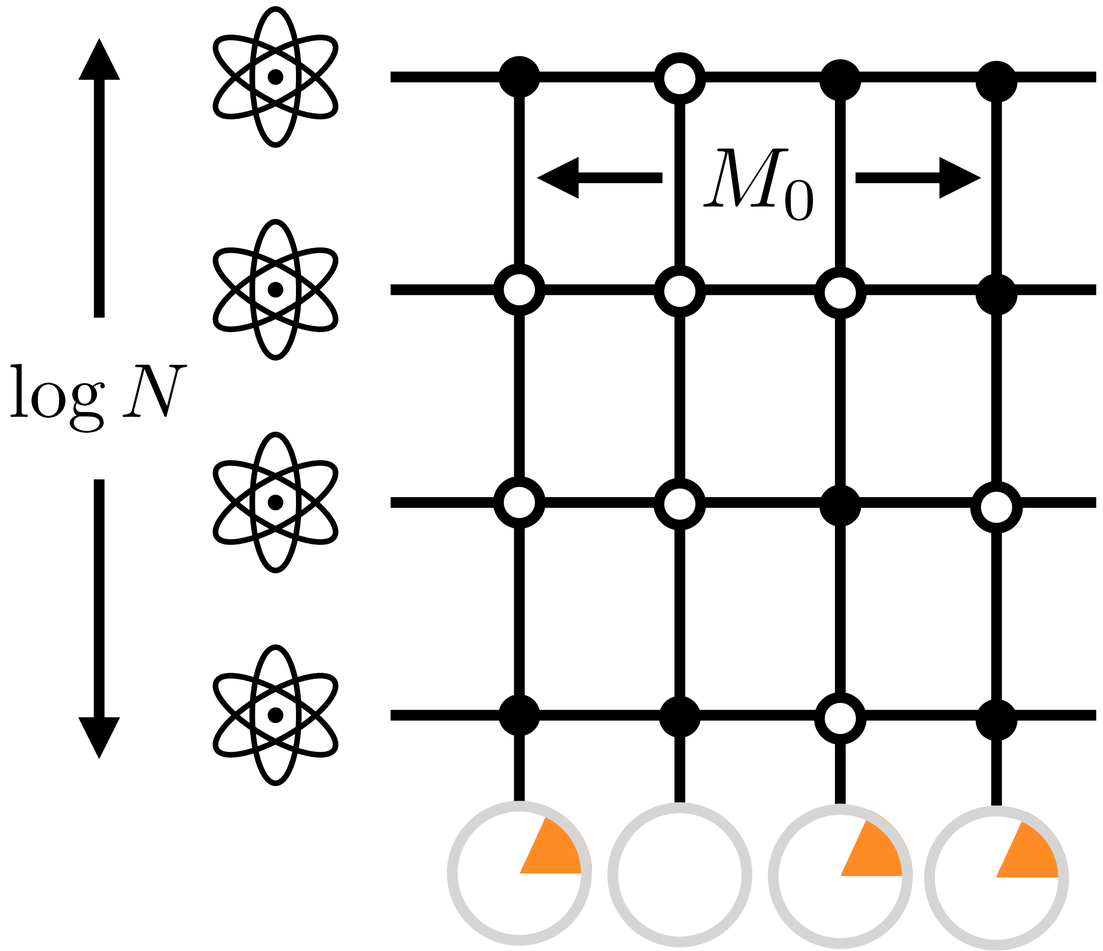
frequency of $x_i = x$

Toy example

QRAM on the fly

Example

Goal: estimate property of $f : [N] \to \{0, 1\}$, given $\{x_i, f(x_i)\}_{i=1}^{M}, p(x) = 1/N$



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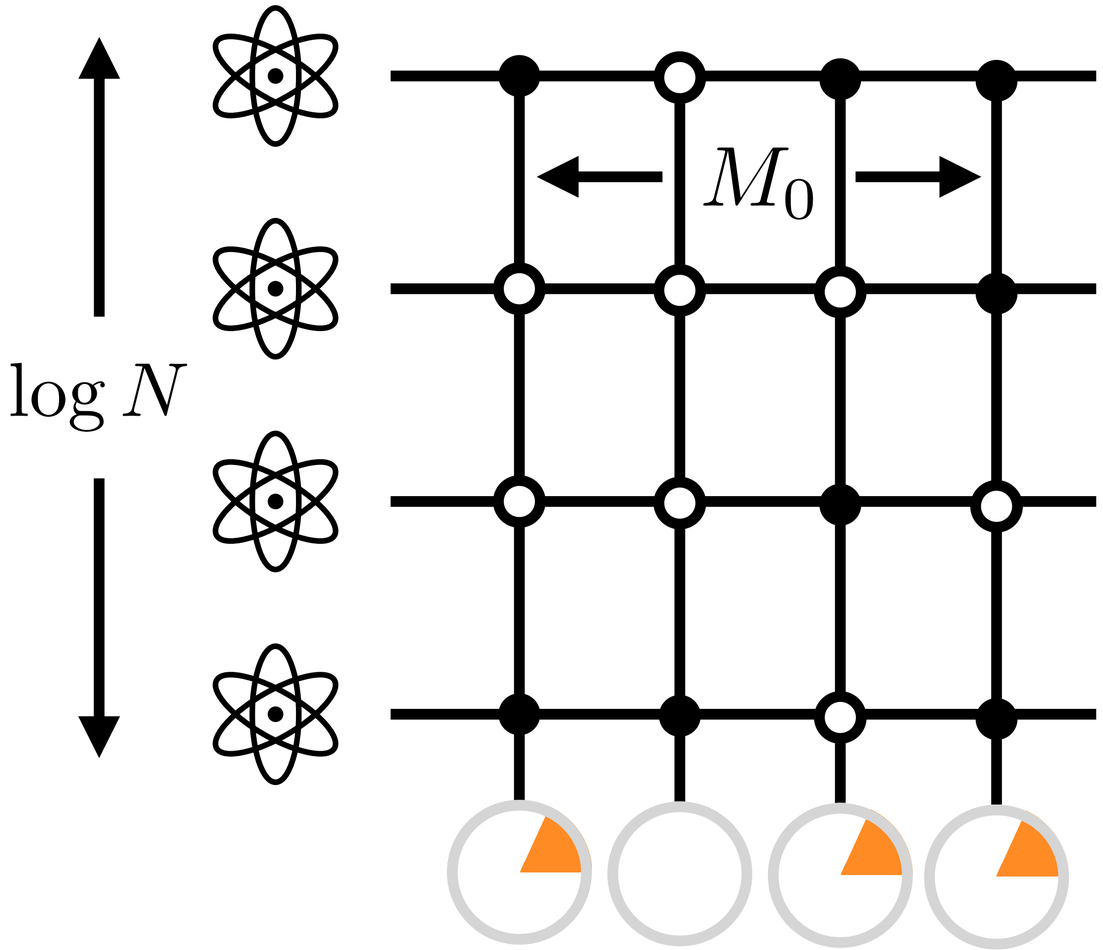


$$V_i = \exp\left(i\frac{t}{M_0} f(x_i)|x_i\rangle\langle x_i|\right)$$

$$V_{M_0} \cdots V_1$$

$$= \exp\left(it \sum_x m_x f(x)|x\rangle\langle x|\right)$$

$t = \pi N$ $\quad m_x \approx p(x) = 1/N$

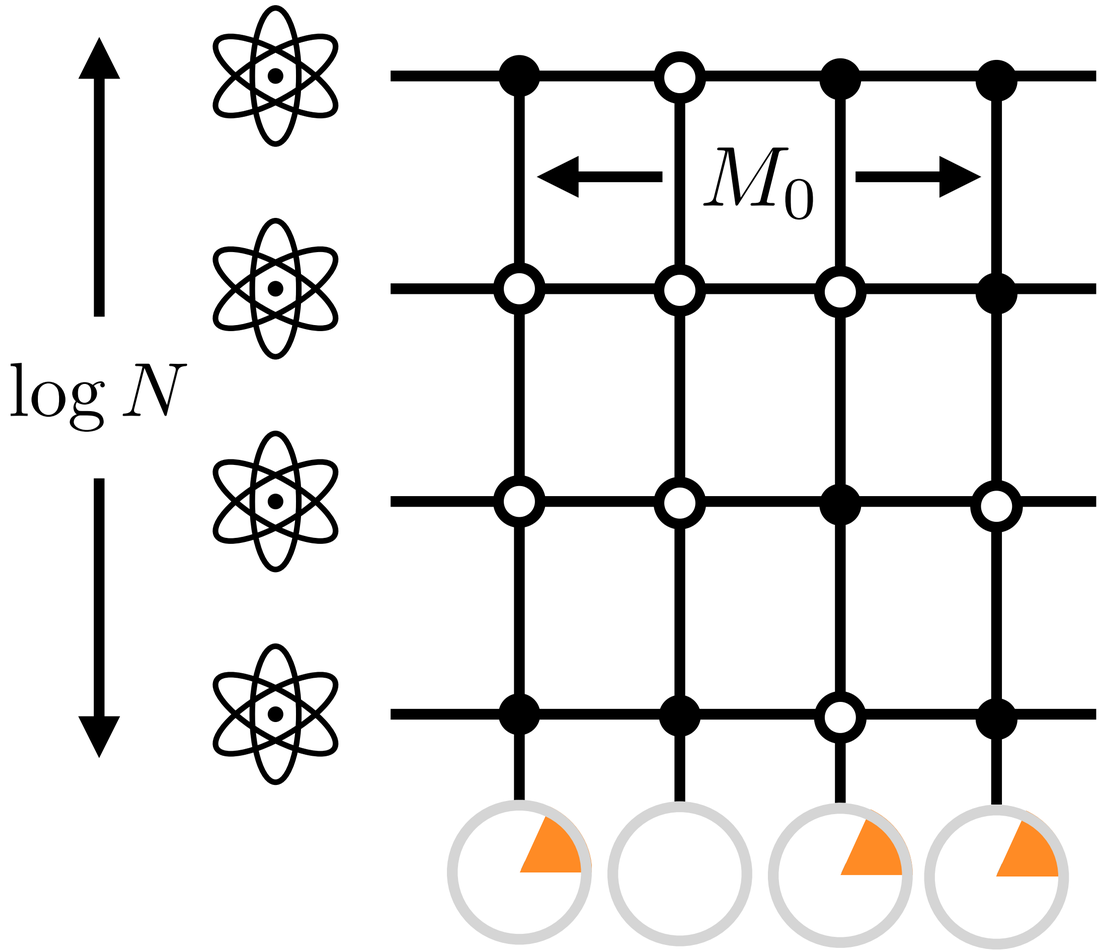
$$\approx \sum_x \exp(i\pi f(x))|x\rangle\langle x|$$

Toy example

QRAM on the fly

Example

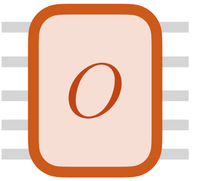
Goal: estimate property of $f : [N] \to \{0, 1\}$, given $\{x_i, f(x_i)\}_{i=1}^{M}, p(x) = 1/N$



$$V_i = \exp\left(i\frac{t}{M_0} f(x_i)|x_i\rangle\langle x_i|\right)$$

$$V_{M_0} \cdots V_1 \approx_{\epsilon} \sum_x (-1)^{f(x)} |x\rangle\langle x|$$

phase oracle



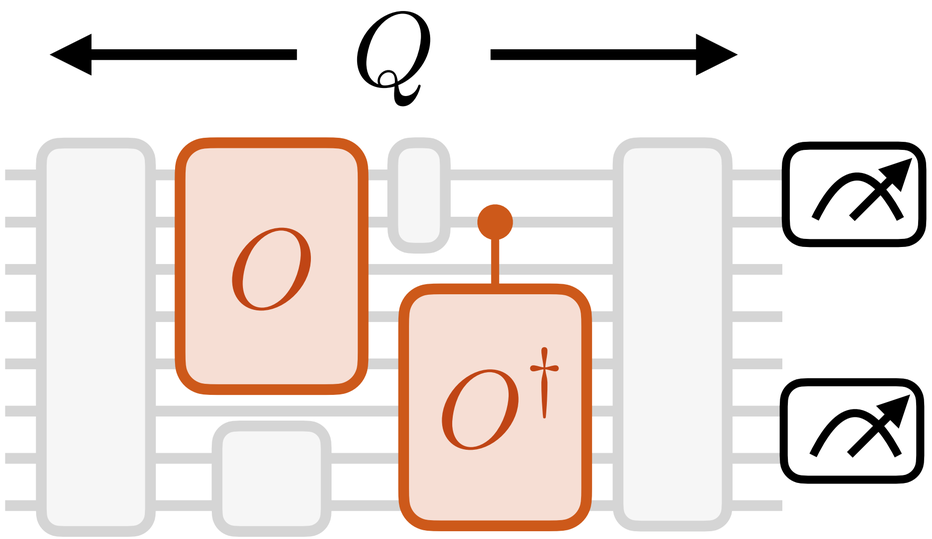
Theorem $M_0 = \Theta(N/\epsilon)$

Toy example

QRAM on the fly

Example

Goal: estimate property of $f : [N] \to \{0, 1\}$, given $\{x_i, f(x_i)\}_{i=1}^{M}, p(x) = 1/N$



any quantum query algorithm

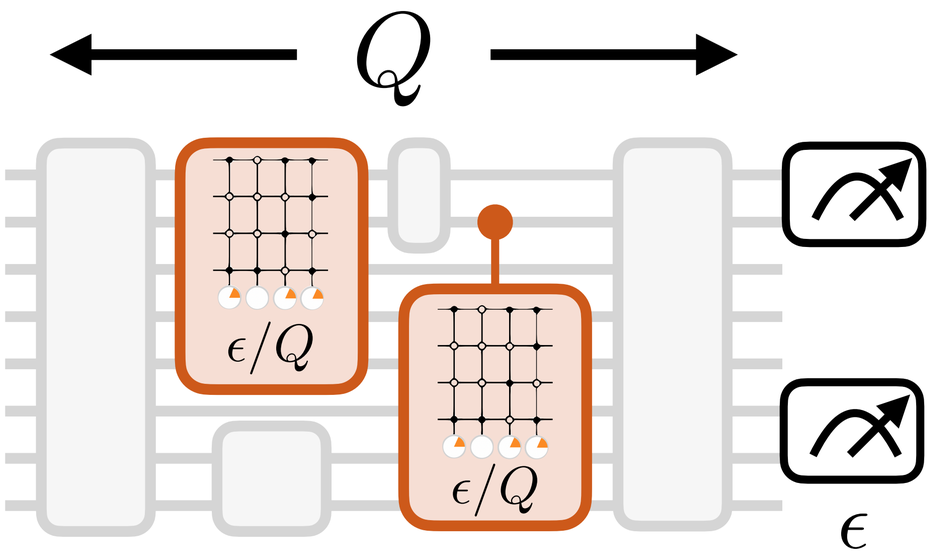
making Q queries

Toy example

QRAM on the fly

Example

Goal: estimate property of $f : [N] \to \{0, 1\}$, given $\{x_i, f(x_i)\}_{i=1}^{M}, p(x) = 1/N$

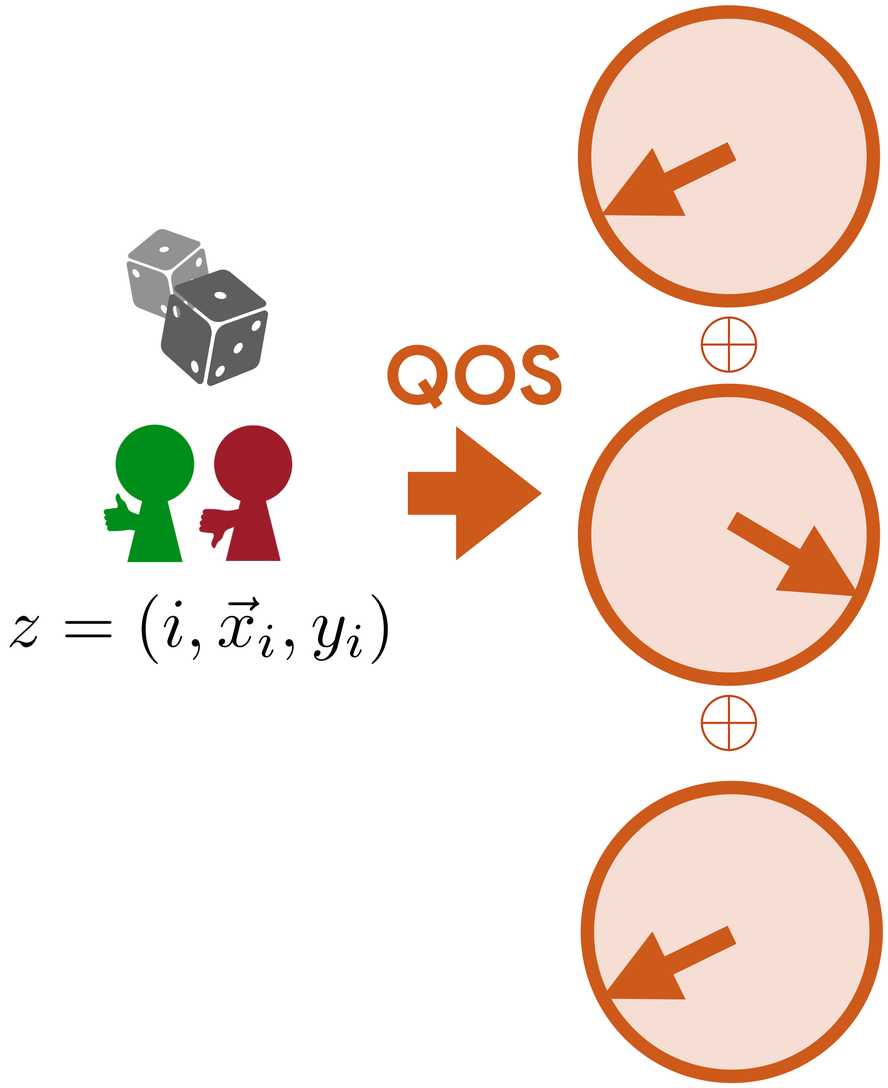


any quantum query algorithm

making Q queries

$$M = Q \cdot \underbrace{O\left(\frac{N}{\epsilon/Q}\right)}_{M_0} = O\left(\frac{NQ^2}{\epsilon}\right)$$

Quantum oracle sketching



Quadratic slowdown in Q is the fundamental price to pay.

incoherent sampling of data, amp^2=prob, HL^2=SQL

Theorem (Quantum Oracle Sketch)

With $\Theta(NQ^2)$ samples, we can emulate Q quantum queries to an oracle of N items with no memory overhead.

Quantum oracle sketching

$z = (x, f(x))$

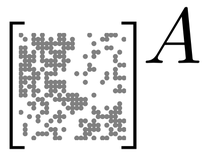
$|x\rangle \to (-1)^{f(x)}|x\rangle$

phase oracle

$|x\rangle|y\rangle \to |x\rangle|y \oplus f(x)\rangle$

XOR oracle

Robust to noise and correlation in data; support flexible data structures.



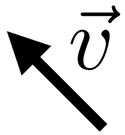
$z = (i, j, A_{ij})$

QOS

$U = \begin{pmatrix} A & \star \\ \star & \star \end{pmatrix}$

block encoding

[cumulative counter oracle + in-place binary search]



$z = (k, v_k)$

$|0\rangle \to |v\rangle$

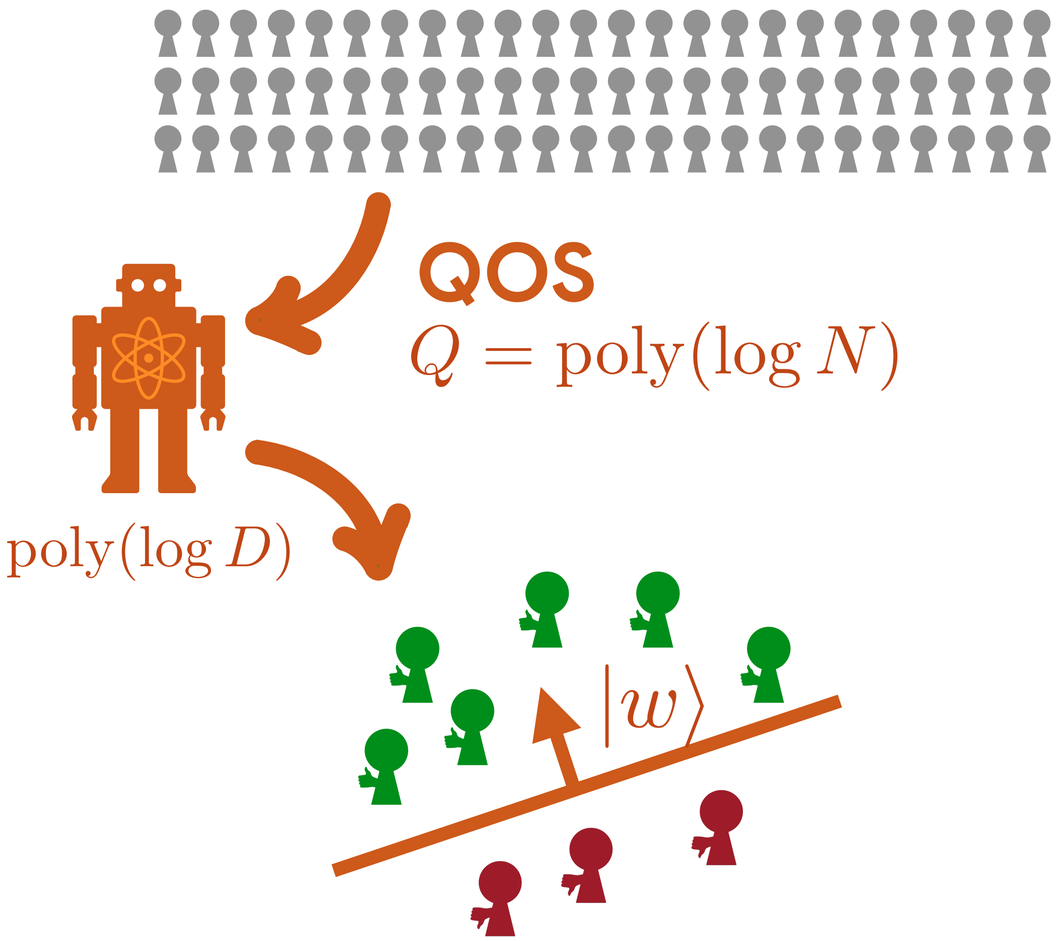
state preparation unitary

Theorem (Quantum Oracle Sketch)

With $\Theta(NQ^2)$ samples, we can emulate Q quantum queries to an oracle of N items with no memory overhead.

[vector flattening w/ randomized signed Hadamard transform]

Quantum oracle sketching



ML tasks can be solved with $\mathrm{poly}(\log D)$ size and $Q = \mathrm{poly}(\log N)$ queries.

SVM: linear system
PCA: ground state prep

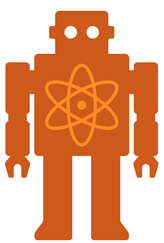
Theorem (Quantum Oracle Sketch)

With $\tilde{\Theta}(N)$ samples, a quantum machine of $\mathrm{poly}(\log D)$ size can solve the ML tasks.

Quantum oracle sketching

QOS

$Q = \mathrm{poly}(\log N)$



$\mathrm{poly}(\log D)$

$|w\rangle$

Classical Shadow

$\{\text{Clifford } C, \text{bitstring } b\}$

$\mathrm{poly}(\log D)$ **size classical model**

Interferometric classical shadow:

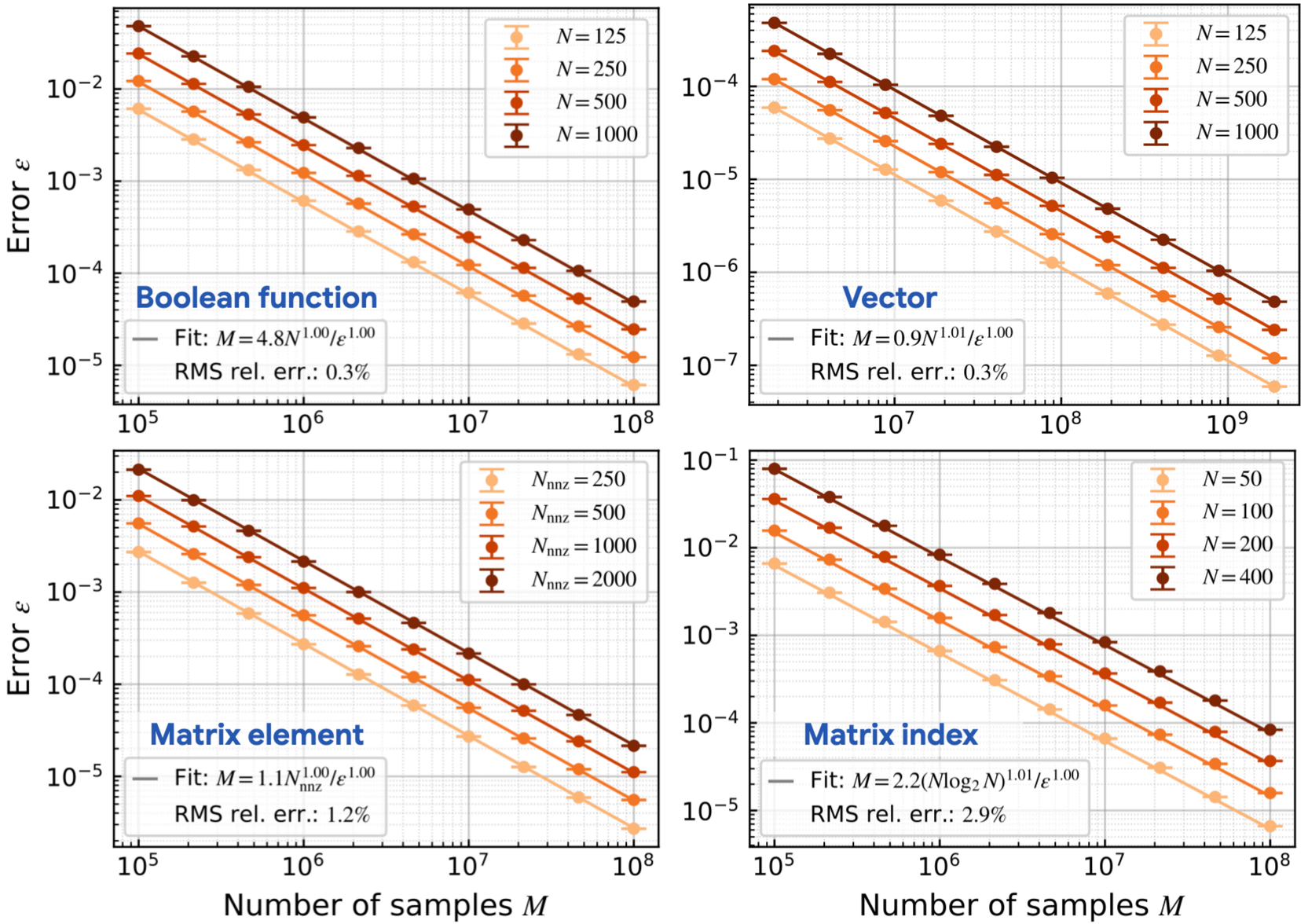
exponentially compact classical model

possible only via quantum technology.

Theorem (Quantum Oracle Sketch)

With $\tilde{\Theta}(N)$ samples, a quantum machine of $\mathrm{poly}(\log D)$ size can solve the ML tasks.

Quantum oracle sketching

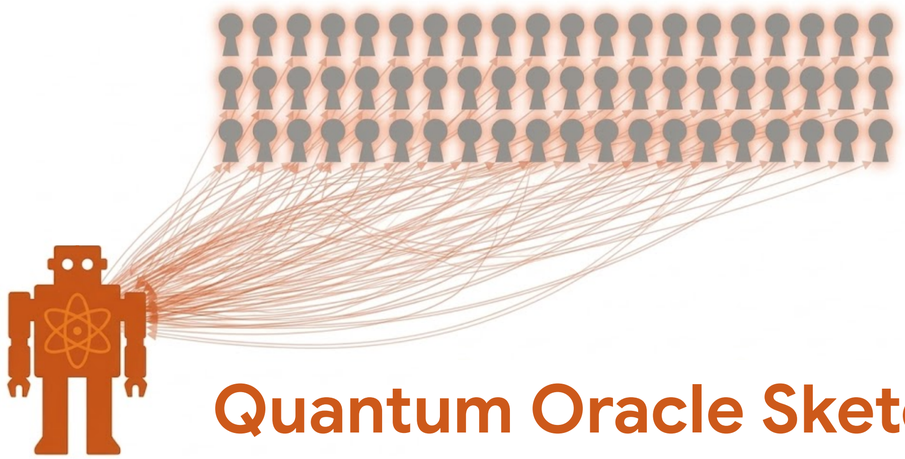


GitHub

haimengzhao/
quantum-oracle-
sketching

JAX-powered,
support GPU/TPU,
auto-differentiation

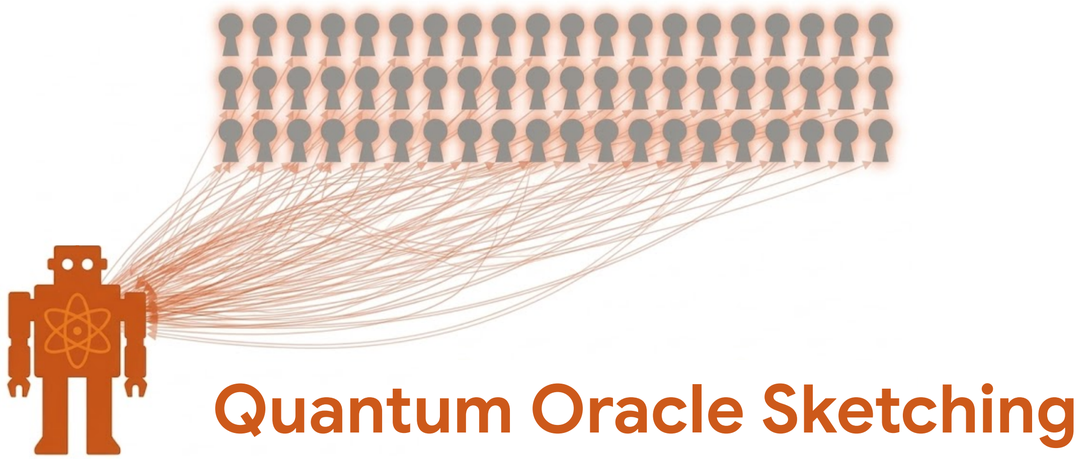
Quantum Algorithm



Quantum Oracle Sketching

query the classical world in quantum superposition

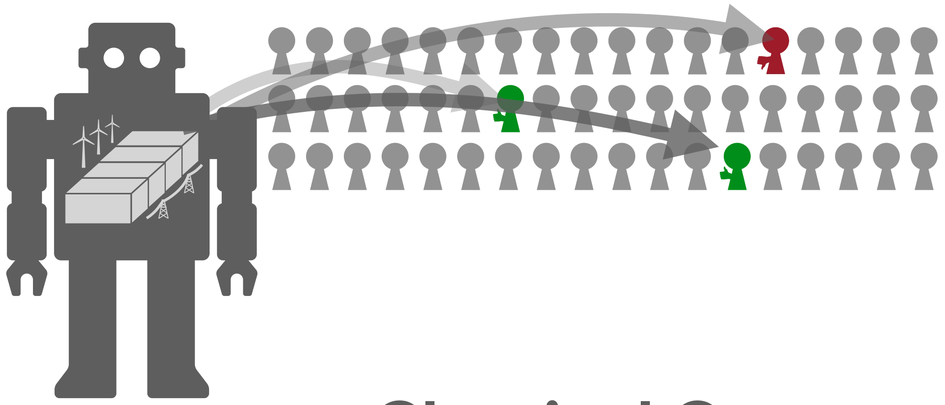
Quantum Algorithm



Quantum Oracle Sketching

query the classical world in quantum superposition

Classical Hardness



Classical Query

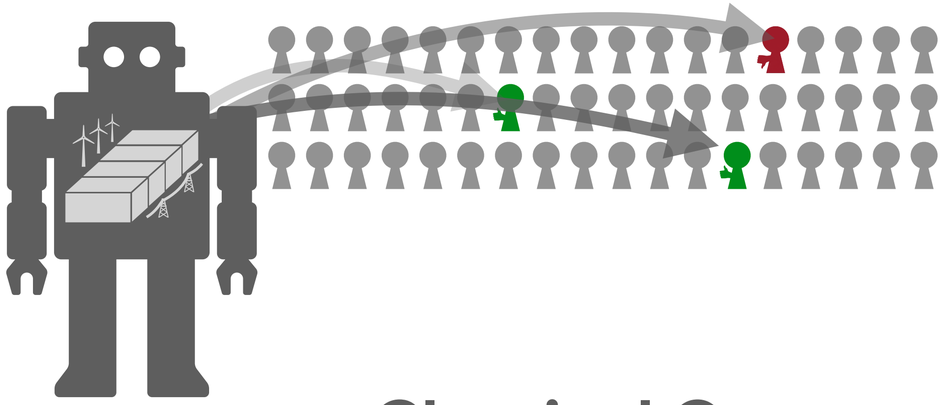
separation in query ability → memory advantage

Classical Hardness

1. Connect query separation to memory advantage

2. Insufficient memory must be compensated by more samples

3. Embed into ML tasks



Classical Query

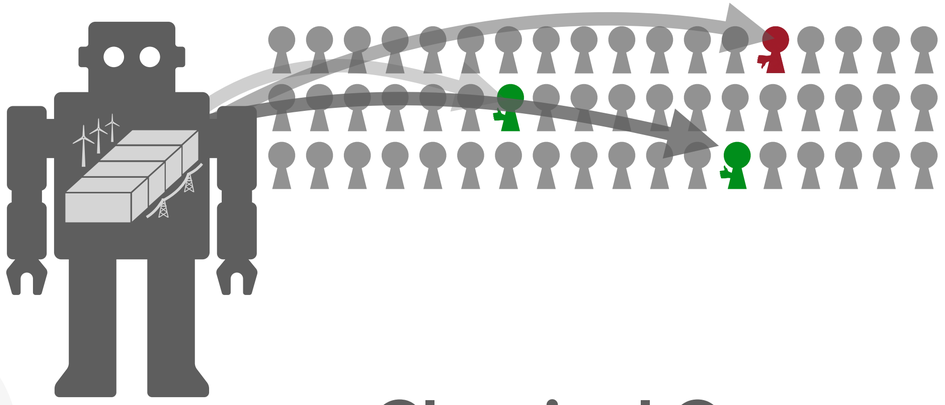
Classical Hardness

1. Connect query separation to memory advantage

Theorem (Classical Hardness)

For any query problem requiring Q quantum queries or Q_C classical queries, to solve an associated learning task with $\Theta(NQ^2)$ samples, any classical machine must have size

$$S \geq \Omega(Q_C/Q^2)$$



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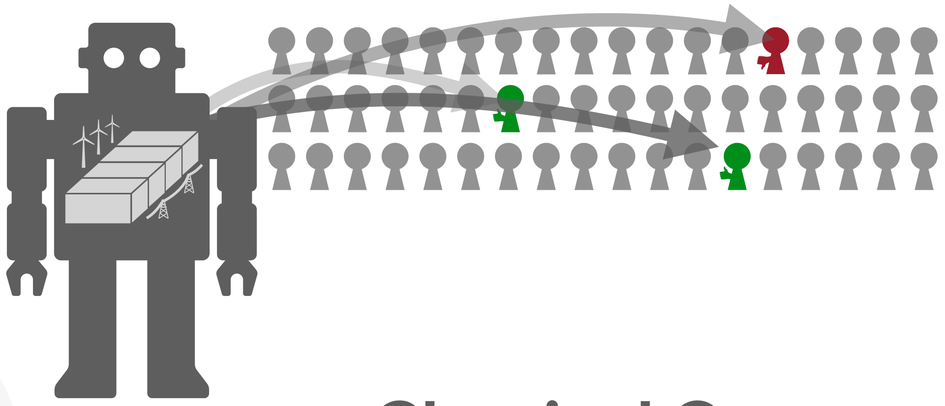

Classical Query

Any super-quadratic query separation,

$$Q = O(N^{0.49}), Q_C = \Theta(N)$$
$$\implies S \geq \Omega(N^{0.02})$$

Classical Hardness

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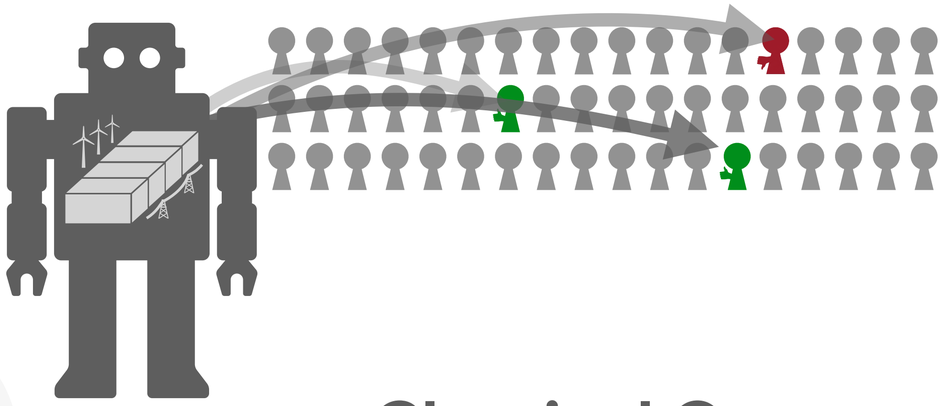

Classical Query

For ML tasks,

$$Q = \mathrm{poly}(\log N), Q_C = \Theta(N^{0.99})$$
$$\implies S \geq \Omega(N^{0.99})$$

Classical Hardness

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Theorem (Classical Hardness)

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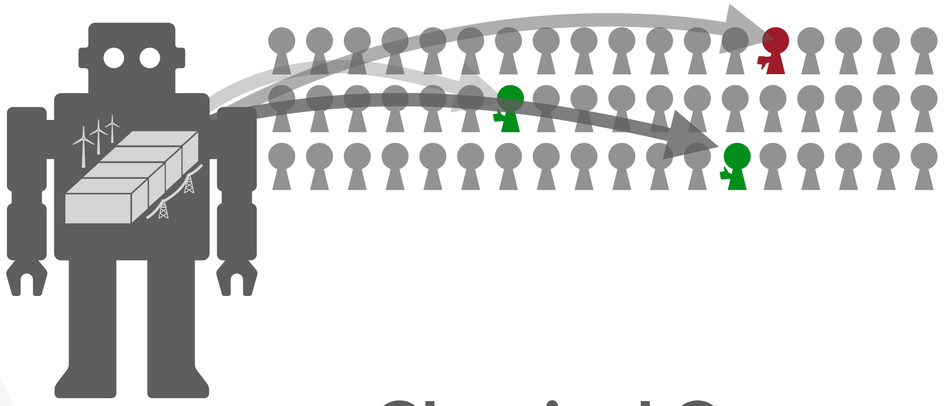


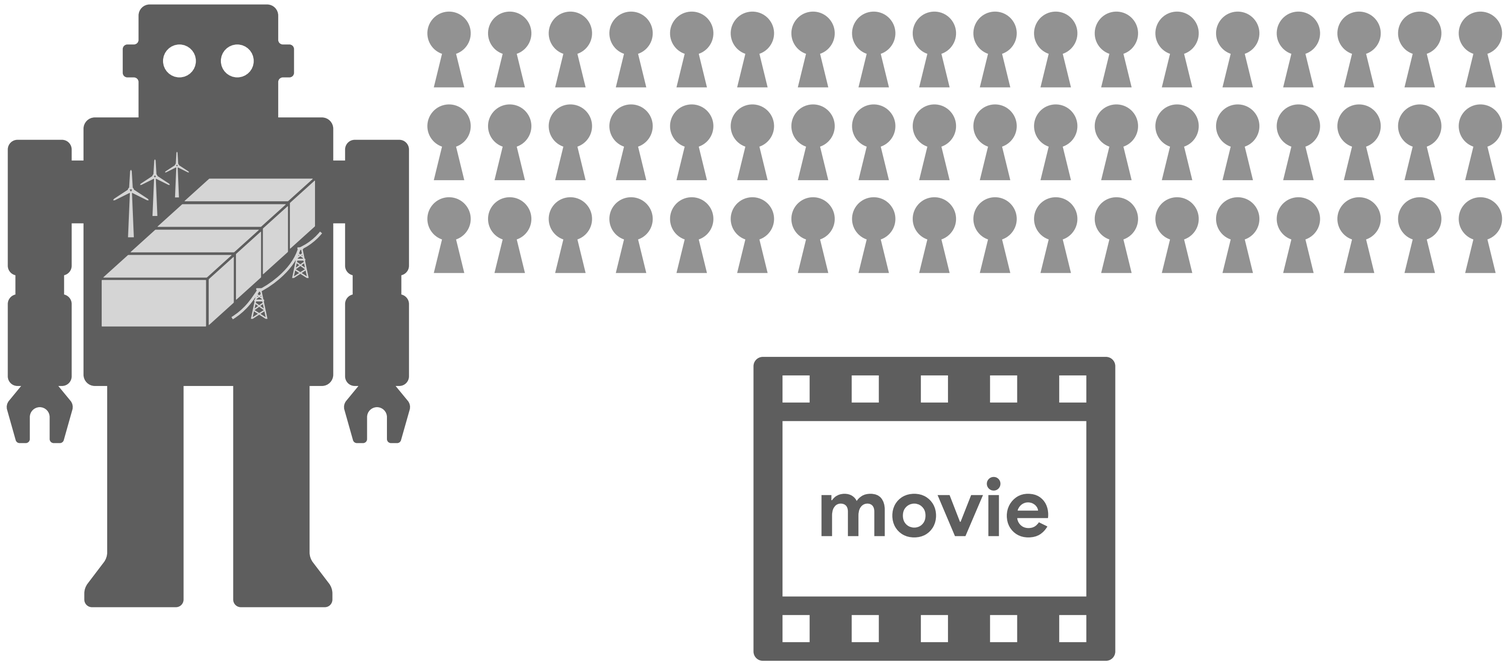
Classical Query

Sample-space lower bound

$$MS \geq \Omega(NQ_C)$$

#sample $M = \Theta(NQ^2)$ from QOS

Query Problem

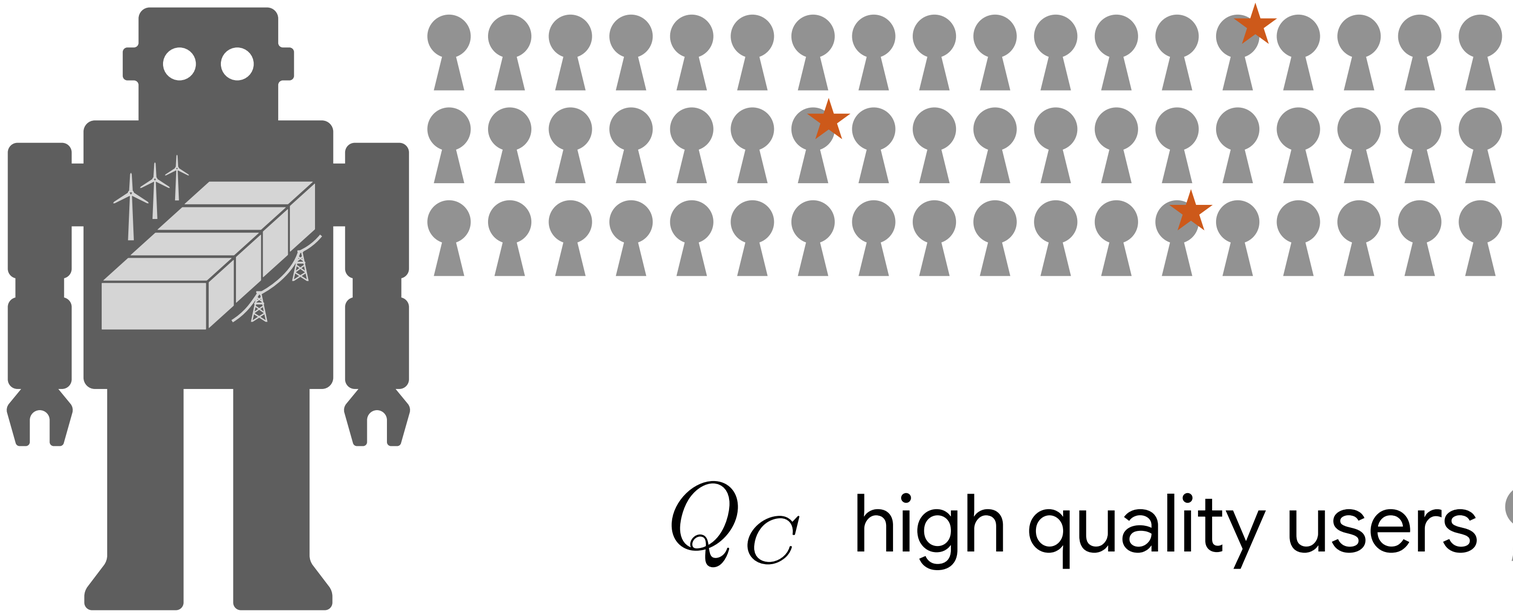


N users

Goal

rate the movie

Query Problem



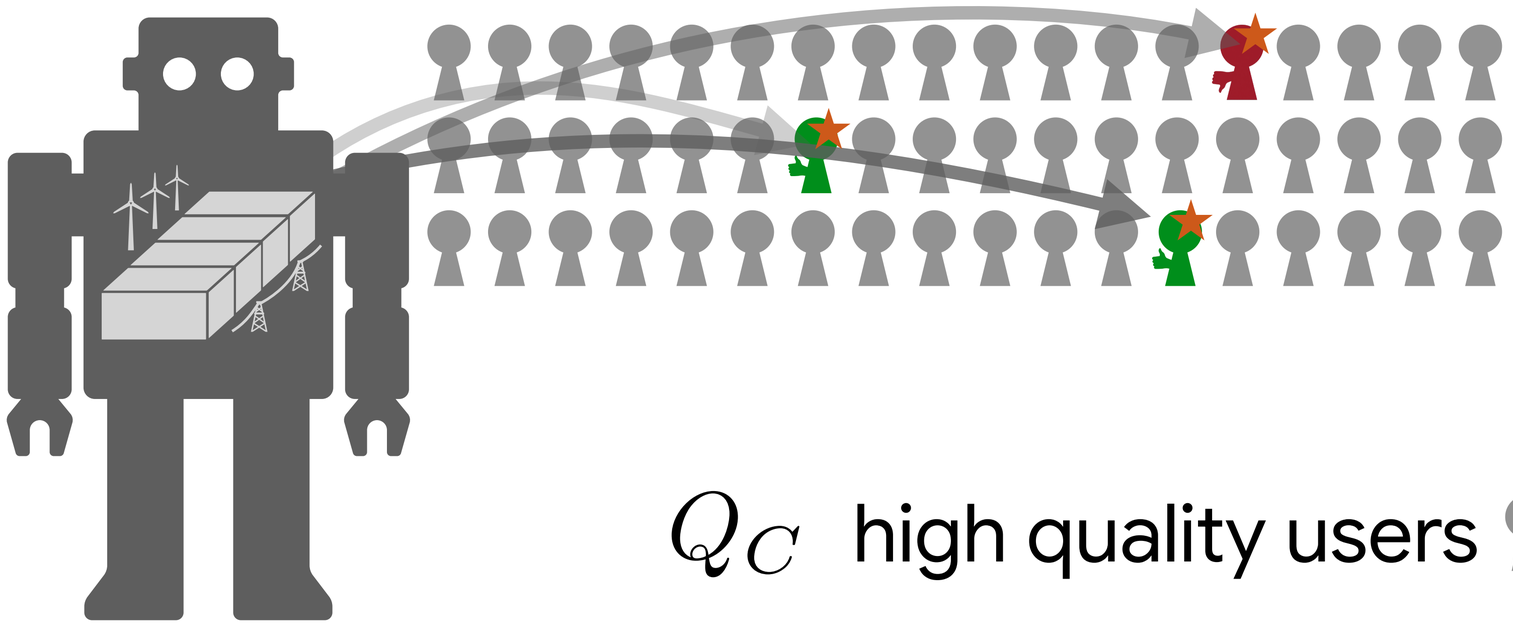
N users

[oracle]

$x \to f(x)$

Q_C high quality users

Query Problem



N users

[oracle]

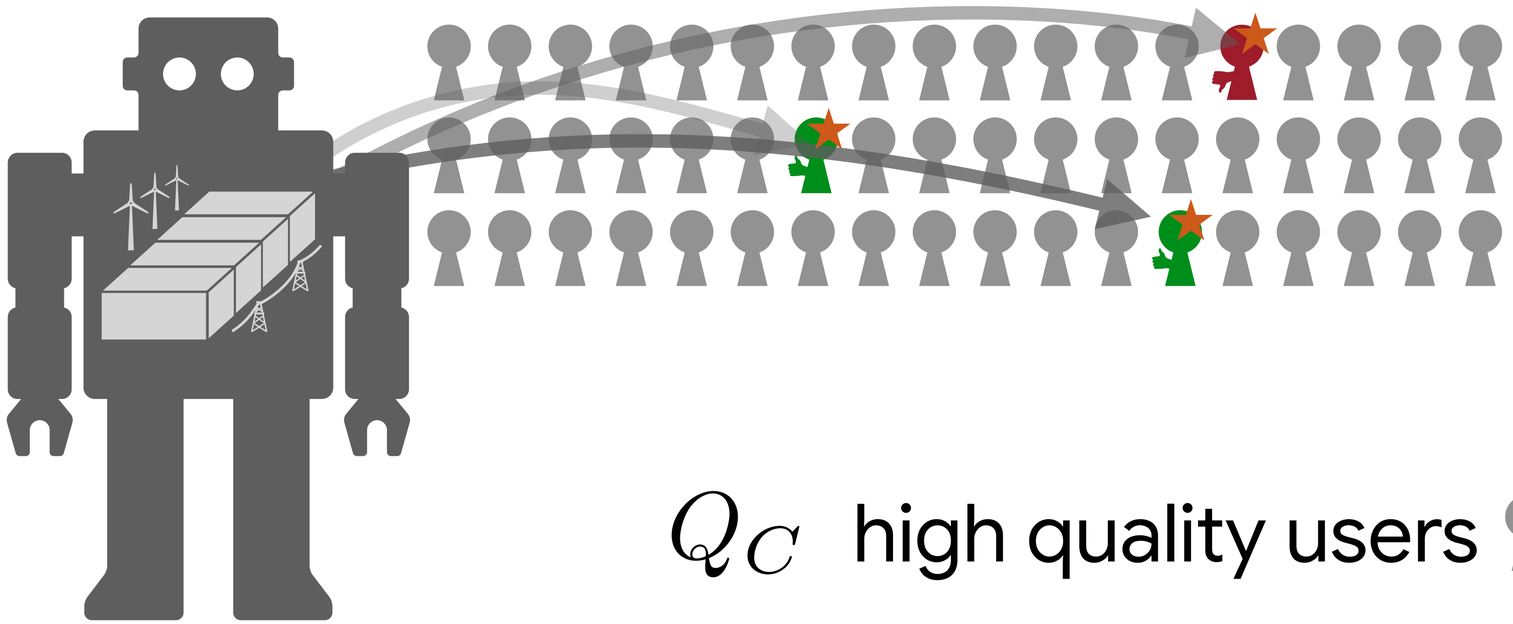
$x \to f(x)$

Q_C high quality users

querying them suffices to determine the goal

[classical query complexity]

Query Problem



N users

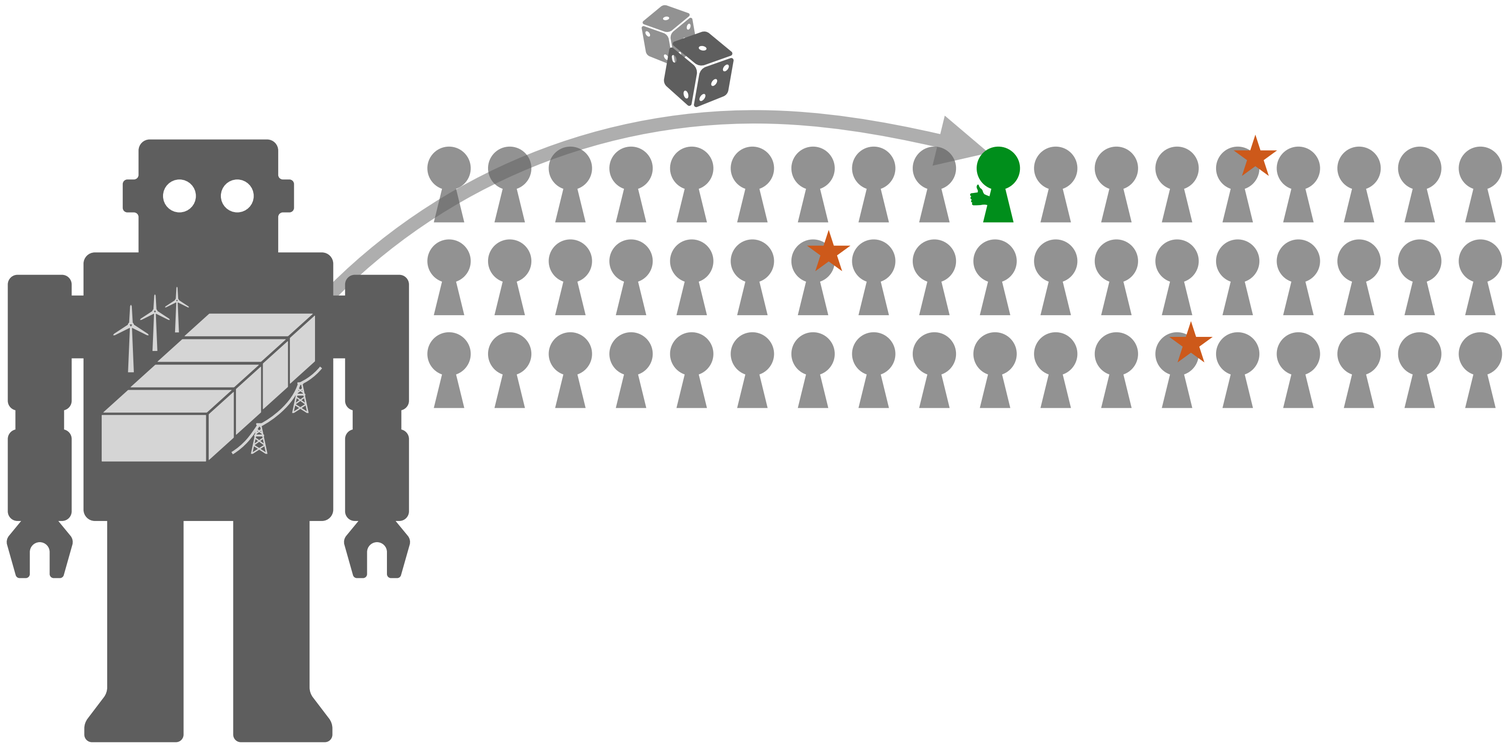
[oracle]

$x \to f(x)$

Q_C high quality users

amount of info needed to estimate some property of the oracle [classical query complexity]

Oracle Property Estimation



random data

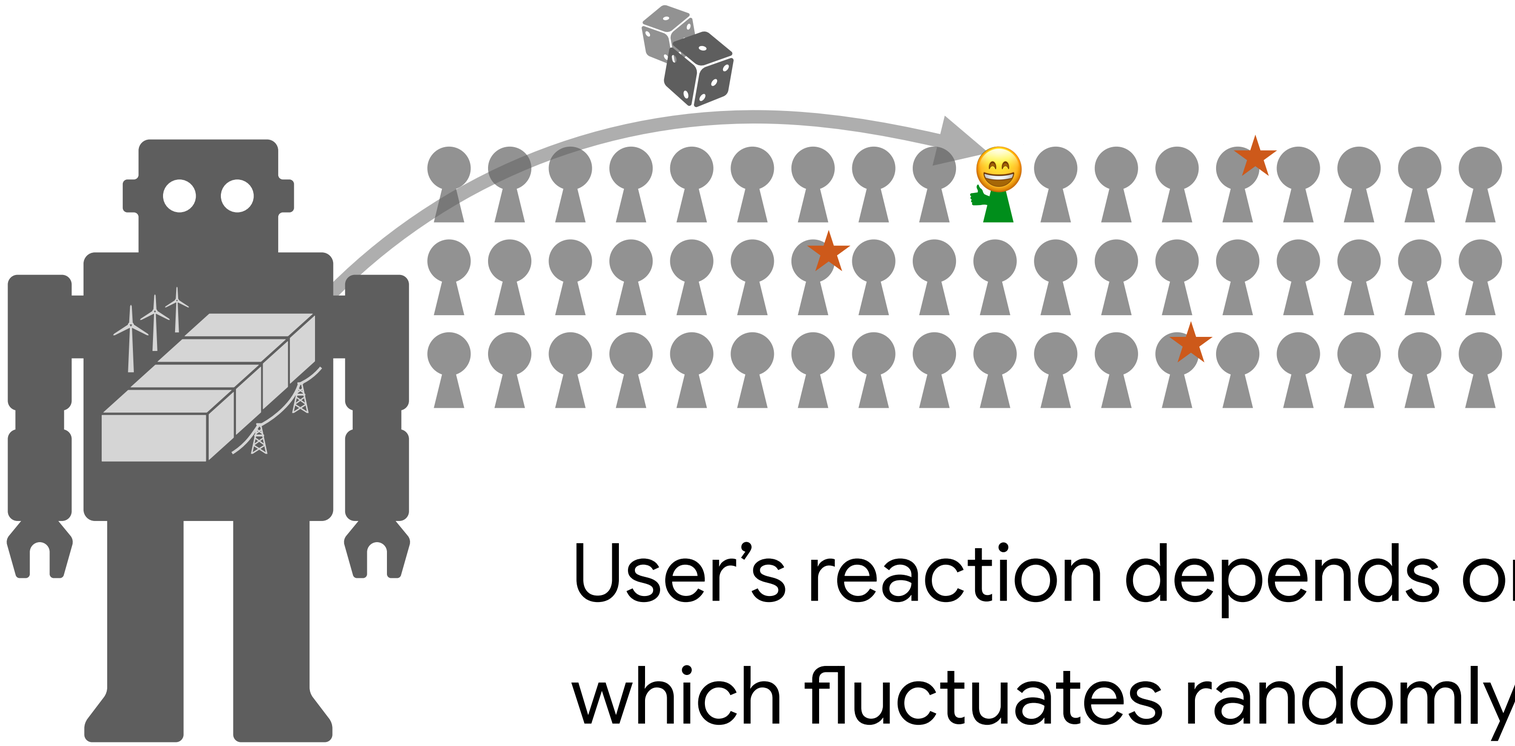
from N users

$z = (x, f(x))$

Q_C

info needed to estimate
oracle property

Noisy Oracle Property Estimation (NOPE)



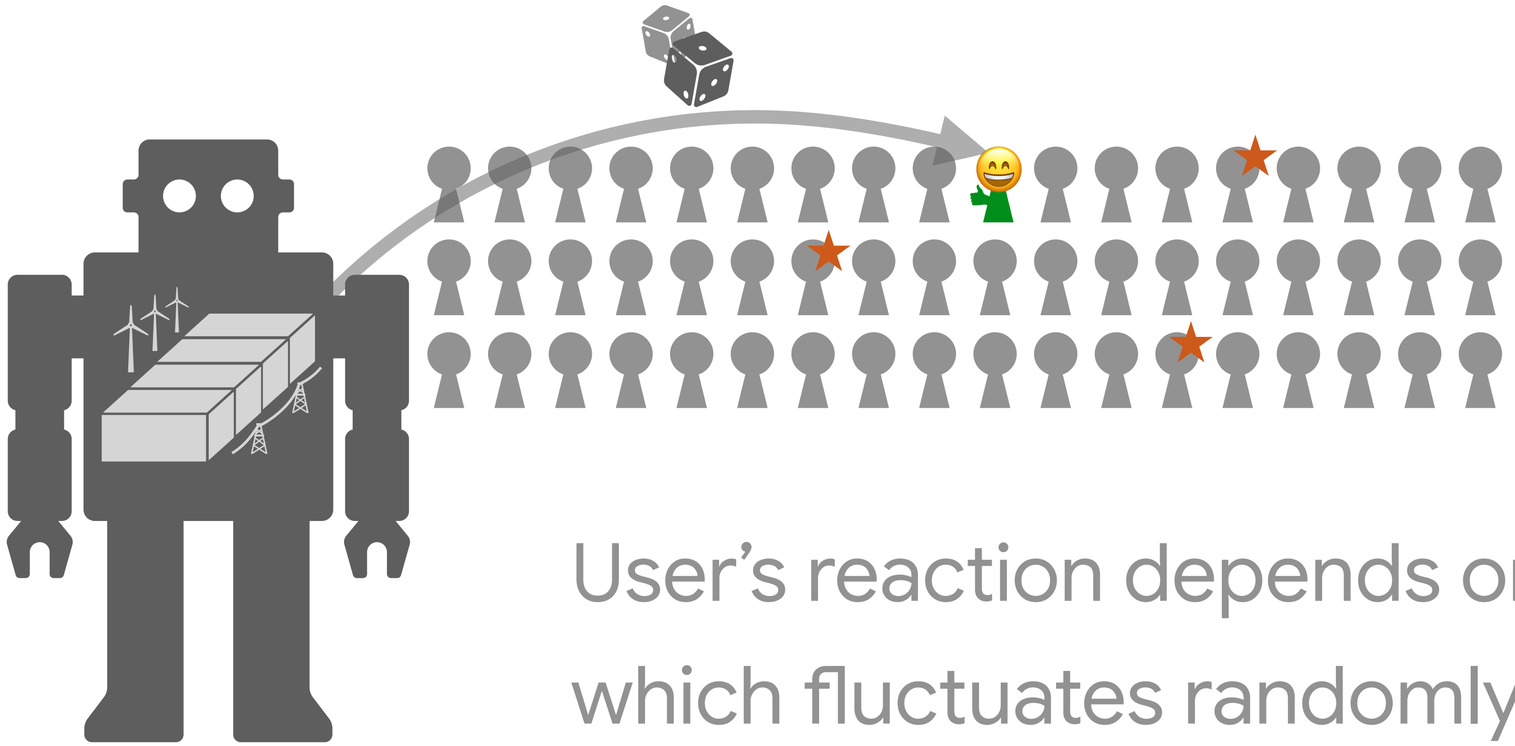
noisy data
from N users
$z = (x, f_\alpha(x), \alpha)$

User's reaction depends on their mood,
which fluctuates randomly over time.

Q_C

info needed to estimate
oracle property

Noisy Oracle Property Estimation (NOPE)



noisy data

from N users

$z = (x, f_\alpha(x), \alpha)$

User's reaction depends on their mood,

which fluctuates randomly over time.

Need to gather both for accurate feedback.

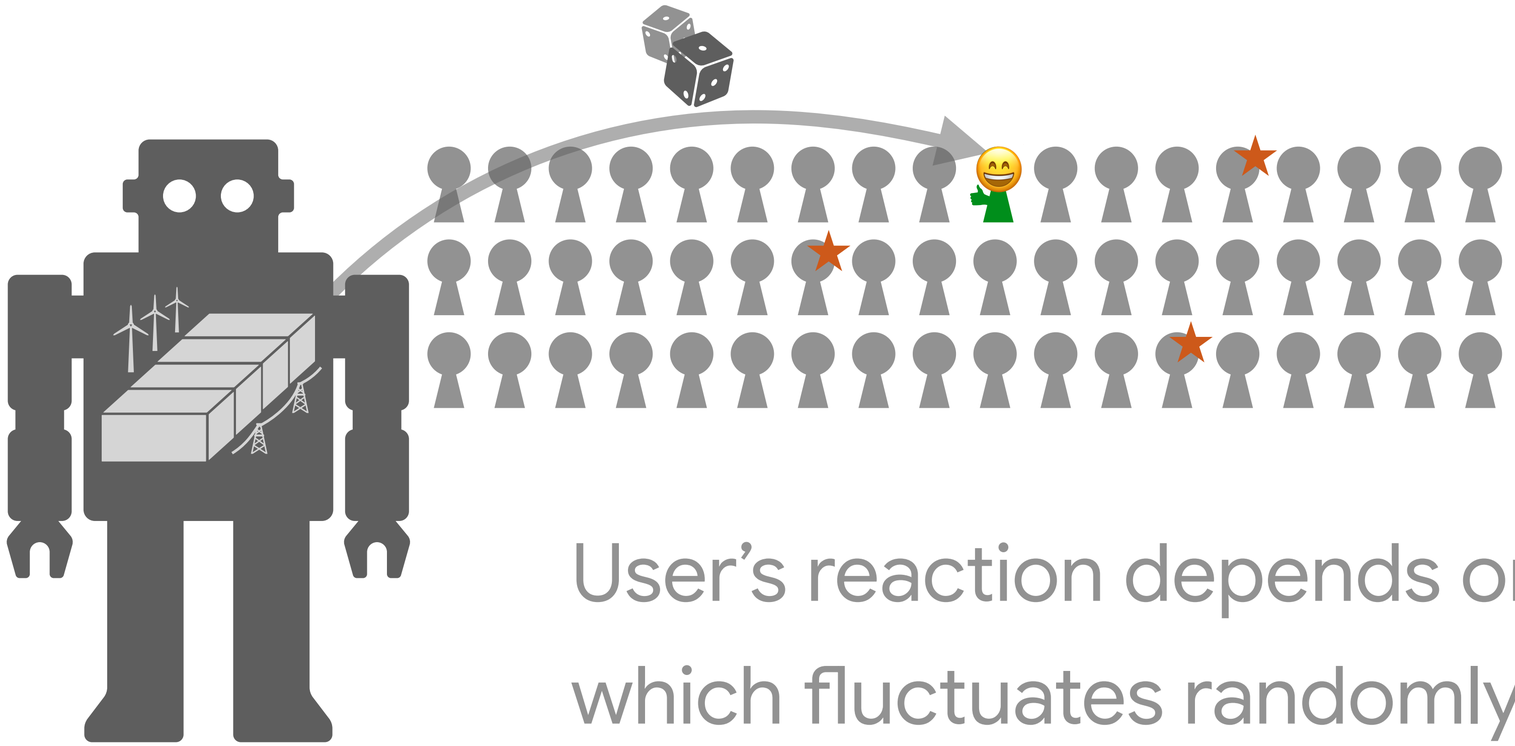
$f_1(x), f_2(x) \to f(x)$

via low-discrepancy encoding

Q_C

info needed to estimate oracle property

Noisy Oracle Property Estimation (NOPE)



noisy data

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User's reaction depends on their mood,

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Need to gather both for accurate feedback.

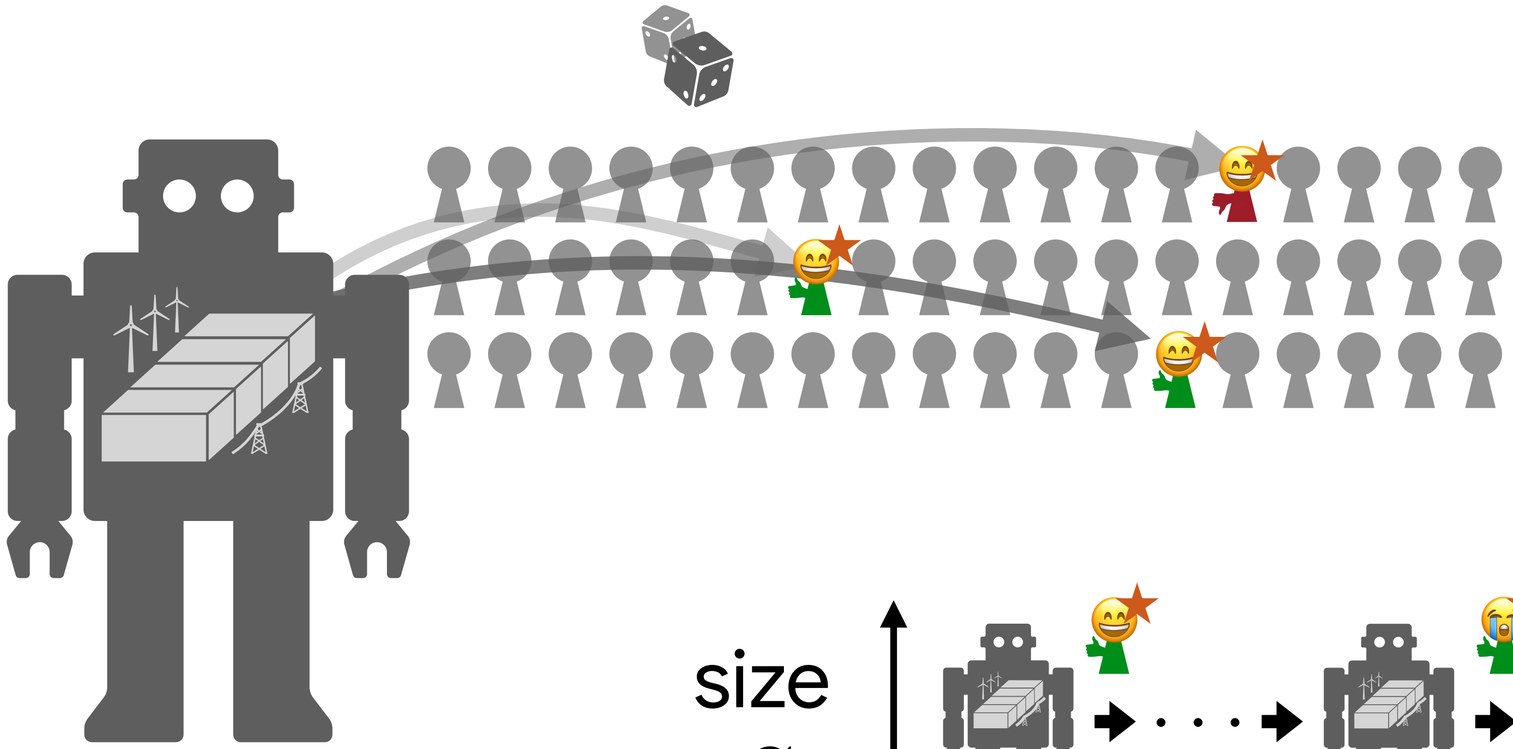
$f_1(x), f_2(x) \to f(x)$

Q_C

info needed to estimate oracle property

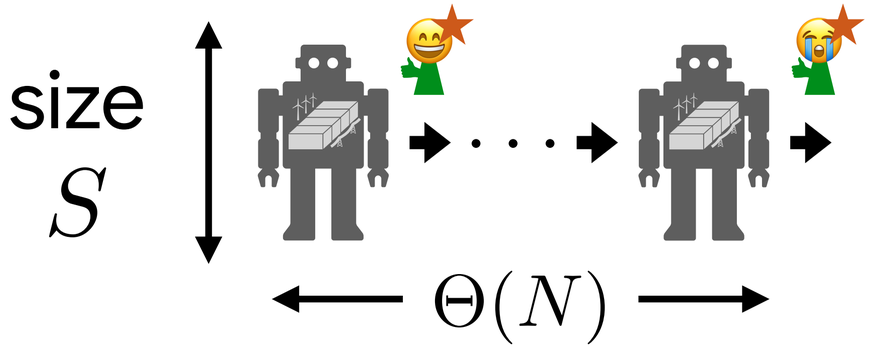
That takes $\Theta(N)$ time/samples. $p(x) = 1/N$

Noisy Oracle Property Estimation (NOPE)



noisy data
from N users

$z = (x, f_\alpha(x), \alpha)$



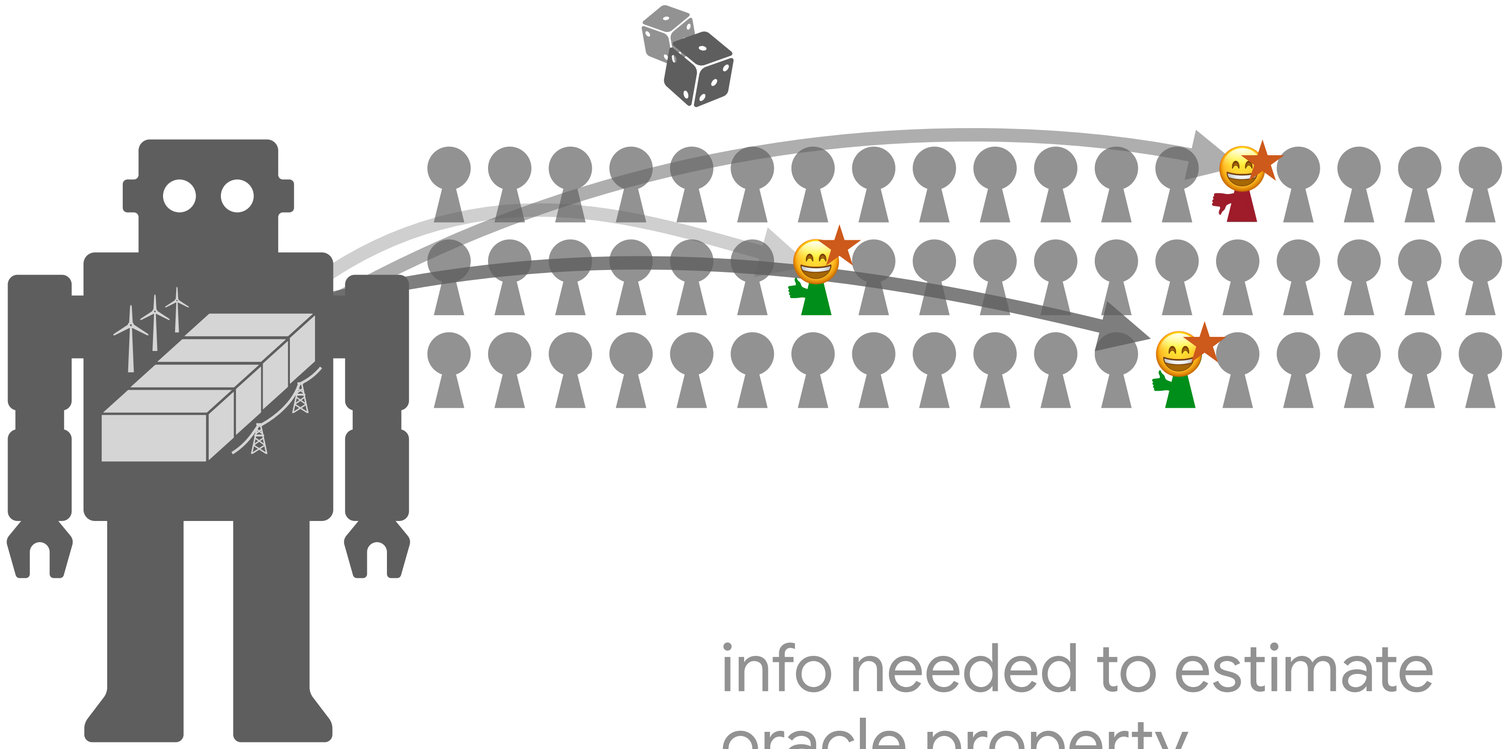
$$\text{info} \leq O\left(\frac{M}{N}\right) \cdot S$$

rounds

Q_C

info needed to estimate
oracle property

Noisy Oracle Property Estimation (NOPE)



noisy data

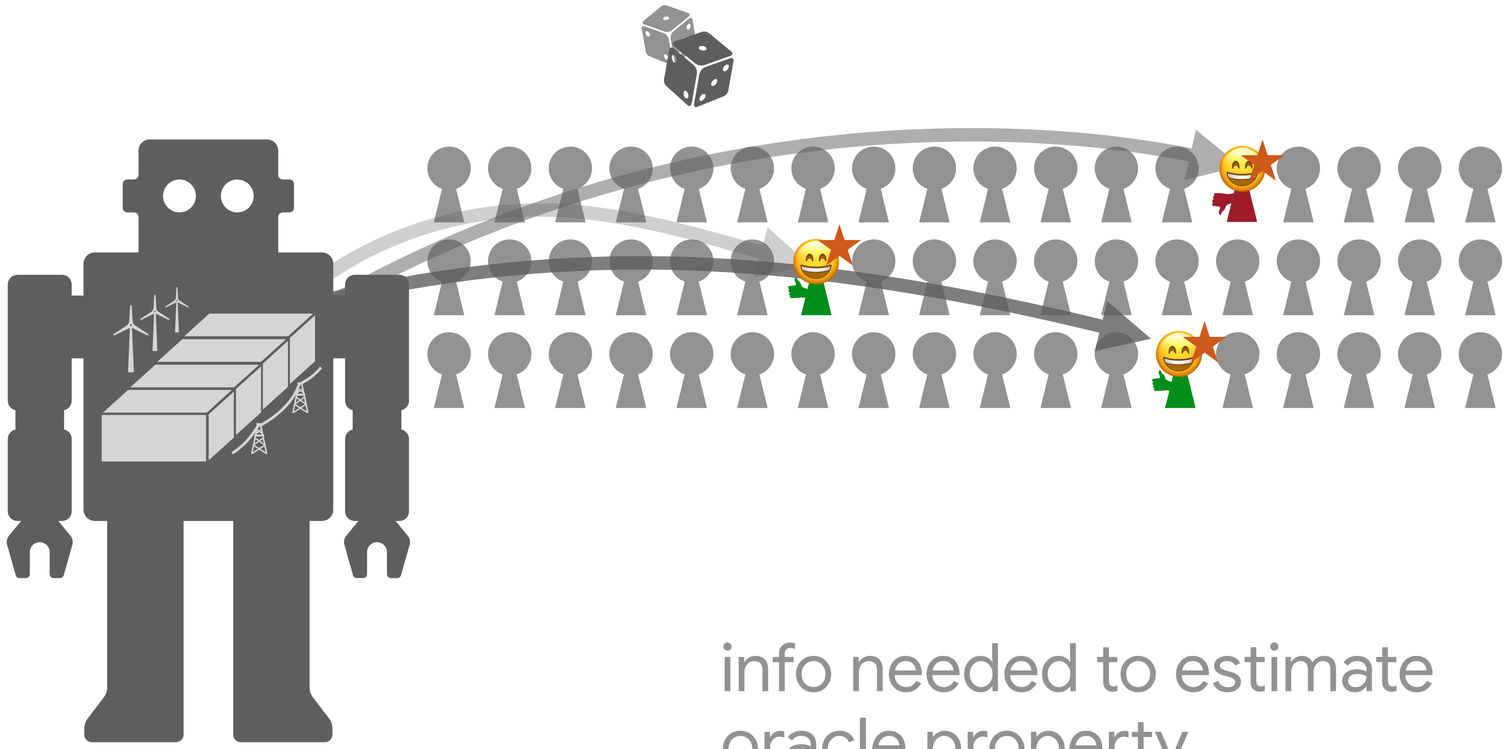
from N users

$z = (x, f_\alpha(x), \alpha)$

info needed to estimate oracle property

$$Q_C \leq \text{info} \leq O\left(\frac{M}{N}\right) \cdot S$$

Noisy Oracle Property Estimation (NOPE)



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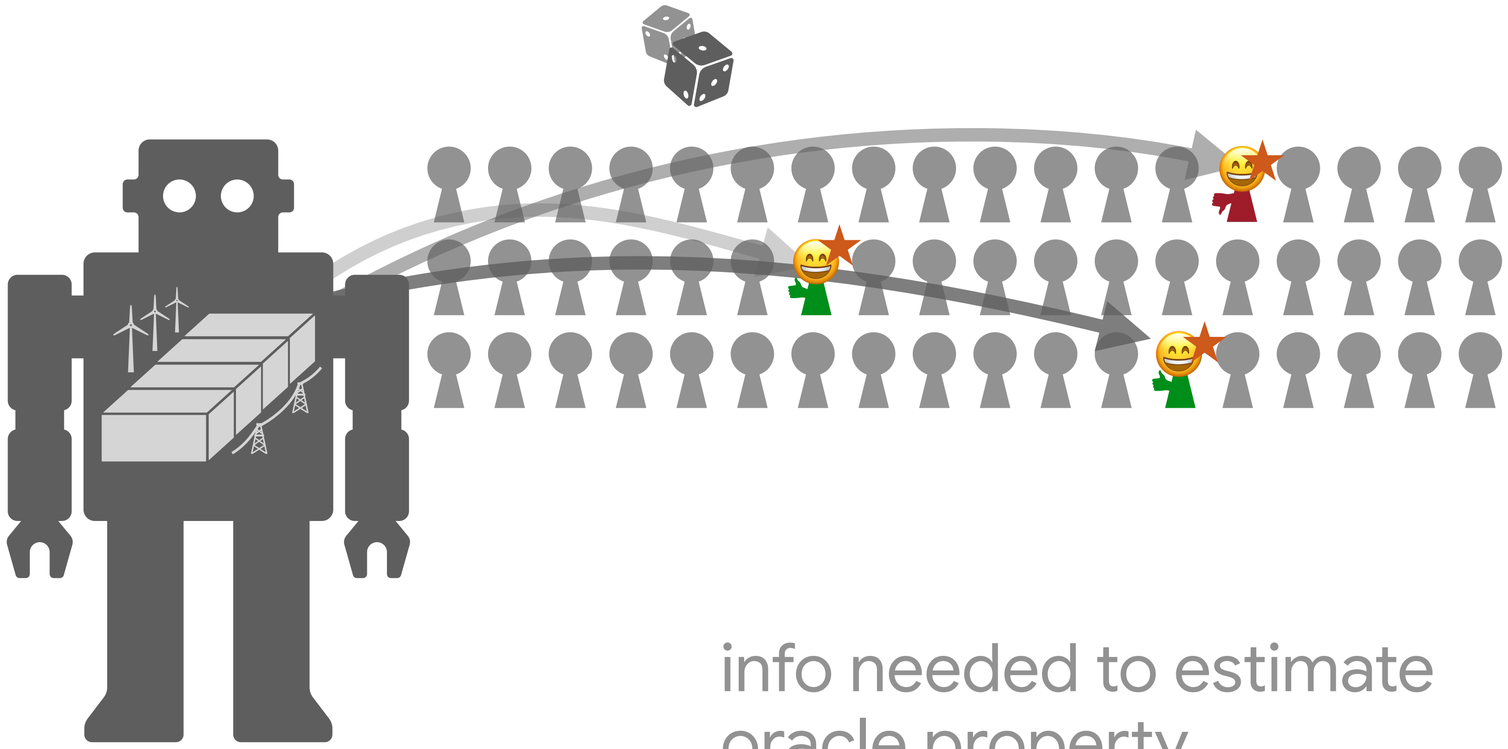
Sample-space lower bound $MS \geq \Omega(NQ_C)$

Noisy Oracle Property Estimation (NOPE)



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$$Q_C \leq \text{info} \leq O\left(\frac{M}{N}\right) \cdot S$$

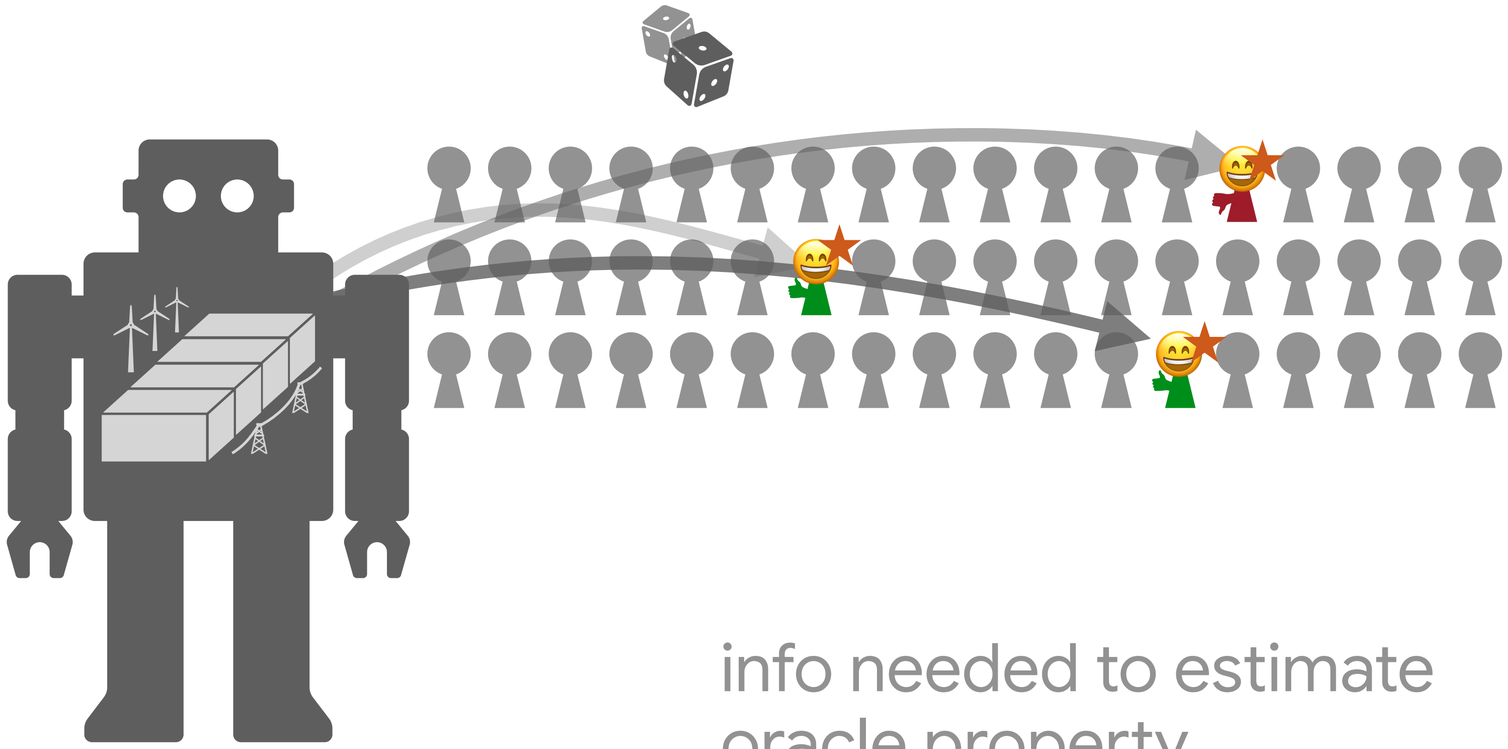
Theorem (Classical Hardness)

$$S \geq \Omega(Q_C/Q^2)$$

$$MS \geq \Omega(NQ_C)$$

#sample $M = \Theta(NQ^2)$ from QOS

Noisy Oracle Property Estimation (NOPE)



noisy data

from N users

$z = (x, f_\alpha(x), \alpha)$

info needed to estimate oracle property

$$Q_C \leq \text{info} \leq O\left(\frac{M}{N}\right) \cdot S$$

Theorem (Classical Hardness)

$$S \geq \Omega(N^{1-\zeta})$$

NOPE of Forrelation

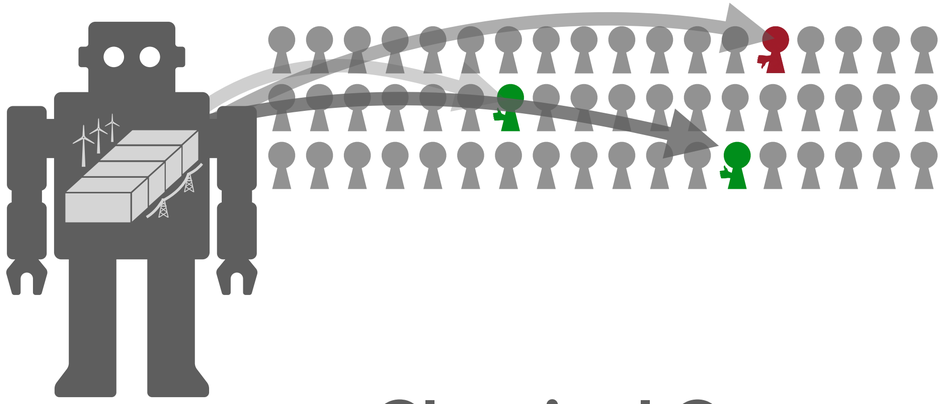
$$Q_C = \Omega(N^{1-\zeta}), Q = O(1)$$

Classical Hardness

1. Connect query separation to memory advantage

$$S \geq \Omega(N^{0.99})$$

2. Insufficient memory must be compensated by more samples



Classical Query

Classical Hardness

1. Connect query separation to memory advantage

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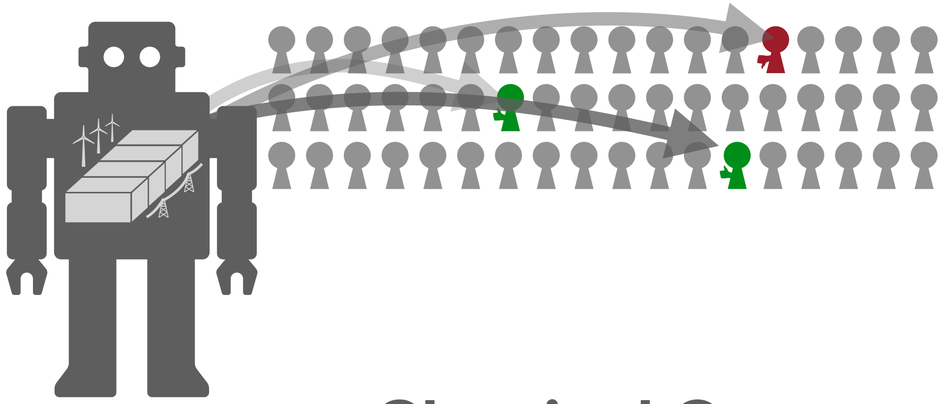

Classical Query

2. Insufficient memory must be compensated by more samples

NOPE: if $S \leq O(N^{0.99})$, then progress ≤ 0.1

Classical Hardness

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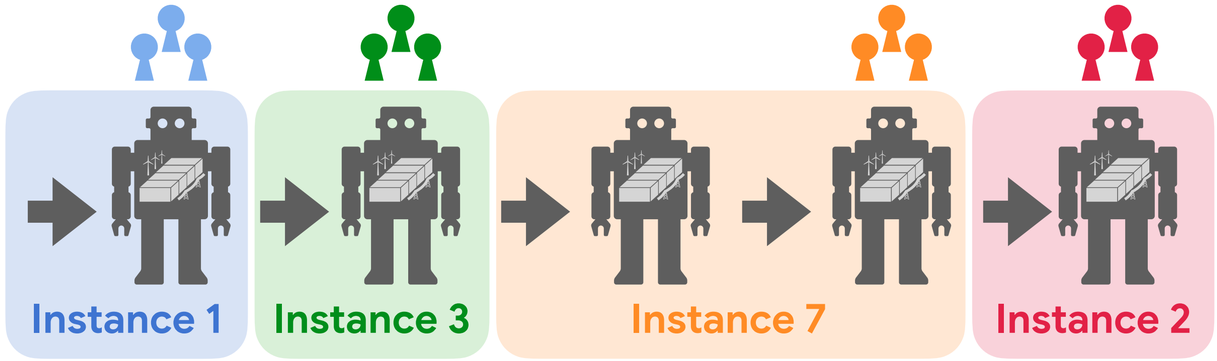
$$S \geq \Omega(N^{0.99})$$



2. Insufficient memory must be compensated by more samples

simultaneously solve
indep. instances of NOPE (the XOR)

NOPE: if $S \leq O(N^{0.99})$, then progress ≤ 0.1

XOR NOPE: if $S \leq O(N^{0.99})$, then progress $\leq (0.1)^{\mathrm{poly}(\log N)} = 1/\mathrm{superpoly}(N)$

Classical Hardness

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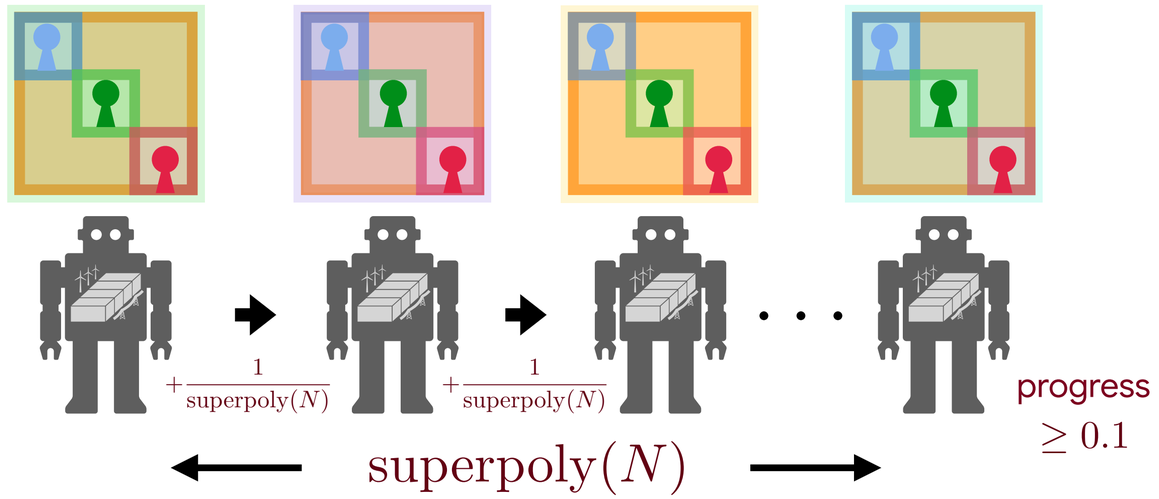
2. Insufficient memory must be compensated by more samples



NOPE: if $S \leq O(N^{0.99})$, then progress ≤ 0.1

XOR NOPE: if $S \leq O(N^{0.99})$, then progress $\leq (0.1)^{\text{poly}(\log N)} = 1/\text{superpoly}(N)$

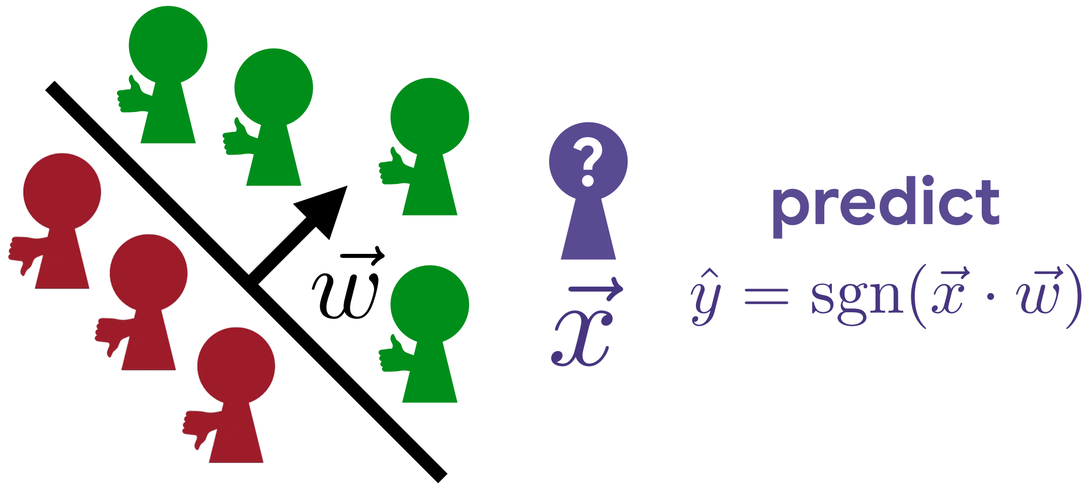
dynamic NOPE: if $S \leq O(N^{0.99})$, then requires $\text{superpoly}(N)$ samples

Classical Hardness

1. Connect query separation to memory advantage

2. Insufficient memory must be compensated by more samples

3. Embed into ML tasks



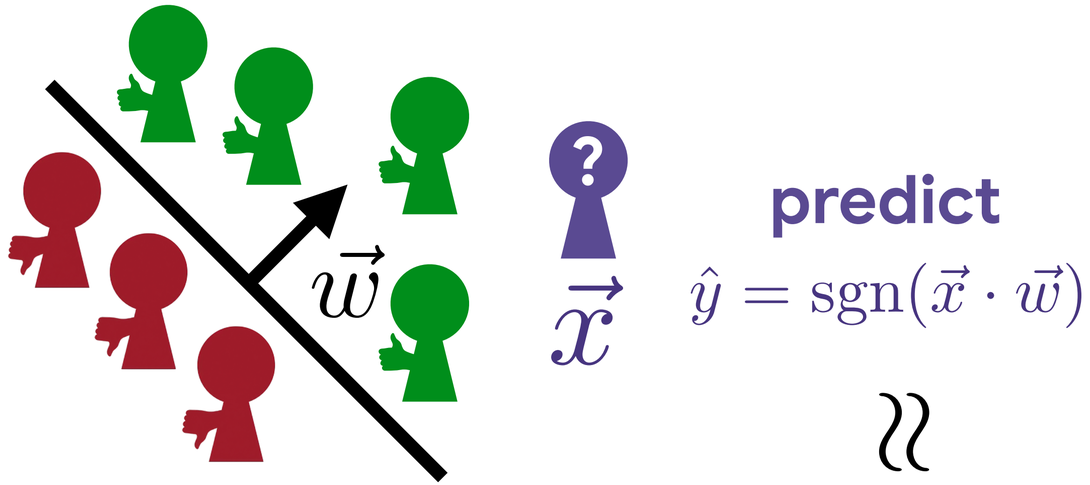
Classical Hardness

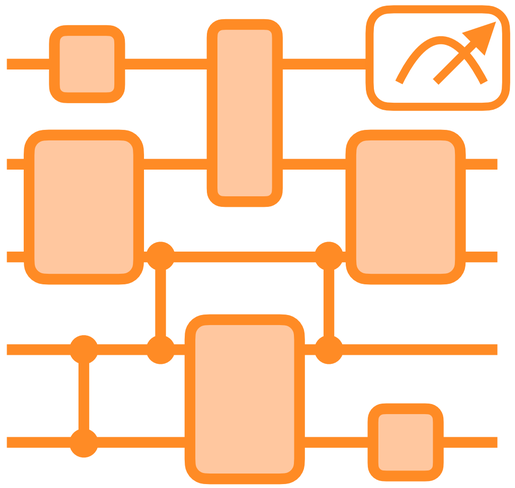
1. Connect query separation to memory advantage

2. Insufficient memory must be compensated by more samples

3. Embed into ML tasks

a. prove BQP hardness of ML tasks





Classical Hardness

1. Connect query separation to memory advantage

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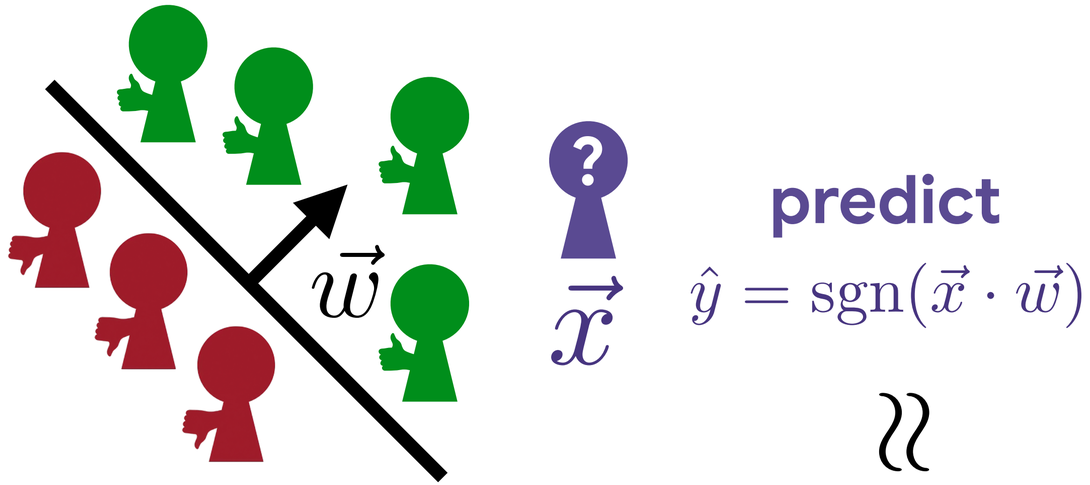
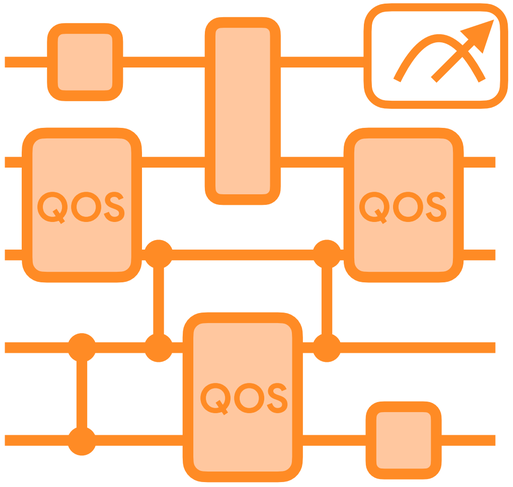


3. Embed into ML tasks

a. prove BQP hardness of ML tasks

b. embed QOS circuit for dynamic NOPE

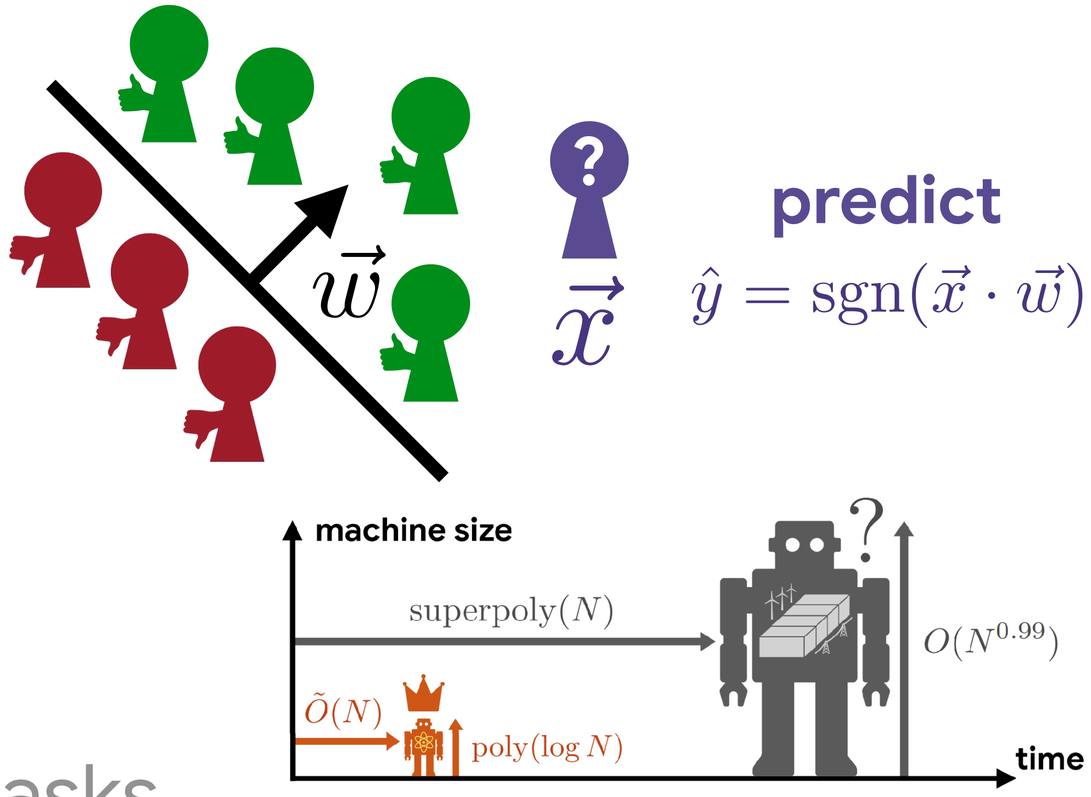


Classical Hardness

1. Connect query separation to memory advantage

2. Insufficient memory must be compensated by more samples

3. **Embed into ML tasks**

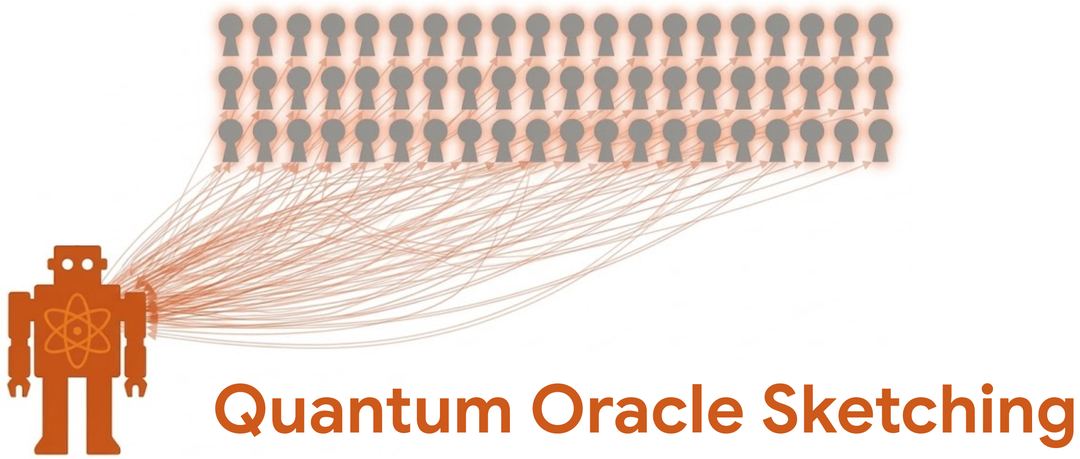


a. prove BQP hardness of ML tasks

b. embed QOS circuit for dynamic NOPE

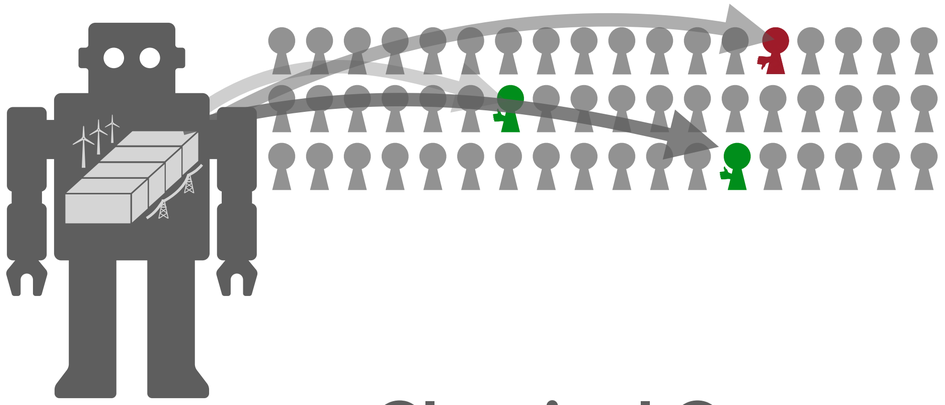
→ dynamic ML is classically hard

Quantum Algorithm



Quantum Oracle Sketching

Classical Hardness

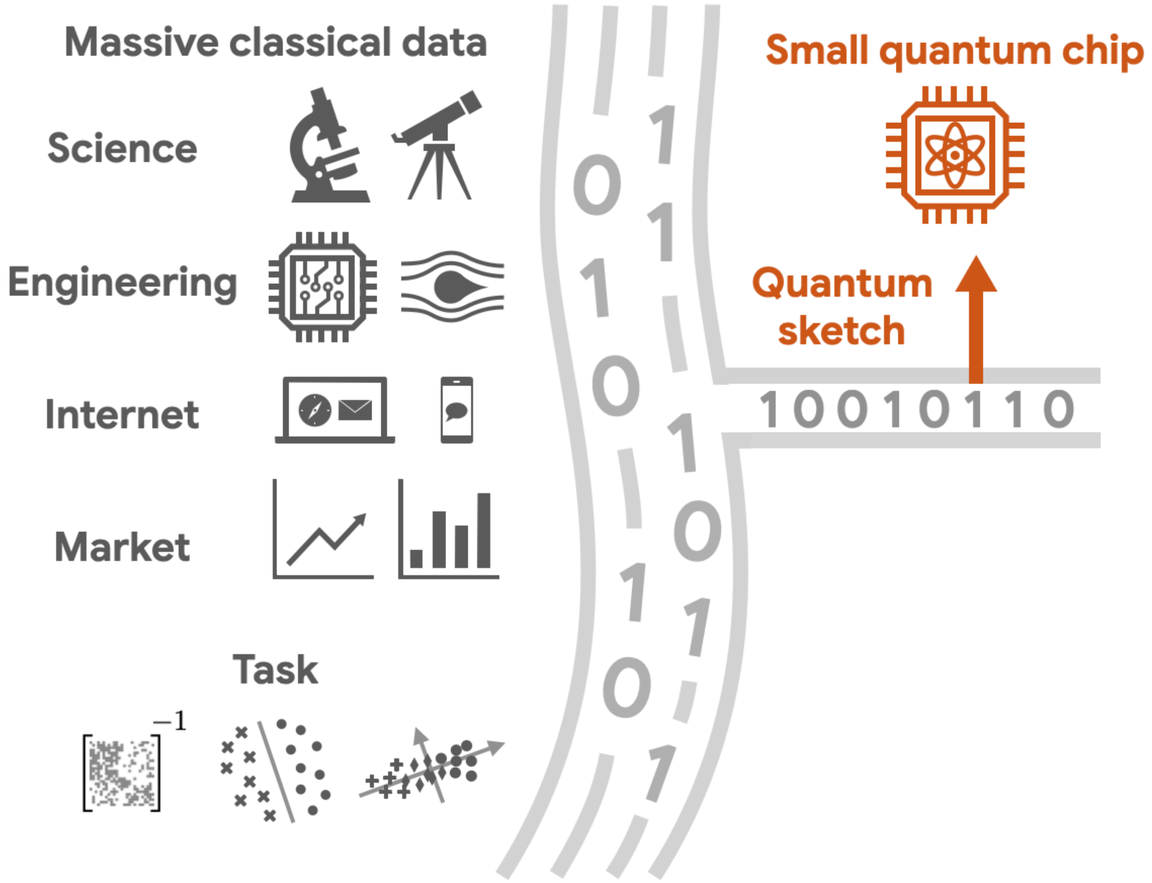


Classical Query

query the classical world in quantum superposition

separation in query ability → memory advantage

Conclusions



We prove that

small quantum machines

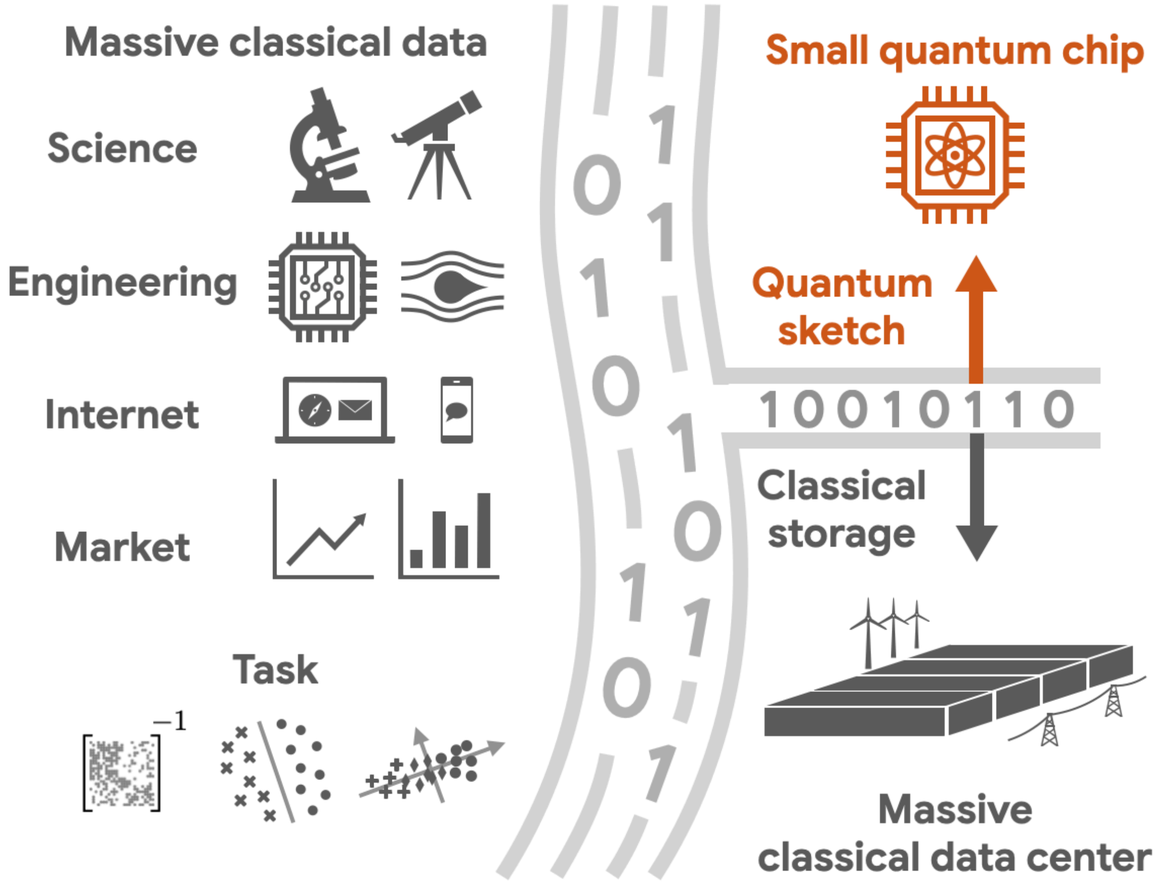
can process

massive classical data

and solve large-scale

machine learning problems.

Conclusions



We prove that

any classical machine

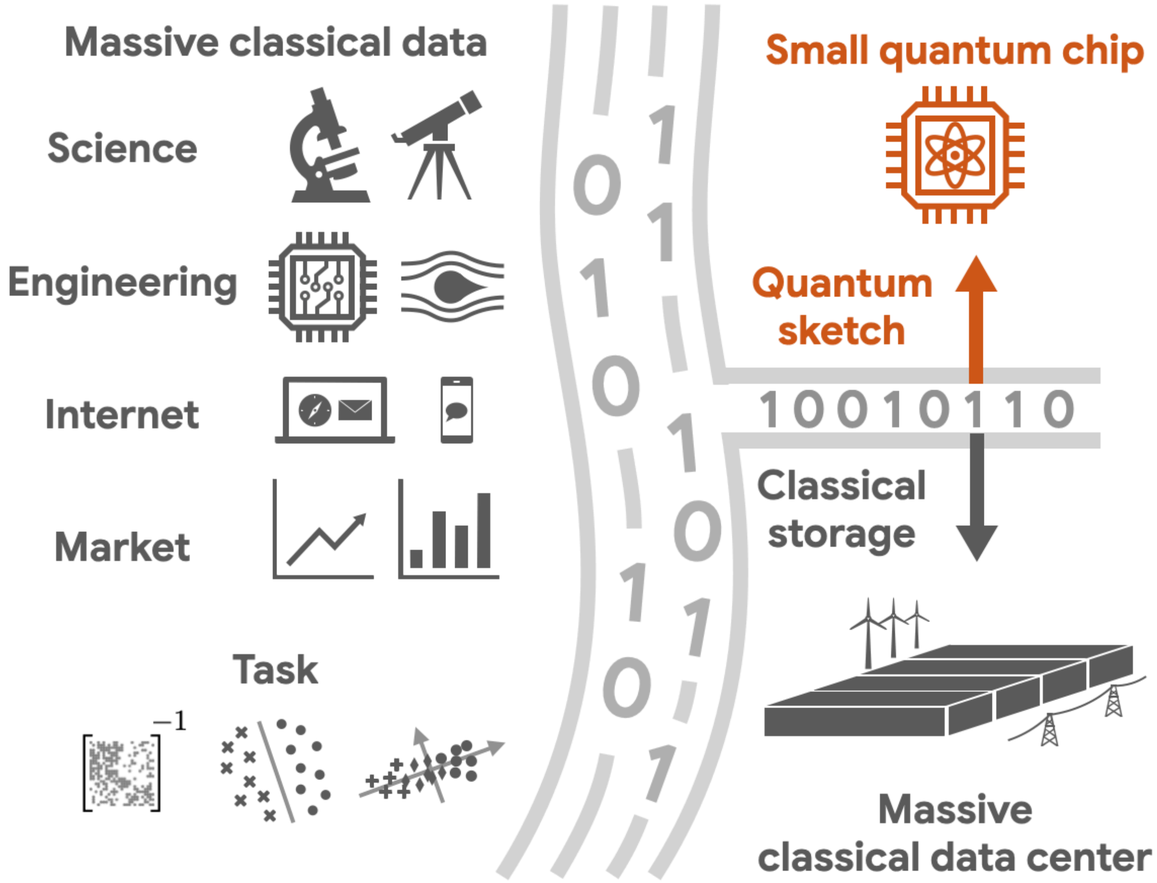
with the same performance

needs **exponentially larger size**

or **superpolynomially**

more samples.

Conclusions



A potential path for quantum computers to be broadly useful in our daily life.

Conclusions

This exponential quantum-classical separation relies solely on the correctness of quantum mechanics, persisting even if P=NP, BPP=BQP, etc.

Given sufficient time

≫

small quantum chip

massive classical data center

Conclusions

This exponential quantum-classical separation relies solely on the correctness of quantum mechanics, persisting even if P=NP, BPP=BQP, etc.

Given sufficient time

$\gg$

hundreds of logical qubits

universe-scale classical computer

Conclusions

Machine learning offers a fundamental test of quantum mechanics at the complexity frontier.

[The convenient illusion of Hilbert space, PRL'11; Quantum information supremacy, 2509.07255]

Given sufficient time

$\gg$

hundreds of logical qubits

universe-scale classical computer

Future directions

More useful other applications in ML and beyond

e.g., [Bouland, Ding, Jain'26], [Wu, Yang'26]: optimization with DQI
[Voronova'26]: atomized refreshable hidden matching

[JPMorganChase, 2610.42069]:
exp space adv in tasks previously dequantized

More flexible trainable, hybrid, & heuristic components

More practical hardware, QEC, and architecture co-design